

The Lung

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The lungs are ingeniously constructed to carry out their cardinal function, the exchange of gases between inspired air and blood. Developmentally, the respiratory system is an outgrowth from the ventral wall of the foregut. The midline trachea develops two lateral outpockets, the lung buds, which eventually divide into branches called lobar bronchi, three on the right and two on the left, giving rise to three lobes on the right and two on the left. The lobar bronchi allow air to pass in and out of the lung. They have firm cartilaginous walls that provide mechanical support and are lined by columnar ciliated epithelium with abundant subepithelial glands that produce mucus, which impedes the entry of microbes. The right mainstem bronchus is more vertical and directly in line with the trachea. Consequently, aspirated foreign materials such as vomitus, blood, and foreign bodies tend to enter the right lung more often than the left.

The right and left lobar bronchi branch to give rise to progressively smaller airways that are accompanied by a dual arterial supply from the pulmonary and bronchial arteries. Distally, the bronchi give way to *bronchioles*, which are distinguished by the lack of cartilage and the presence of submucosal glands within their walls. Further branching of bronchioles leads to the *terminal bronchioles*, which are less than 2 mm in diameter. Beyond the terminal bronchiole is the *acinus*, a roughly spherical structure with a diameter of about 7 mm. An acinus is composed of a *respiratory bronchiole* (which gives off several alveoli from its sides), *alveolar ducts*, and *alveolar sacs*, the blind ends of the respiratory passages, whose walls are formed entirely of alveoli, the site of gas exchange. A cluster of three to five terminal bronchioles, each with its appended acinus, is referred to as the *pulmonary lobule*.

Except for the true vocal cords, which are covered by stratified squamous epithelium, the entire respiratory tree, including the larynx, trachea, and bronchioles, is normally lined mainly by tall, columnar, pseudostratified, ciliated epithelial cells and a smaller population of nonciliated cells called club cells that secrete several protective substances into the airway. There are also scattered cells called ionocytes that express high levels of the cystic fibrosis transmembrane conductance regulator (CFTR) and appear to modulate the ion content and viscosity of bronchial secretions. In addition, the bronchial mucosa contains neuroendocrine cells with neurosecretory-type granules that can release a variety of factors including serotonin, calcitonin, and gastrin-releasing peptide (bombesin). Numerous mucus-secreting goblet cells and submucosal glands are also dispersed throughout the walls of the trachea and bronchi (but not the bronchioles).

The microscopic structure of the alveolar walls (or alveolar septa) consists of (Fig. 15.1):

- An intertwining network of *anastomosing capillaries* lined with endothelial cells.
- *Basement membrane and surrounding interstitium*, which separates the endothelial cells from the alveolar lining epithelial cells. In thin portions of the alveolar septum the basement membranes of epithelium and endothelium are fused, whereas in thicker portions they are separated by an interstitial space (*pulmonary interstitium*) containing fine elastic fibers, small bundles of collagen, a few fibroblast-like interstitial cells, smooth muscle cells, mast cells, and rare lymphocytes and monocytes.
- *Alveolar epithelium*, a continuous layer of two cell types: Flat, plate-like *type I pneumocytes*, covering 95% of the alveolar surface, and rounded *type II pneumocytes*. Type II cells synthesize *surfactant* (which forms a very thin layer over the alveolar cell membranes) and are involved in the repair of alveolar damage through their ability to proliferate and give rise to type I cells.

The alveolar walls are perforated by numerous *pores of Kohn*, which permit the passage of bacteria and exudate between adjacent alveoli

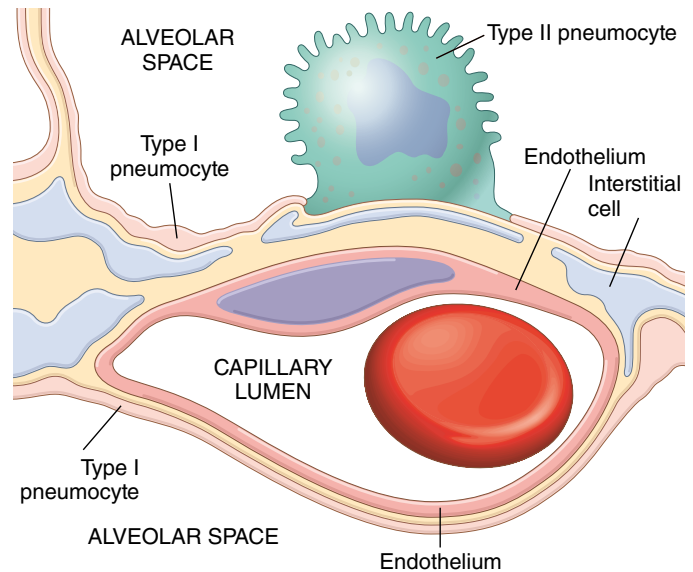


Fig. 15.1 Microscopic structure of the alveolar wall. Portions of interstitial cells are shown.

(see Fig. 15.30B). Scattered resident *alveolar macrophages* are also present, either loosely attached to epithelial cells or lying free within the alveolar spaces.

CONGENITAL ANOMALIES

Developmental anomalies of the lung are rare. The most common of these include:

- *Pulmonary hypoplasia* is a defect in the development of both lungs (one may be more affected than the other) that results in decreased lung size. It is caused by abnormalities that compress the lung or impede lung expansion in utero, such as congenital diaphragmatic hernia and oligohydramnios. Severe hypoplasia is fatal in the early neonatal period.
- *Foregut cysts* arise from abnormal detachments of primitive foregut and are most often located in the hilum or middle mediastinum. Depending on the wall structure, these cysts are classified as bronchogenic (most common), esophageal, or enteric. A *bronchogenic cyst* is rarely connected to the tracheobronchial tree. It is lined by ciliated pseudostratified columnar epithelium and has a wall containing bronchial glands, cartilage, and smooth muscle, essentially recapitulating normal airway microanatomy. Although commonly incidental findings, they may come to attention due to symptoms resulting from compression of nearby structures or superimposed infection.
- *Pulmonary sequestration* is a discrete area of lung tissue that (1) is not connected to the airways and (2) has an abnormal blood supply arising from the aorta or its branches. *Extralobar sequestration* is external to the lung and most commonly presents in infants as a mass lesion. It may be associated with other congenital anomalies. *Intralobar sequestration* occurs within the lung. It usually presents in older children, often due to recurrent localized infections or bronchiectasis.

Other, less common congenital abnormalities include tracheal and bronchial anomalies (atresia, stenosis, tracheoesophageal fistula), vascular anomalies, congenital pulmonary airway malformation, and congenital lobar overinflation (emphysema).

ATELECTASIS (COLLAPSE)

Atelectasis refers either to incomplete expansion of the lungs (neonatal atelectasis) or to the collapse of previously inflated lung, and results in areas of poorly aerated pulmonary parenchyma. The main types of acquired atelectasis, which is encountered principally in adults, are (Fig. 15.2):

- *Resorption atelectasis* stems from obstruction of an airway. Over time, air is resorbed from distal alveoli, which subsequently collapse. Since lung volume is diminished, the mediastinum shifts *toward* the atelectatic lung. Airway obstruction is most often caused by excessive secretions (e.g., mucus plugs) or exudates within smaller bronchi, as may occur in bronchial asthma, chronic bronchitis, bronchiectasis, and postoperative states. Aspiration of foreign bodies and intrabronchial tumors may also lead to airway obstruction and atelectasis.
- *Compression atelectasis* results whenever significant volumes of fluid (transudate, exudate, or blood), tumor, or air (*pneumothorax*) accumulate within the pleural cavity. With compression atelectasis, the mediastinum shifts *away* from the affected lung.
- *Contraction atelectasis* occurs when focal or generalized pulmonary or pleural fibrosis prevents full lung expansion.

Significant atelectasis reduces oxygenation and predisposes to infection. Except in cases caused by fibrosis, atelectasis is a reversible disorder.

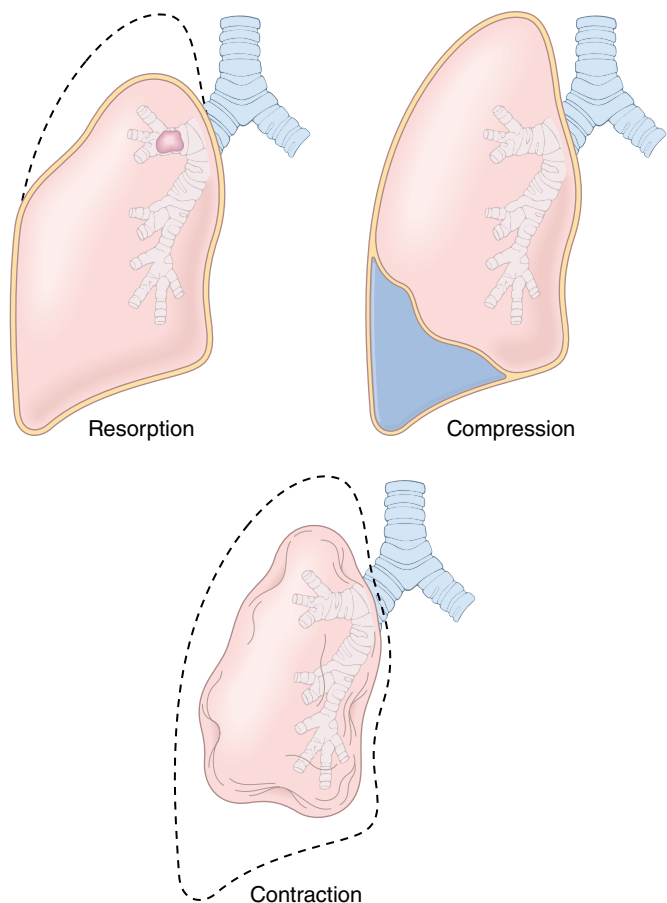


Fig. 15.2 Various forms of acquired atelectasis. *Dashed lines* indicate normal lung volume.

PULMONARY EDEMA

Pulmonary edema (excessive interstitial fluid in the alveoli) can result from hemodynamic disturbances (cardiogenic pulmonary edema) or increased capillary permeability due to microvascular injury (noncardiogenic pulmonary edema) (Table 15.1). A general consideration of edema is given in Chapter 4, and pulmonary congestion and edema are described briefly in the context of congestive heart failure (Chapter 12). Whatever the clinical setting, pulmonary congestion and edema produce heavy, wet lungs. Therapy and outcome depend on the underlying etiology.

Hemodynamic Pulmonary Edema

Hemodynamic pulmonary edema is caused by increased hydrostatic pressure and occurs most commonly in left-sided congestive heart failure. Edema accumulates initially in the basal regions of the lower lobes because hydrostatic pressure is greatest in these sites (dependent edema). Histologically, the alveolar capillaries are engorged, and there is an intraalveolar transudate that appears as finely granular pale pink material. Alveolar microhemorrhages and *hemosiderin-laden macrophages* (“heart failure” cells) may be present. In long-standing pulmonary congestion (e.g., as seen in mitral stenosis), hemosiderin-laden macrophages are abundant, and fibrosis and thickening of the alveolar walls cause the soggy lungs to become firm and brown (*brown induration*). These changes not only impair respiratory function but also predispose to infection.

TABLE 15.1 Classification and Causes of Pulmonary Edema

Hemodynamic Edema
Increased Hydrostatic Pressure (Increased Pulmonary Venous Pressure)
Left-sided heart failure (common)
Volume overload
Pulmonary vein obstruction
Decreased Oncotic Pressure (Less Common)
Hypoalbuminemia
Nephrotic syndrome
Liver disease
Protein-losing enteropathies
Lymphatic Obstruction (Rare)
Edema Due to Alveolar Wall Injury (Microvascular or Epithelial Injury)
Direct Injury
Infections: Bacterial pneumonia
Inhaled gases: High concentrations of oxygen, smoke
Liquid aspiration: Gastric contents, near-drowning
Radiation
Lung trauma
Indirect Injury
Systemic inflammatory response syndrome (e.g., associated with sepsis, burns, pancreatitis, extensive trauma)
Blood transfusion-related
Drugs and chemicals: Chemotherapeutic agents (bleomycin), other medications (methadone, amphotericin B), heroin, cocaine, kerosene, paraquat
Edema of Undetermined Origin
High altitude
Neurogenic (central nervous system trauma)

Edema Caused by Microvascular (Alveolar) Injury

Noncardiogenic pulmonary edema is caused by injury of the alveolar septa. Primary injury to the vascular endothelium or damage to alveolar epithelial cells (with secondary microvascular injury) produces an inflammatory exudate that leaks into the interstitial space and, in more severe cases, into the alveoli. Injury-related alveolar edema is an important feature of a serious and often fatal condition, *acute respiratory distress syndrome* (discussed below).

ACUTE LUNG INJURY AND ACUTE RESPIRATORY DISTRESS SYNDROME (DIFFUSE ALVEOLAR DAMAGE)

Acute lung injury (ALI) is characterized by the abrupt onset of hypoxemia and bilateral pulmonary edema in the absence of cardiac failure (noncardiogenic pulmonary edema). Acute respiratory distress syndrome (ARDS) is the clinical manifestation of severe ALI. ALI/ARDS is associated with increases in pulmonary vascular permeability, edema, and epithelial cell death. These processes result in a pattern of histologic changes called *diffuse alveolar damage* (DAD).

ALI/ARDS is a well-recognized complication of diverse conditions including both pulmonary and systemic disorders (Table 15.2). In

TABLE 15.2 Conditions Associated with Development of Acute Respiratory Distress Syndrome

Infection
Sepsis ^a
Diffuse pulmonary infections ^a
Viral (e.g. SARS-CoV-2), Mycoplasma, and pneumocystis pneumonia; miliary tuberculosis
Gastric aspiration ^a
Physical/Injury
Mechanical trauma including head injuries ^a
Pulmonary contusions
Near-drowning
Fractures with fat embolism
Burns
Ionizing radiation
Inhaled Irritants
Oxygen toxicity
Smoke
Irritant gases and chemicals
Chemical Injury
Heroin or methadone overdose
Acetylsalicylic acid
Barbiturate overdose
Paraquat
Hematologic Conditions
Transfusion-associated lung injury
Disseminated intravascular coagulation
Pancreatitis
Uremia
Cardiopulmonary Bypass
Hypersensitivity Reactions
Organic solvents
Drugs

^aMore than 50% of cases of acute respiratory distress syndrome are associated with these four conditions.

many cases, several predisposing conditions are present (e.g., shock, oxygen therapy, and sepsis). In other uncommon instances, ALI appears acutely in the absence of known triggers and follows a rapidly progressive clinical course, a condition known as *acute interstitial pneumonia*. The distinction between ALI/ARDS and acute interstitial pneumonia is entirely clinical and relates primarily to the presence of an identifiable trigger.

Pathogenesis

ALI/ARDS is initiated by injury of pneumocytes and pulmonary endothelium, setting in motion a vicious cycle of increasing inflammation and pulmonary damage (Fig. 15.3).

- **Endothelial activation** is an important early event. In some instances, endothelial activation is secondary to pneumocyte injury, which is sensed by resident alveolar macrophages. In response, these immune sentinels secrete mediators such as tumor necrosis factor (TNF) that act on the neighboring endothelium. Alternatively, circulating inflammatory mediators may activate pulmonary endothelium directly in the setting of severe tissue injury or sepsis. Some of these mediators injure endothelial cells, while others (notably cytokines) induce endothelial cells to express increased levels of adhesion molecules, procoagulant proteins, and chemokines.
- **Adhesion and extravasation of neutrophils.** Neutrophils adhere to the activated endothelium and migrate into the interstitium and the alveoli, where they degranulate and release inflammatory mediators including proteases, reactive oxygen species, and cytokines. Experimental evidence suggests that neutrophil extracellular traps (NETs) are released and also contribute directly to lung damage. These injuries and associated proinflammatory factors create the feedback loop of inflammation and endothelial damage that lies at the heart of ALI/ARDS.
- **Accumulation of intraalveolar fluid and formation of hyaline membranes.** Endothelial activation and injury make pulmonary capillaries leaky, allowing interstitial and intraalveolar edema fluid to form. Damage and necrosis of type II alveolar pneumocytes lead to surfactant abnormalities, further compromising alveolar gas exchange. Ultimately the inspissated protein-rich edema fluid and debris from dead alveolar epithelial cells organize into hyaline membranes, the characteristic histologic feature of DAD in ALI/ARDS.
- **Resolution of injury** is impeded in ALI/ARDS due to epithelial necrosis and inflammatory damage that impairs the ability of remaining cells to assist with edema resorption. Eventually, however, if the triggers of inflammation abate, macrophages remove intraalveolar debris and release fibrogenic cytokines such as transforming growth factor β (TGF- β) and platelet-derived growth factor. These factors stimulate fibroblast growth and collagen deposition, leading to fibrosis of alveolar walls. Some patients who survive this fibroproliferative phase enter an extended phase of resolution that may last several months. Residual type II pneumocytes proliferate to replace type I pneumocytes, reconstituting the alveolar lining. Endothelial restoration occurs through proliferation of uninjured capillary endothelium. In other cases, however, disease progression causes impaired cardiopulmonary function.

The lesions in ALI/ARDS are not evenly distributed, and as a result there are typically areas that are stiff and poorly aerated and regions that have nearly normal levels of compliance and ventilation. Because poorly aerated regions continue to be perfused, there is a mismatch of ventilation and perfusion, a phenomenon that exacerbates hypoxemia and cyanosis.

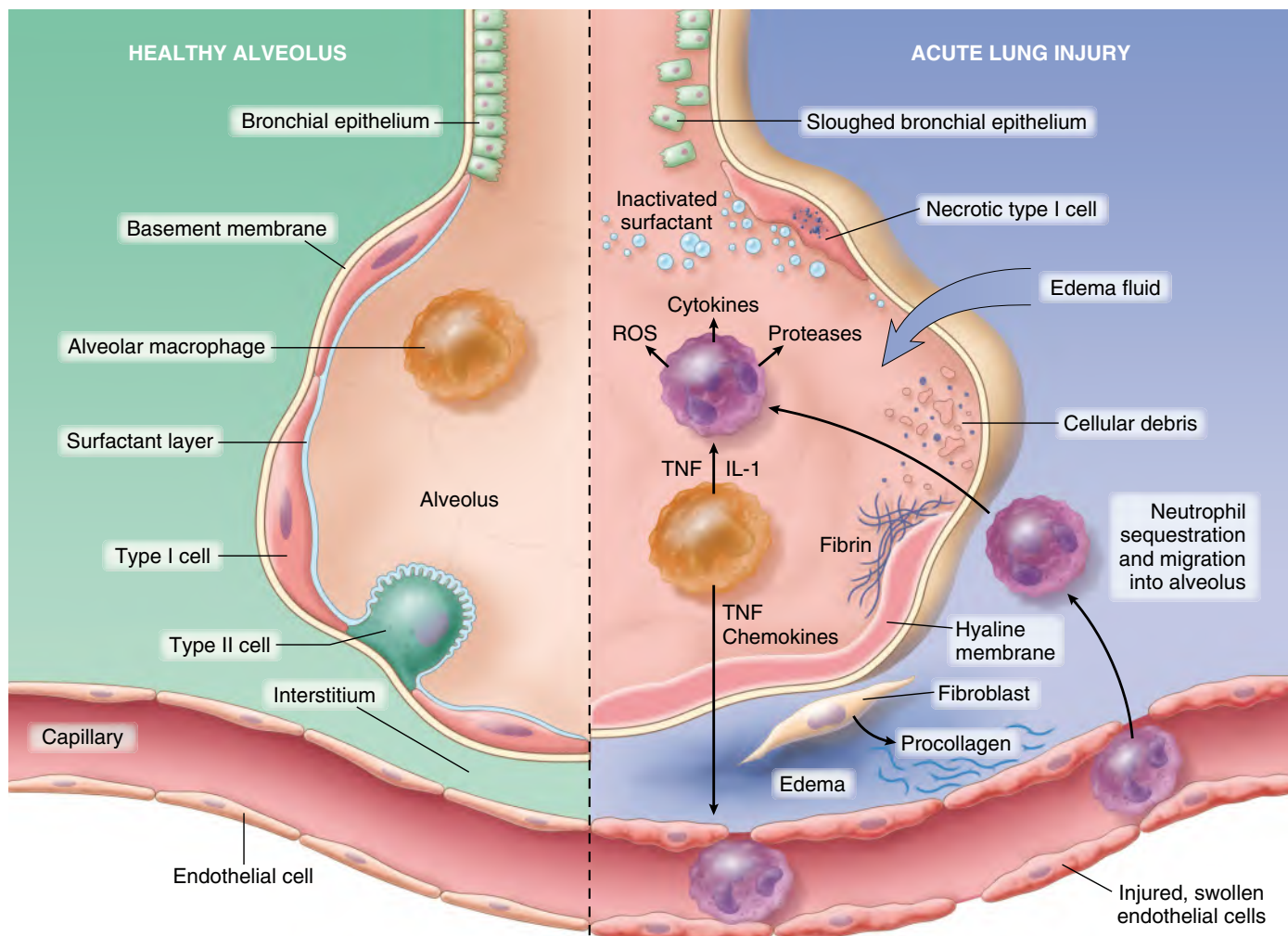


Fig. 15.3 The normal alveolus (*left*) compared with the injured alveolus (*right*) in the early phase of acute lung injury and acute respiratory distress syndrome. IL-1, Interleukin-1; ROS, reactive oxygen species; TNF, tumor necrosis factor. (Modified from Matthay MA, Ware LB, Zimmerman GA. The acute respiratory distress syndrome. *J Clin Invest*. 2012;122:2731-2740.)

Epidemiologic studies have shown that ALI/ARDS is more common and has a worse prognosis in individuals with alcohol use disorder and in smokers. Genome-wide association studies have identified several genetic variants that increase the risk of ARDS, some of which map to genes linked to inflammation and coagulation.

MORPHOLOGY

In the acute exudative stage, the lungs are heavy, firm, red, and boggy. They exhibit congestion, interstitial and intraalveolar edema, inflammation, fibrin deposition, and **diffuse alveolar damage**. This manifests radiologically as patchy ground-glass opacities with or without consolidation (Fig. 15.4A). By histology, the alveolar walls become lined with waxy **hyaline membranes** (Fig. 15.4B) that are morphologically similar to those seen in hyaline membrane disease of neonates (Chapter 10). Alveolar hyaline membranes consist of fibrin-rich edema fluid mixed with the remnants of necrotic epithelial cells. In the proliferative or organizing stage, type II pneumocytes proliferate, and granulation tissue forms in the alveolar walls and spaces (Fig. 15.4C).

Clinical Features

Profound dyspnea and tachypnea herald ALI/ARDS, followed by increasing respiratory failure, hypoxemia, cyanosis, and the appearance of diffuse bilateral infiltrates on radiographic examination. Hypoxemia may be refractory to oxygen therapy due to ventilation-perfusion mismatch, and respiratory acidosis can develop. Early in the course, the lungs become stiff due to loss of functional surfactant, leading to the need for intubation and high ventilatory pressures to maintain adequate gas exchange.

There are no proven specific treatments for ALI/ARDS, which is common in acutely ill patients and continues to take a high toll, even in patients receiving state-of-the-art supportive care. In a 2016 study of intensive care units in 50 countries, the incidence of ARDS was 10.4%, and mortality rates were 35% for mild, 40% for moderate, and 46% for severe ARDS. Epidemiological studies suggest that there are approximately 190,000 cases of ARDS in the United States each year. Most deaths are attributable to sepsis, multiorgan failure, or severe lung injury. Survivors usually recover pulmonary function, but in a minority of patients lung damage results in interstitial fibrosis and chronic pulmonary disease.

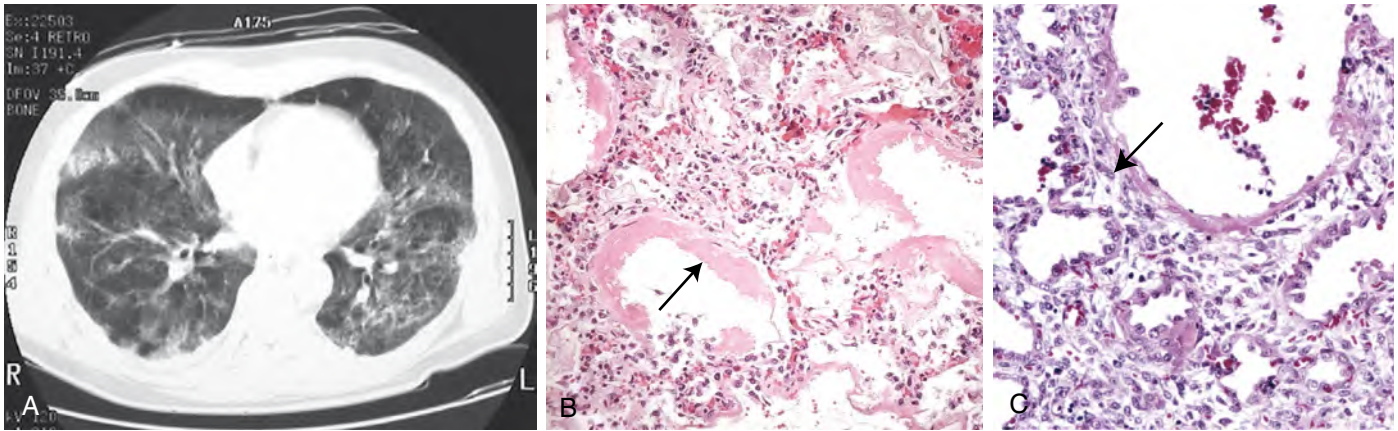


Fig. 15.4 Diffuse alveolar damage (acute respiratory distress syndrome [ARDS]). (A) Radiography showing multifocal opacities typical of ARDS. (B–C) High-power images of acute and organizing diffuse alveolar damage with characteristic histologic changes. Some of the alveoli are collapsed, while others are distended. Many are lined by hyaline membranes (*arrow*). With time, the interstitium begins to organize (C) and shows increased immature fibroblasts in the interstitial space (*arrow*).

KEY CONCEPTS

Acute Respiratory Distress Syndrome

- ARDS is a clinical syndrome of progressive respiratory insufficiency in the setting of sepsis, severe trauma, diffuse pulmonary infection, or other identifiable triggers.
- Damage to endothelial and alveolar epithelial cells and secondary inflammation are the key initiating events and result in a pattern of injury known as diffuse alveolar damage.
- The characteristic histologic finding in diffuse alveolar damage is hyaline membranes lining alveolar walls accompanied by edema, scattered neutrophils and macrophages, and epithelial cell necrosis.
- With improved care, in most cases complete recovery occurs but a small number progress to diffuse fibrosis.

OBSTRUCTIVE AND RESTRICTIVE LUNG DISEASES

Obstructive lung diseases are characterized by an increase in resistance to airflow due to diffuse airway disease, which may affect any level of the respiratory tract. By contrast, restrictive diseases are marked by reduced expansion of lung parenchyma and decreased total lung capacity. The clinical distinction between these diseases is based

primarily on pulmonary function tests. In individuals with diffuse obstructive disorders, pulmonary function tests show decreased maximal airflow rates during forced expiration, usually expressed as forced expiratory volume at 1 second (FEV_1) over forced ventilatory capacity (FVC). An FEV_1/FVC ratio of less than 0.7 generally indicates obstructive disease. Expiratory airflow obstruction may be caused by a variety of conditions (Table 15.3), each with characteristic pathologic changes and different mechanisms of airflow obstruction. As discussed later, however, the distinctions between these forms of obstructive lung disease are not absolute, and many patients have diseases with overlapping features. By contrast, restrictive diseases are associated with proportionate decreases in both total lung capacity and FEV_1 , such that the FEV_1/FVC ratio remains normal. Restrictive defects occur in two broad kinds of conditions: (1) *chest wall disorders* (e.g., severe obesity, pleural diseases, kyphoscoliosis, and neuromuscular diseases such as poliomyelitis) and (2) *chronic interstitial and infiltrative diseases*, such as pneumoconioses and interstitial fibrosis.

OBSTRUCTIVE LUNG DISEASES

Common obstructive lung diseases include chronic obstructive pulmonary disease (COPD), asthma, and bronchiectasis (Table 15.3).

TABLE 15.3 Disorders Associated with Airflow Obstruction: The Spectrum of Chronic Obstructive Pulmonary Disease

Clinical Term	Anatomic Site	Major Pathologic Changes	Etiology	Signs/Symptoms
Chronic bronchitis	Bronchus	Mucous gland hyperplasia, hypersecretion	Tobacco smoke, air pollutants	Cough, sputum production
Bronchiectasis	Bronchus	Airway dilation and scarring	Persistent or severe infections	Cough, purulent sputum, fever
Asthma	Bronchus	Smooth muscle hyperplasia, excess mucus, inflammation	Immunologic or undefined causes	Episodic wheezing, cough, dyspnea
Emphysema	Acinus	Airspace enlargement; wall destruction	Tobacco smoke	Dyspnea
Small airways disease, bronchiolitis ^a	Bronchiole	Inflammatory scarring/obliteration	Tobacco smoke, air pollutants, miscellaneous	Cough, dyspnea

^aCan be seen with any form of obstructive lung disease or as an isolated finding.

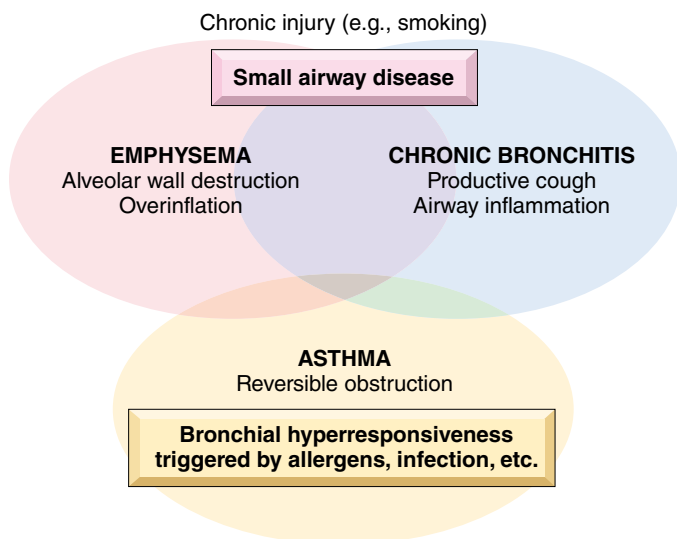


Fig. 15.5 Schematic representation of overlap between chronic obstructive lung diseases.

COPD has two major clinicopathologic manifestations, emphysema and chronic bronchitis, which are often found together in the same patient almost certainly because they share the same major etiologic factor—cigarette smoking. While asthma is distinguished from chronic bronchitis and emphysema by the presence of reversible bronchospasm, some patients with otherwise typical asthma also develop an irreversible component (Fig. 15.5). Conversely, some patients with otherwise typical COPD have a reversible component. Clinicians commonly label such patients as having COPD/asthma.

Chronic Obstructive Pulmonary Disease

COPD, a major public health problem, is defined by the World Health Organization (WHO) as “a common, preventable and treatable disease that is characterized by persistent respiratory symptoms and airflow limitation that is due to airway and/or alveolar abnormalities caused by exposure to noxious particles or gases.” It is currently the fourth leading cause of death in the world and is projected to rank third by 2020 due to increases in cigarette smoking in countries such as China. There is a strong association between heavy cigarette smoking and COPD. Overall, 35% to 50% of heavy smokers develop COPD and about 80% of COPD is attributable to smoking. Additional risk factors include poor lung development early in life, exposure to environmental and occupational pollutants, airway hyperresponsiveness, and certain genetic polymorphisms.

Recognizing that emphysema and chronic bronchitis often occur together in patients with COPD, it is still useful to discuss these patterns of lung injury and associated functional abnormalities individually to highlight the pathophysiologic basis of different causes of airflow obstruction.

Emphysema

Emphysema is defined by irreversible enlargement of the airspaces distal to the terminal bronchiole, accompanied by destruction of alveolar walls. Emphysema is classified according to its anatomic distribution within the lobule. Recall that the lobule is a cluster of acini, the terminal respiratory units. Based on the segments of the respiratory units that are involved, emphysema is subdivided into four major types: (1) *centriacinar*, (2) *panacinar*, (3) *paraseptal*, and

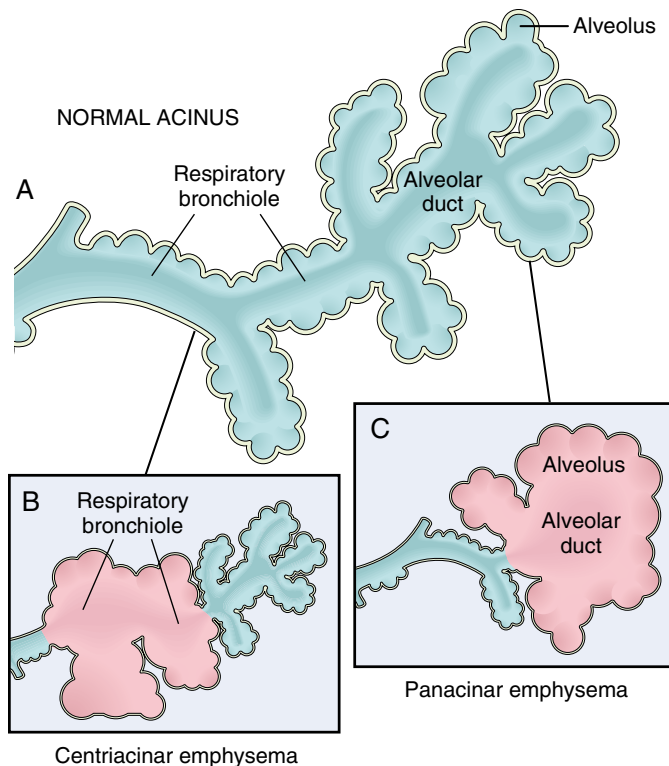


Fig. 15.6 Clinically significant patterns of emphysema. (A) Structure of the normal acinus. (B) Centriacinar emphysema with dilation that initially affects the respiratory bronchioles. (C) Panacinar emphysema with initial distention of the alveolus and alveolar duct.

(4) *irregular*. Of these, only the first two cause clinically significant airflow obstruction (Fig. 15.6). Subtle but functionally important small airway fibrosis (distinct from chronic bronchitis) is also usually present and is a significant contributor to airflow obstruction.

- **Centriacinar (centrilobular) emphysema.** Centriacinar emphysema is the most common form, constituting more than 95% of clinically significant cases. It occurs predominantly in heavy smokers with COPD. In this type of emphysema the central or proximal parts of the acini, formed by respiratory bronchioles, are affected, whereas distal alveoli are spared (Figs. 15.6B and 15.7B). Thus, both emphysematous and normal airspaces exist within the same acinus and lobule. The lesions are more common and usually more pronounced in the upper lobes, particularly in the apical segments. In severe centriacinar emphysema, the distal acinus may also be involved, making differentiation from panacinar emphysema difficult.
- **Panacinar (panlobular) emphysema.** Panacinar emphysema is associated with α_1 -antitrypsin deficiency (Chapter 18) and is exacerbated by smoking. In this type, the acini are uniformly enlarged from the level of the respiratory bronchiole to the terminal blind alveoli (Figs. 15.6C and 15.7C). In contrast to centriacinar emphysema, panacinar emphysema tends to occur more commonly in the lower zones and in the anterior margins of the lung, and it is usually most severe at the bases.
- **Distal acinar (paraseptal) emphysema.** Distal acinar emphysema probably underlies many cases of spontaneous pneumothorax in young adults. In this type, the proximal portion of the acinus is normal and the distal part is predominantly involved. The emphysema is more striking adjacent to the pleura, along the lobular

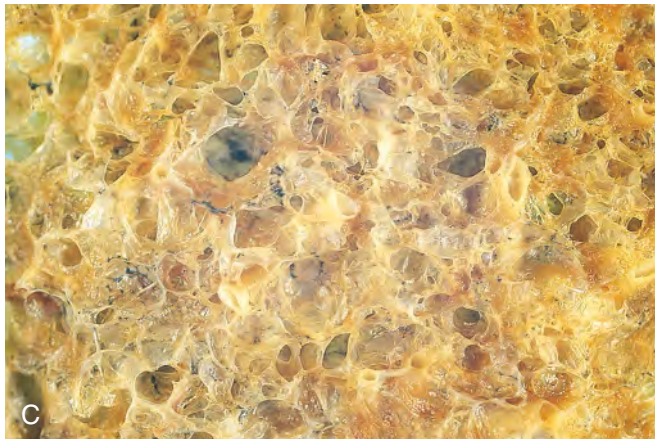
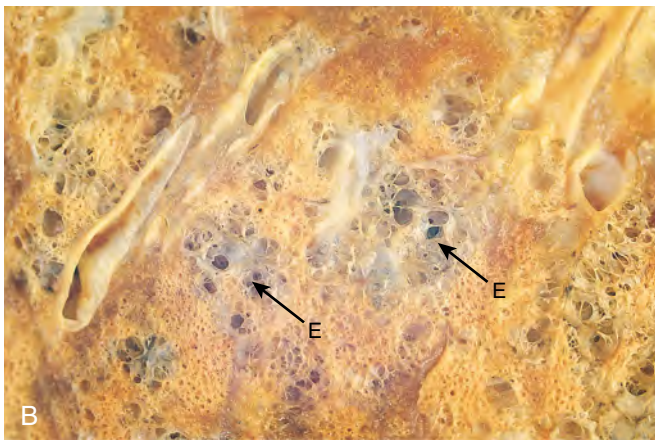
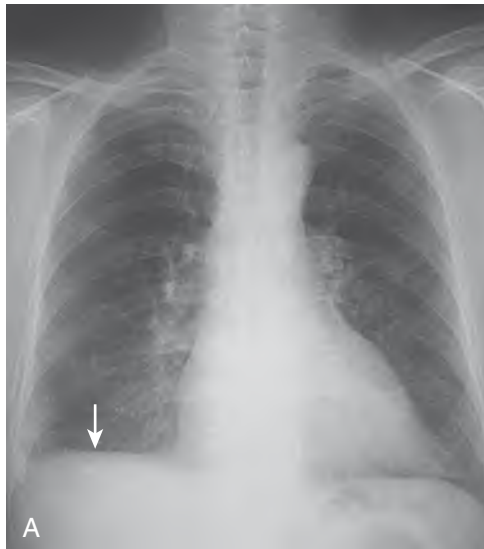


Fig. 15.7 Emphysema. (A) Lung radiograph of patient with advanced emphysema showing flattened diaphragm (arrow). (B) Centriacinar emphysema. Central areas show marked emphysematous damage (E) surrounded by relatively spared alveolar spaces. (C) Panacinar emphysema involving the entire pulmonary lobule.

connective tissue septa, and at the margins of the lobules. It occurs adjacent to areas of fibrosis, scarring, or atelectasis and is usually more severe in the upper half of the lungs. The characteristic finding is multiple enlarged airspaces, ranging from less than 0.5 cm to more than 2.0 cm in diameter, which sometimes form cyst-like structures.

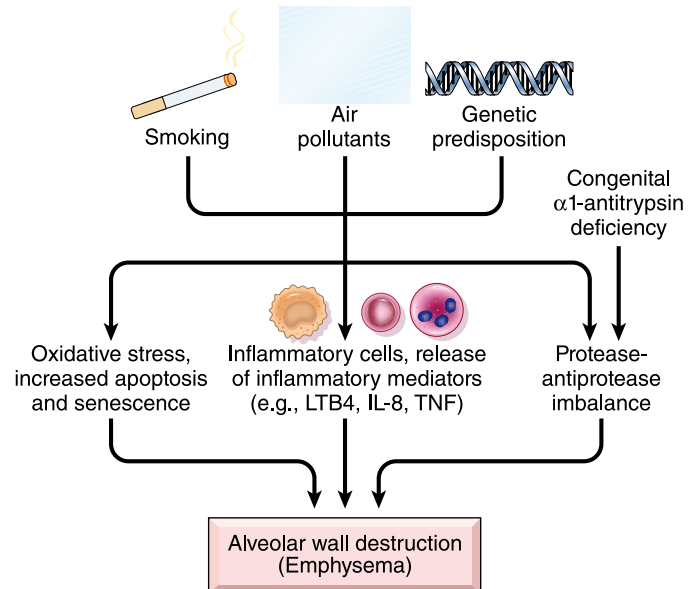


Fig. 15.8 Pathogenesis of emphysema. See text for details.

- *Irregular (paracicatricial) emphysema.* Irregular emphysema, so named because the acinus is irregularly involved, is almost invariably associated with scarring. In most instances it occurs in small foci and is clinically insignificant.

Pathogenesis

Clinically significant emphysema is largely confined to smokers and to patients with α_1 -antitrypsin deficiency, highlighting the importance of these two etiologic factors. Mechanisms relating to these factors that contribute to the development of emphysema include (Fig. 15.8):

- *Toxic injury and inflammation.* Inhaled cigarette smoke and other noxious particles damage respiratory epithelium and cause inflammation, which results in variable degrees of parenchymal destruction. A wide variety of inflammatory mediators (including leukotriene B_4 , interleukin [IL]-8, TNF, and others) are increased in the affected parts of the lung. These mediators are released by resident epithelial cells and macrophages and variously attract inflammatory cells from the circulation (chemotactic factors), amplify the inflammatory process (proinflammatory cytokines), and induce structural changes (growth factors). Chronic inflammation also leads to the accumulation of T and B cells in affected parts of the lung, but the role of adaptive immunity in emphysema is currently uncertain.
- *Protease-antiprotease imbalance.* Several proteases are released from the inflammatory cells and epithelial cells that break down connective tissue components. In patients who develop emphysema, there is a relative deficiency of protective antiproteases, which in some instances has a genetic basis (discussed later).
- *Oxidative stress.* Substances in tobacco smoke, alveolar damage, and inflammatory cells all produce oxidants, which may cause tissue damage, endothelial dysfunction, and inflammation. The role of oxidants is supported by studies of mice in which the *NFE2L2* gene is inactivated. *NFE2L2* encodes NRF2, a transcription factor that serves as a sensor for oxidants in many cell types including alveolar epithelial cells. Intracellular oxidants activate NRF2, which upregulates the expression of genes that protect cells from oxidant damage. Mice without NRF2 are significantly more sensitive to

tobacco smoke than normal mice, and genetic variants in NRF2, NRF2 regulators, and NRF2 target genes are all associated with smoking-related lung disease in humans.

- **Infection.** Although infection is not thought to play an initiating role in the tissue destruction, bacterial and/or viral infections may acutely exacerbate existing disease.

The idea that proteases are important is based in part on the observation that patients with a genetic deficiency of the antiprotease α_1 -antitrypsin have a markedly enhanced tendency to develop emphysema that is compounded by smoking. About 1% of all patients with emphysema have this defect. α_1 -Antitrypsin, normally present in serum, tissue fluids, and macrophages, is a major inhibitor of proteases (particularly elastase) secreted by neutrophils during inflammation. It is encoded by the proteinase inhibitor (*Pi*) locus on chromosome 14. The *Pi* locus is polymorphic, and approximately 0.012% of the US population is homozygous for the Z allele, a genotype associated with very low serum levels of α_1 -antitrypsin. More than 80% of ZZ individuals develop symptomatic panacinar emphysema, which occurs at an earlier age and is of greater severity if the individual smokes. It is postulated that any injury (e.g., that induced by smoking) that increases the activation and influx of neutrophils into the lung leads to local release of proteases, which in the absence of α_1 -antitrypsin activity result in excessive digestion of elastic tissue and, with time, emphysema.

Several other genetic variants have also been linked to risk of emphysema. Among these are variants of the nicotinic acetylcholine receptor that are hypothesized to influence the addictiveness of tobacco smoke and thus the behavior of smokers. Not surprisingly, the same variants are also linked to lung cancer risk, emphasizing the importance of smoking in both diseases.

Several factors contribute to airway obstruction in emphysema. Small airways are normally held open by the elastic recoil of the lung parenchyma. The loss of elastic tissue in the walls of alveoli that surround respiratory bronchioles reduces radial traction, leading to collapse of respiratory bronchioles during expiration and functional airflow obstruction in the absence of mechanical obstruction. In addition, even young smokers often have changes related to small airway inflammation that also contribute to airway narrowing and obstruction (described later).

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Advanced emphysema produces voluminous lungs, which often overlap the heart anteriorly and flatten the diaphragm (Fig. 15.7A). Generally, in patients with smoking-related disease, the upper two-thirds of the lungs are more severely affected. Large alveoli can easily be seen on the cut surface of fixed lungs (Fig. 15.7B and C). Apical blebs or bullae characteristic of irregular emphysema may appear in patients with advanced disease (Fig. 15.9).

Microscopically, abnormally large alveoli are separated by thin septa with focal centriacinar fibrosis. There is loss of attachments between alveoli and the outer wall of small airways. The pores of Kohn are so large that septa appear to be floating or protrude blindly into alveolar spaces with a club-shaped end. As alveolar walls are destroyed, there is a decrease in the capillary bed area. With advanced disease, there are even larger abnormal airspaces and possibly blebs or bullae, which often deform and compress the respiratory bronchioles and vasculature of the lung. Inflammatory changes in small airways are often superimposed (described next under chronic bronchitis), as are vascular changes related to pulmonary hypertension stemming from local hypoxemia and loss of capillary beds.



Fig. 15.9 Bullous emphysema. Note the large subpleural bullae (upper left).

Chronic Bronchitis

Chronic bronchitis is defined clinically as persistent cough with sputum production for at least 3 months in at least 2 consecutive years in the absence of any other identifiable cause. Long-standing chronic bronchitis is associated with progressive lung dysfunction, which may be so severe as to lead to hypoxemia, pulmonary hypertension, and cor pulmonale.

Pathogenesis

The primary or initiating factor in the genesis of chronic bronchitis is exposure to noxious or irritating inhaled substances such as tobacco smoke (90% of those affected are smokers) and dust from grain, cotton, and silica. Several factors contribute to its pathogenesis:

- **Mucus hypersecretion.** The earliest feature of chronic bronchitis is hypersecretion of mucus in the large airways, associated with enlargement of the submucosal glands in the trachea and bronchi. The basis for mucus hypersecretion is incompletely understood, but it appears to involve inflammatory mediators such as histamine and IL-13. With time, there is also a marked increase in goblet cells in small airways—small bronchi and bronchioles—leading to excessive mucus production that contributes to airway obstruction. It is thought that both the enlargement of submucosal glands and the increase in numbers of goblet cells are protective reactions against tobacco smoke or other pollutants (e.g., sulfur dioxide and nitrogen dioxide).
- **Acquired cystic fibrosis transmembrane conductance regulator (CFTR) dysfunction.** There is substantial evidence that smoking leads to acquired CFTR dysfunction, which in turn causes the secretion of abnormal, dehydrated mucus that exacerbates the severity of chronic bronchitis.
- **Inflammation.** Inhalants that induce chronic bronchitis cause cellular damage, eliciting both acute and chronic inflammatory responses involving neutrophils, lymphocytes, and macrophages.

Long-standing inflammation and accompanying fibrosis involving small airways (small bronchi and bronchioles, less than 2 to 3 mm in diameter) can also lead to chronic airway obstruction.

- **Infection.** Infection does not initiate chronic bronchitis but is probably significant in maintaining it and may be critical in producing acute exacerbations.

Cigarette smoke predisposes to chronic bronchitis in several ways. Not only does it damage airway-lining cells, leading to chronic inflammation, but it also interferes with the ciliary action of the respiratory epithelium, preventing the clearance of mucus and increasing the risk of infection.

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Grossly, there is hyperemia, swelling, and edema of the mucous membranes, frequently accompanied by excessive mucinous or mucopurulent secretions. Sometimes, heavy casts of secretions and pus fill the bronchi and bronchioles. The characteristic microscopic features are chronic inflammation of the airways (predominantly lymphocytes and macrophages); thickening of the bronchiolar wall due to smooth muscle hypertrophy, deposition of extracellular matrix in the muscle layer, and peribronchial fibrosis; goblet cell hyperplasia; and enlargement of the mucus-secreting glands of the trachea and bronchi. Of these, the most striking change is an increase in the size of the mucous glands. This increase can be assessed by the ratio of the thickness of the mucous gland layer to the thickness of the wall between the epithelium and the cartilage (**Reid index**). The Reid index (normally 0.4) is increased in chronic bronchitis, usually in proportion to the severity and duration of the disease. The mucus plugging, inflammation, and fibrosis may lead to marked narrowing of bronchioles, and in the most severe cases, there may be obliteration of the lumen due to fibrosis (**bronchiolitis obliterans**). The bronchial epithelium may also exhibit squamous metaplasia and dysplasia due to the irritating and mutagenic effects of substances in tobacco smoke.

Clinical Features

Most affected patients have a smoking history of 40 pack-years or greater. COPD often presents insidiously with slowly increasing dyspnea on exertion and chronic cough with sputum production, slight at first but increasing over time. Other patients present with exacerbations caused by superimposed infection that can lead to confusion with other disorders, such as asthma (due to wheezing). The most important diagnostic test is spirometry, which typically shows an FEV₁/FVC ratio of less than 0.7.

Once COPD appears, symptoms often wax and wane over time and are generally worse in the morning. The clinical picture varies according to the severity of the disease and the relative contributions of emphysematous and bronchitic changes (Table 15.4). At one extreme end of the spectrum lie “pink puffers,” patients in whom emphysema dominates. Classically, the patient is barrel-chested and dyspneic, with obviously prolonged expiration, sits forward in a hunched-over position, and breathes through pursed lips. Cough is often slight, overdistention of the lungs is severe, diffusion capacity is low, and blood gas values are relatively normal at rest. Weight loss is common and can be so severe as to suggest an occult cancer. At the other end of the spectrum lie patients with pure chronic bronchitis, ingloriously referred to as “blue bloaters.” Their cardinal symptom is a persistent cough productive of sputum, coupled with hypercapnia, hypoxemia, and mild cyanosis. Most patients are somewhere in the middle, with signs and symptoms stemming from both bronchitic and emphysematous changes.

TABLE 15.4 Predominant Features of Emphysema and Chronic Bronchitis

	Bronchitis	Emphysema
Age (years)	40–45	50–75
Dyspnea	Mild; late	Severe; early
Cough	Early; copious sputum	Late; scanty sputum
Infections	Common	Occasional
Respiratory insufficiency	Early, periodic	End-stage
Cor pulmonale	Common	Uncommon, end-stage
Airway resistance	Increased	Normal or slightly increased
Elastic recoil	Normal	Low
Chest radiograph	Prominent vessels; large heart size	Hyperinflation; normal heart size
Appearance	Cyanotic	Well oxygenated

Treatment options include smoking cessation, oxygen therapy, long-acting bronchodilators with inhaled corticosteroids, antibiotics, physical therapy, bullectomy, and (in selected patients) lung volume reduction surgery and lung transplantation. Even with intervention, however, COPD often progresses and frequently proves fatal. Long-standing COPD, particularly in patients with a bronchitic component, commonly leads to pulmonary hypertension, cor pulmonale, and death due to heart failure. Death may also result from acute respiratory failure due to acute infections superimposed on COPD. In patients with emphysematous changes, subpleural blebs may rupture, leading to fatal pneumothorax. The development of *mucus plugs* has also been shown to result in increased all-cause mortality for patients with COPD. The best hope for a major change in this dire picture is more effective programs aimed at prevention of smoking and other environmental exposures.

KEY CONCEPTS

Chronic Obstructive Pulmonary Disease

- Most common in long-standing tobacco smokers (typically >40 pack-years); air pollutants also contribute
- Underlying pulmonary pathology usually includes both chronic bronchitis and emphysema
- Often fatal due to development of heart failure, or respiratory failure due to superimposed infection

Emphysema

- In COPD, usually follows a centriacinar distribution characterized by permanent enlargement of airspaces distal to terminal bronchioles
- Particularly severe in patients with α_1 -antitrypsin deficiency, in which a panacinar pattern of emphysematous change may be seen
- Tissue destruction is caused by elastases and oxidants released from inflammatory cells, particularly neutrophils, which are responding to cellular injury caused by tobacco smoke and pollutants

Chronic Bronchitis

- Defined as persistent productive cough for at least 3 consecutive months in at least 2 consecutive years
- Dominant pathologic features are mucus hypersecretion due to enlargement of mucus-secreting glands and chronic inflammation associated with bronchiolar wall fibrosis

Other Forms of Emphysema. In addition to emphysema occurring in the setting of COPD, several other conditions may be associated with lung overinflation or focal emphysematous change and are mentioned here in brief.

- **Compensatory hyperinflation.** This term is used to designate dilation of alveoli in response to loss of lung substance elsewhere, for example, following surgical removal of a lung or lobe with cancer.
- **Obstructive overinflation.** In this condition the lung expands because air is trapped within it. A common cause is subtotal obstruction of an airway by a tumor or foreign object. In infants, it may be caused by *congenital lobar overinflation*, which most often results from hypoplasia of bronchial cartilage. Overinflation occurs either (1) because of an obstruction that acts as ball valve, allowing air to enter on inspiration while preventing its exodus on expiration, or (2) because collaterals bring in air behind the obstruction. These collaterals consist of the *pores of Kohn* and other direct accessory bronchioloalveolar connections (the *canals of Lambert*). Obstructive overinflation can be life-threatening if the affected portion distends sufficiently to compress the adjacent uninvolved lung.
- **Bullous emphysema.** This is a descriptive term for large subpleural blebs or bullae (spaces greater than 1 cm in diameter in the distended state) that can occur in any form of emphysema (Fig. 15.9), often near the apex. Rupture of the bullae may give rise to pneumothorax.
- **Interstitial emphysema.** Entrance of air into the connective tissue stroma of the lung, mediastinum, or subcutaneous tissue produces *interstitial emphysema*. In most instances, it is caused by alveolar tears that occur in patients with pulmonary emphysema due to transient increases in intraalveolar pressure, for example, during coughing. Less commonly, chest wounds or fractured ribs that puncture the lung provide the portal for entrance of air into surrounding soft tissues.

Asthma

Asthma is a heterogeneous disease, usually characterized by chronic airway inflammation and variable expiratory airflow obstruction. It produces episodic wheezing, shortness of breath, chest tightness, and cough, symptoms that vary over time and in intensity. Symptomatic episodes are most likely to occur at night or in the early morning and are produced by bronchoconstriction. Between attacks, patients may be virtually asymptomatic. Rarely, an unremitting attack called *acute severe asthma* (formerly known as *status asthmaticus*) may prove fatal; usually, such patients have a long history of asthma. Of note, there has been a significant increase in the incidence of asthma in the Western world over the past 40 to 50 years, a trend that has now started to abate. However, the prevalence of asthma continues to increase in low-resource countries and in some ethnic groups in which its prevalence was previously low.

Asthma has several distinct clinical phenotypes, each with different underlying pathogenic mechanisms. It may be categorized as *atopic* (evidence of allergen sensitization and immune activation, often in a patient with allergic rhinitis or eczema) or *nonatopic* (no evidence of allergen sensitization), of which several subtypes exist. In all types, episodes of bronchospasm may have diverse triggers, such as respiratory infections (especially viral infections), irritants (e.g., smoke, fumes), cold air, stress, and exercise. One biologically meaningful and clinically useful way to classify asthma is based on its triggers. We will first briefly describe the various major subtypes of asthma and then delve more deeply into its pathogenesis.

Atopic Asthma

This type of asthma is a classic example of an IgE-mediated (type I) hypersensitivity reaction (Chapter 6). The disease usually begins in childhood and is triggered by environmental allergens, such as dusts, pollens, cockroach or animal dander, and foods, which most frequently act in synergy with other proinflammatory environmental cofactors, most notably respiratory viral infections. A positive family history of asthma is common, and a skin test with the offending antigen in these patients results in an immediate wheal-and-flare reaction. Atopic asthma may also be diagnosed based on high total serum IgE levels or evidence of allergen sensitization by serum radioallergen sorbent tests (RASTs), which can detect the presence of IgE antibodies that are specific for individual allergens.

Nonatopic Asthma

Individuals with nonatopic asthma do not have evidence of allergen sensitization, and skin test results are usually negative. A positive family history of asthma is less common in these patients. Respiratory infections due to viruses (e.g., rhinovirus, parainfluenza virus, and respiratory syncytial virus) are common triggers. Inhaled air pollutants such as tobacco smoke, sulfur dioxide, ozone, and nitrogen dioxide may also contribute to the chronic airway inflammation and hyperactivity in some cases. As already mentioned, in some instances attacks may be triggered by seemingly innocuous events, such as exposure to cold and even exercise.

Drug-Induced Asthma

Several pharmacologic agents provoke asthma. Aspirin-sensitive asthma is an uncommon type associated with recurrent rhinitis and nasal polyps. Affected individuals are exquisitely sensitive to small doses of aspirin and other nonsteroidal antiinflammatory medications, which trigger urticaria as well as asthmatic attacks. Aspirin and related drugs cause asthma in these patients by inhibiting the cyclooxygenase pathway of arachidonic acid metabolism, leading to a rapid decrease in prostaglandin E₂. Normally prostaglandin E₂ inhibits enzymes that generate proinflammatory mediators such as leukotrienes B₄, C₄, D₄, and E₄, which are believed to have central roles in aspirin-induced asthma.

Occupational Asthma

This form of asthma may be triggered by fumes (epoxy resins, plastics), organic and chemical dusts (wood, cotton, platinum), gases (toluene), or other chemicals (formaldehyde, penicillin products). Only minute quantities of chemicals are required to induce the attack, which usually occurs after repeated exposure. The underlying mechanisms vary according to stimulus and include type I reactions, direct liberation of bronchoconstrictor substances, and hypersensitivity responses of unknown origin.

Pathogenesis

Atopic asthma, the most common form of the disease, is caused by a Th2-mediated IgE response to environmental allergens in genetically predisposed individuals. Airway inflammation is central to the disease pathophysiology and causes airway dysfunction partly through the release of potent inflammatory mediators and partly through remodeling of the airway wall (Fig. 15.10). As the disease becomes more severe, there is increased local secretion of growth factors, which induce mucous gland enlargement, smooth muscle proliferation, angiogenesis, and fibrosis. Varying combinations of these processes help explain the different asthma subtypes, their response to treatment, and their natural history over a person's lifetime.

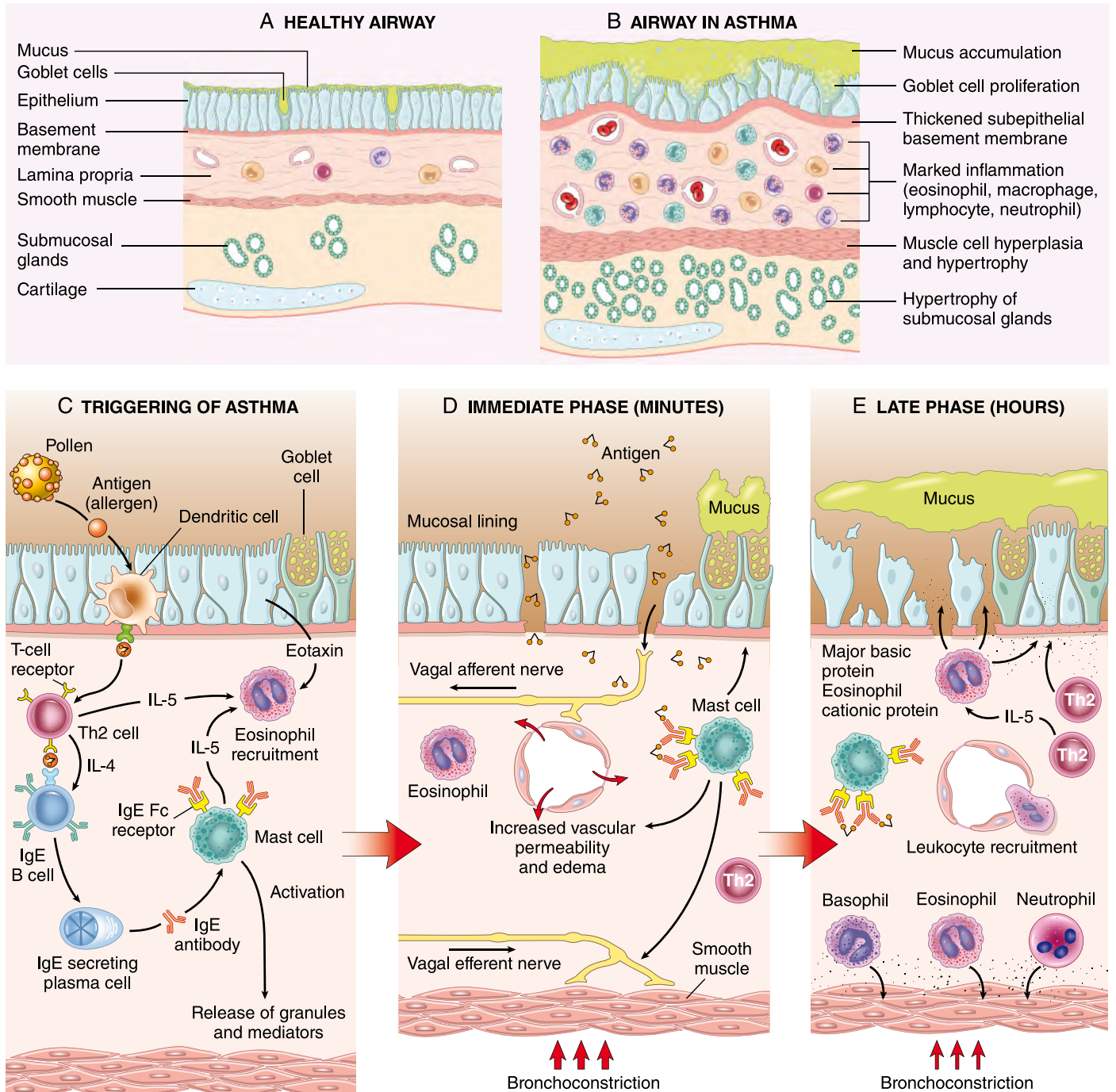


Fig. 15.10 (A–B) Comparison of a healthy airway and an airway involved by asthma. The asthmatic airway is marked by accumulation of mucus in the bronchial lumen secondary to an increase in the number of mucus-secreting goblet cells in the mucosa and hypertrophy of submucosal glands; intense chronic inflammation due to recruitment of eosinophils, macrophages, and other inflammatory cells; thickened basement membrane; and hypertrophy and hyperplasia of smooth muscle cells. (C) Inhaled allergens (antigen) elicit a Th2-dominated response favoring IgE production and eosinophil recruitment. (D) On re-exposure to antigen, the immediate reaction is triggered by antigen-induced cross-linking of IgE bound to Fc receptors on mast cells. These cells release preformed mediators that directly and via neuronal reflexes induce bronchospasm, increased vascular permeability, mucus production, and recruitment of leukocytes. (E) Leukocytes recruited to the site of reaction (neutrophils, eosinophils, and basophils; lymphocytes and monocytes) release additional mediators that initiate the late-phase reaction. Several factors released from eosinophils (e.g., major basic protein, eosinophil cationic protein) also cause damage to the epithelium. *IgE*, Immunoglobulin E; *IL-5*, interleukin-5.

The contributions of the immune response, genetics, and environment are discussed separately below, although they are closely intertwined.

Th2 Responses, IgE, and Inflammation. A fundamental abnormality in asthma is an exaggerated Th2 response to normally harmless environmental antigens. Th2 cells secrete cytokines that promote inflammation and stimulate B cells to produce IgE and other antibodies. These cytokines include IL-4, which stimulates the production of IgE; IL-5, which activates locally recruited eosinophils; and IL-13, which stimulates mucus secretion from bronchial submucosal glands and promotes IgE production by B cells. The T cells and epithelial cells secrete chemokines that recruit more T cells and eosinophils, thus exacerbating the reaction. As in other allergic reactions (Chapter 6), IgE binds to the Fc receptors on submucosal mast cells and repeat exposure to the allergen triggers the mast cells to release granule contents and produce cytokines and other mediators, which collectively induce the early phase (immediate hypersensitivity) reaction and the late-phase reaction (Fig. 15.10C–E). Tissue-resident group 2 innate lymphoid cells (ILCs) produce many of the same cytokines as Th2 cells and may contribute to the initiation of asthma. ILCs are activated not by antigens (they lack antigen receptors) but by cytokines produced from damaged epithelial cells. Because Th2 cells and ILC2s participate in the same types of allergic reactions, this type of immune response is called *type 2 immunity*.

The *early phase reaction* is dominated by bronchoconstriction, increased mucus production, variable degrees of vasodilation, and increased vascular permeability (see Fig. 15.10C). Bronchoconstriction is triggered by direct stimulation of subepithelial vagal (parasympathetic) receptors through both central and local reflexes triggered by mediators produced by mast cells and other cells in the reaction (see Fig. 15.10D). The *late-phase reaction* is dominated by recruitment of leukocytes, notably eosinophils, neutrophils, and more T cells (see Fig. 15.10E). Although Th2 cells are the dominant lymphoid cell type involved in the disease, there is evidence that other cell types, such as innate lymphoid cells, also produce factors that contribute to the inflammation.

Many mediators produced by leukocytes and epithelial cells have been implicated in the asthmatic response. The long list of “suspects” in acute asthma can be ranked based on the clinical efficacy of pharmacologic intervention with antagonists of specific mediators.

- Mediators whose role in bronchospasm is clearly supported by efficacy of pharmacologic intervention are (1) *leukotrienes C₄, D₄, and E₄*, which cause prolonged bronchoconstriction as well as increased vascular permeability and increased mucus secretion; (2) *acetylcholine*, released from intrapulmonary parasympathetic nerves, which can cause airway smooth muscle constriction by directly stimulating muscarinic receptors; (3) *IL-5*, antagonists of which are effective in treating severe forms of asthma that are associated with peripheral blood eosinophilia; and (4) *galectin-10* (GAL10), which is released from eosinophils and forms crystals known as *Charcot-Leyden crystals*. Long recognized as a feature of asthma, recent studies have shown that these crystals are strong inducers of inflammation and mucus production.
- A second group of agents are present at the “scene of the crime” but seem to have relatively minor contributions because of lack of efficacy of potent antagonists or synthesis inhibitors. These include (1) *histamine*, a potent bronchoconstrictor; (2) *prostaglandin D₂*, which elicits bronchoconstriction and vasodilation; and (3) *platelet-activating factor*, which causes aggregation of platelets and release of serotonin from their granules. These mediators might yet prove important in certain types of chronic or nonallergic asthma.

- Finally, a large third group comprises “suspects” for whom specific antagonists or inhibitors are not available or have been insufficiently studied as yet.

It is thus clear that multiple mediators contribute to the acute asthmatic response. Moreover, the composition of this “mediator soup” likely varies among individuals or different types of asthma. The appreciation of the importance of inflammatory cells and mediators in asthma has led to greater emphasis on antiinflammatory drugs, such as corticosteroids, in its treatment.

Genetic Susceptibility. Susceptibility to atopic asthma is multigenic and often associated with increased incidence of other allergic disorders, such as allergic rhinitis (hay fever) and eczema. Genetic polymorphisms linked to asthma and other allergic disorders were described in Chapter 6. Suffice it to say here that many of these are likely to influence immune responses and the subsequent inflammatory reaction. Some of the genetic variants most strongly associated with asthma include:

- *A susceptibility locus for asthma located on chromosome 5q near the gene cluster encoding the cytokines IL-3, IL-4, IL-5, IL-9, and IL-13 and the IL-4 receptor.* Among the genes in this cluster, polymorphisms in the *IL13* gene have the strongest and most consistent associations with asthma or allergic disease, while *IL-4* receptor gene variants are associated with atopy, elevated total serum IgE, and asthma.
- *Particular HLA class II alleles* linked to production of IgE antibodies against some antigens, such as ragweed pollen.
- *Variants associated with the genes encoding IL-33*, a member of the IL-1 family of cytokines, and its receptor, ST2, which induce the production of Th2 cytokines.
- *Variants associated with the gene encoding thymic stromal lymphopoietin (TSLP)*, a cytokine produced by epithelium that may have a role in initiating allergic reactions.

Environmental Factors. Asthma is a disease of high-resource societies where most people live in cities. Two ideas, neither wholly satisfying, have been proposed to explain this association. First, industrialized environments contain many airborne pollutants that can serve as allergens to initiate the Th2 response. Second, city life tends to limit the exposure of very young children to certain antigens, particularly microbial antigens, and exposure to such antigens may protect children from asthma and atopy. The idea that microbial exposure during early life reduces the later incidence of allergic (and some autoimmune) diseases has been popularized as the hygiene hypothesis. Although the underlying mechanisms of this protective effect are unclear, it has spurred trials of probiotics and intentional early exposure of children to putative allergens to decrease their risk of later developing allergies.

Infections do not cause asthma by themselves but may be important cofactors. Young children with aeroallergen sensitization who develop lower respiratory tract viral infections (rhinovirus type C, respiratory syncytial virus) have a 10- to 30-fold increased risk of developing persistent and/or severe asthma. Both viral and bacterial infections (identified by cultures and nonculture tools) are associated with acute exacerbations of the disease.

Over time, repeated bouts of allergen exposure and immune reactions result in structural changes in the bronchial wall, referred to as *airway remodeling*. These changes, described later in greater detail, include hypertrophy and hyperplasia of bronchial smooth muscle, epithelial injury, increased airway vascularity, subepithelial mucous gland enlargement, and subepithelial fibrosis.

A small subset of patients with asthma, many of whom have severe disease that is refractory to glucocorticoids, has inflammatory infiltrates

that are enriched for neutrophils rather than eosinophils. This form of the disease may be driven by a Th17 T-cell response to chronic bacterial colonization of the lung.

MORPHOLOGY

In patients dying of acute severe asthma (status asthmaticus), the lungs are overinflated and contain small areas of atelectasis. The most striking gross finding is occlusion of bronchi and bronchioles by thick, tenacious mucus plugs, which often contain shed epithelium. A characteristic finding in sputum or bronchoalveolar lavage specimens of patients with atopic asthma is **Curschmann spirals**, which may result from extrusion of mucus plugs from subepithelial mucous gland ducts or bronchioles. Also present are numerous eosinophils and **Charcot-Leyden crystals** (Fig. 15.11B) composed of the eosinophil-derived protein galectin-10. The other characteristic histologic findings of asthma, collectively called **airway remodeling** (Figs. 15.10B and 15.11), include:

- **Thickening of airway wall**
- **Sub-basement membrane fibrosis** (due to deposition of type I and III collagens)
- **Increased vascularity**
- **Increase in the size of the submucosal glands and number of airway goblet cells**
- **Hypertrophy and/or hyperplasia of bronchial wall smooth muscle**

While acute airflow obstruction is primarily attributed to muscular bronchoconstriction, acute edema, and mucus plugging, airway remodeling may contribute to chronic irreversible airway obstruction.

Clinical Features

A classic acute asthmatic attack lasts up to several hours. In some patients, however, the cardinal symptoms of chest tightness, dyspnea, wheezing, and coughing (with or without sputum production) are present at a low level constantly. In its most severe form, acute severe asthma, the paroxysm persists for days or even weeks, sometimes causing airflow obstruction so extreme that marked cyanosis or even death ensues.

The diagnosis is based on demonstration of an increase in airflow obstruction (from baseline levels); difficulty with exhalation (prolonged

expiration, wheeze); and (in those with atopic asthma) identification of eosinophilia in the peripheral blood, and eosinophils, Curschmann spirals, and Charcot-Leyden crystals in the sputum. In the usual case with intervals of freedom from respiratory difficulty, the disease is more discouraging and disabling than lethal, and most individuals manage to maintain a productive life.

Therapy is based on the severity of the disease. The centerpieces of standard therapy are bronchodilators, glucocorticoids, and leukotriene antagonists. For severe and difficult-to-treat asthma in adolescents and adults, biologic therapies targeting inflammatory mediators such as the IL-4 receptor, IL-5, and TSLP are available. Up to 50% of childhood asthma remits in adolescence only to return in adulthood in a significant number of patients. In other cases, there is a variable decline in baseline lung function over time.

KEY CONCEPTS

Asthma

- Asthma is characterized by reversible bronchoconstriction caused by airway hyperresponsiveness to a variety of stimuli.
- Atopic asthma is caused by a Th2 and IgE-mediated immunologic reaction to environmental allergens characterized by acute-phase (immediate) and late-phase reactions. The Th2 cytokines IL-4, IL-5, and IL-13 are important mediators.
- Triggers for nonatopic asthma are less clear but include viral infections and inhaled air pollutants, which can also trigger atopic asthma.
- Eosinophils are key inflammatory cells in atopic asthma; other inflammatory cells implicated in its pathogenesis include mast cells, neutrophils, and T lymphocytes.
- Airway remodeling (sub-basement membrane fibrosis, hypertrophy of bronchial glands, and smooth muscle hyperplasia) adds an irreversible component to the obstructive disease.

Bronchiectasis

Bronchiectasis is a disorder in which destruction of smooth muscle and elastic tissue by inflammation stemming from persistent or severe infections leads to permanent dilation of bronchi and

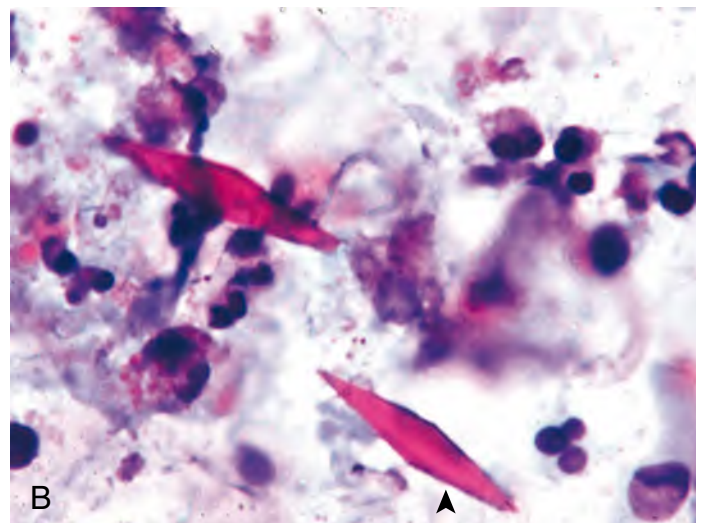
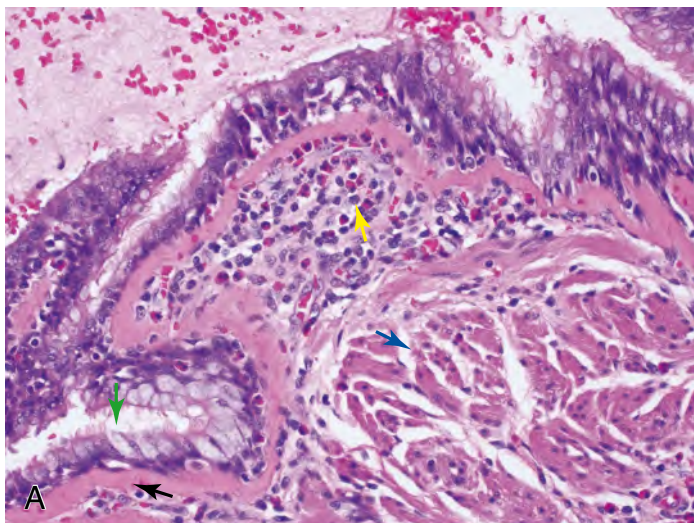


Fig. 15.11 (A) Bronchus from an asthmatic patient showing goblet cell hyperplasia (green arrow), sub-basement membrane fibrosis (black arrow), eosinophilic inflammation (yellow arrow), and muscle hypertrophy (blue arrow). (B) Sputum sample showing the presence of eosinophilic Charcot-Leyden crystals.

bronchioles. Because of better control of lung infections, bronchiectasis is now uncommon but may still develop in association with:

- *Congenital or hereditary conditions that predispose to chronic infections*, including cystic fibrosis, intralobar sequestration of the lung, immunodeficiency states, primary ciliary dyskinesia, and Kartagener syndrome.
- *Severe necrotizing pneumonia* caused by bacteria, viruses, or fungi; this may be a single severe episode or recurrent infections.
- *Bronchial obstruction*, due to tumor, foreign body, or mucus impaction; in each instance the bronchiectasis is localized to the obstructed lung segment.
- *Immune disorders*, including rheumatoid arthritis, systemic lupus erythematosus, inflammatory bowel disease, and the posttransplant setting (chronic rejection after lung transplant and chronic graft-versus-host disease after hematopoietic stem cell transplantation).
- Up to 50% of cases are *idiopathic*, in which there appears to be dysfunctional host immunity to infectious agents that leads to chronic inflammation.

Pathogenesis

Obstruction and infection are the major conditions associated with bronchiectasis. The infections that lead to bronchiectasis are usually the result of defective airway clearance. Sometimes this defect stems from airway obstruction, leading to distal pooling of secretions.

Both mechanisms are readily apparent in a severe form of bronchiectasis that is associated with cystic fibrosis (Chapter 10). In cystic fibrosis the primary defect in ion transport results in thick viscous secretions that perturb mucociliary clearance and lead to airway obstruction. This sets the stage for chronic bacterial infections, which cause widespread damage to airway walls. With destruction of supporting smooth muscle and elastic tissue, the bronchi become markedly dilated, while smaller bronchioles are progressively obliterated by fibrosis (*bronchiolitis obliterans*).

Primary ciliary dyskinesia is an autosomal recessive disease with a frequency of 1 in 10,000 to 20,000 births. The disease-causing mutations affect ciliary motor proteins (e.g., mutations involving dynein) and result in ciliary dysfunction, again preventing mucociliary clearance, setting the stage for recurrent infections that lead to bronchiectasis. Ciliary function is also necessary during embryogenesis to ensure proper rotation of the developing organs in the chest and abdomen. As a result, approximately half of patients with primary ciliary dyskinesia have *Kartagener syndrome*, marked by situs inversus or a partial lateralizing abnormality associated with bronchiectasis and sinusitis. Males with this condition also tend to be infertile due to sperm dysmotility.

Allergic bronchopulmonary aspergillosis occurs in patients with asthma or cystic fibrosis and frequently leads to the development of bronchiectasis. It is a hyperimmune response to the fungus *Aspergillus fumigatus*. Sensitization to *Aspergillus* leads to activation of Th2 helper T cells, which release cytokines that recruit eosinophils and other leukocytes. Characteristically, there are high serum IgE levels, serum antibodies to *Aspergillus*, intense airway inflammation with eosinophils, and formation of mucus plugs, which play a primary role in the development of bronchiectasis.

MORPHOLOGY

Bronchiectasis usually affects the lower lobes bilaterally, particularly air passages that are vertical, and is most severe in the more distal bronchi and bronchioles. When tumor or aspiration of foreign bodies leads to bronchiectasis, the involvement is localized. **The airways are dilated, sometimes**

up to four times normal size. Characteristically, the bronchi and bronchioles are so dilated that they can be followed almost to the pleural surfaces. By contrast, in the normal lung the bronchioles cannot be seen by eye beyond a point 2 to 3 cm from the pleural surface. On the cut surface of the lung, the dilated bronchi appear cystic and are filled with mucopurulent secretions (Fig. 15.12).

The histologic findings vary with the activity and chronicity of the disease. In the full-blown, active case there is an intense acute and chronic inflammatory exudate within the walls of the bronchi and bronchioles associated with desquamation of the lining epithelium and extensive ulceration. There may also be squamous metaplasia of the remaining epithelium in response to chronic inflammation, further diminishing mucociliary clearance. In some instances, necrosis destroys the bronchial or bronchiolar walls and forms an abscess. Fibrosis of the bronchial and bronchiolar walls and peribronchiolar fibrosis develop in more chronic cases, leading to varying degrees of subtotal or total obliteration of bronchiolar lumens.

Haemophilus influenzae is found in approximately 50% and *Pseudomonas aeruginosa* in 12% to 30% of sputum cultures from patients with bronchiectasis, with four other types of bacteria (including nontuberculous mycobacteria) making up most of the remaining cases in patients from various geographic locations. This observation suggests that the inflamed, mucoid, sometimes anaerobic bronchiectatic microenvironment supports colonization by relatively few microbial species. In **allergic bronchopulmonary aspergillosis**, fungal hyphae can be seen on special stains within the mucoinflammatory contents of the dilated segmental bronchi. In late stages the fungus may infiltrate the bronchial wall.

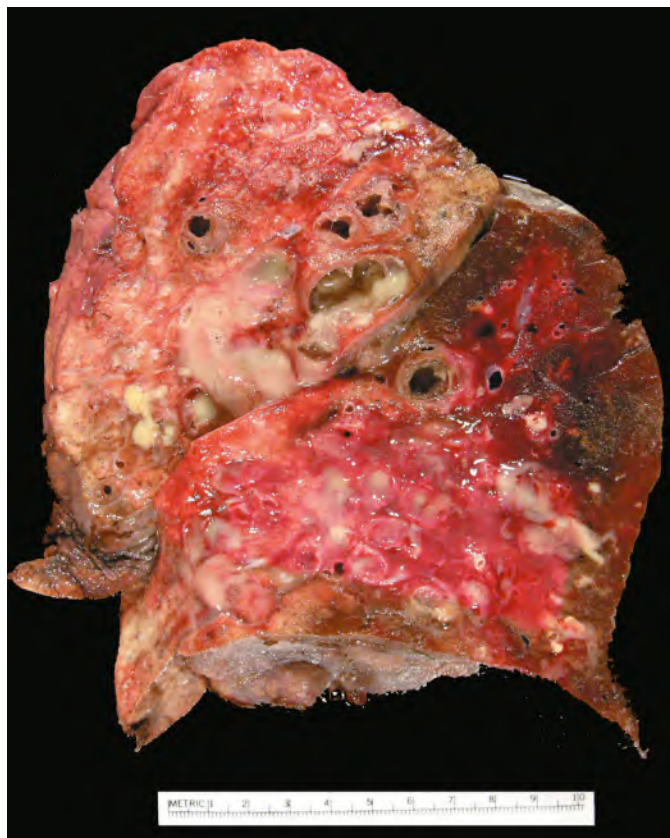


Fig. 15.12 Bronchiectasis in a patient with cystic fibrosis who underwent lung transplantation. Cut surface of lung shows markedly distended peripheral bronchi filled with mucopurulent secretions.

Clinical Features

Bronchiectasis causes severe persistent cough; expectoration of foul-smelling, sometimes bloody sputum; dyspnea and orthopnea in severe cases; and, on occasion, hemoptysis, which may be massive. Symptoms are often episodic and may be precipitated by upper respiratory tract infections or the introduction of new pathogenic agents. Paroxysms of cough are particularly frequent when the patient rises in the morning, as the change in position causes collections of pus and secretions to drain into the bronchi. Obstructive respiratory insufficiency may lead to marked dyspnea and cyanosis. However, the availability of more effective antibiotics combined with physical therapy have improved outcomes considerably, and almost doubled life expectancy. Hence, cor pulmonale, brain abscesses, and amyloidosis are less frequent complications of bronchiectasis now than in the past.

CHRONIC DIFFUSE INTERSTITIAL (RESTRICTIVE) DISEASES

Restrictive lung disorders fall into two general categories: (1) *chronic interstitial and infiltrative diseases*, such as pneumoconioses and interstitial fibrosis of unknown etiology; and (2) *chest wall disorders* (e.g., neuromuscular diseases such as poliomyelitis, severe obesity, pleural diseases, and kyphoscoliosis), which are not discussed here.

Chronic interstitial pulmonary diseases are a heterogeneous group of disorders characterized predominantly by inflammation and fibrosis of the lung interstitium and pulmonary function studies indicative of restrictive lung disease. Diffuse restrictive diseases are categorized based on histology and clinical features (Table 15.5). Many of the entities are of unknown cause and pathogenesis, and some have an intraalveolar as well as an interstitial component. Patients have dyspnea, tachypnea, end-inspiratory crackles, and eventual cyanosis, without wheezing or other evidence of airway obstruction. Typical functional abnormalities are reductions in diffusion capacity, lung volume, and lung compliance. Chest radiographs show bilateral lesions in the form of small nodules, irregular lines, or *ground-glass shadows*, all corresponding to areas of interstitial fibrosis. Although specific entities often have distinguishing features in their early stages, advanced

forms are hard to differentiate because all lead to diffuse scarring of the lung, often referred to as *end-stage lung* or *honeycomb lung*. Eventually, secondary pulmonary hypertension and right-sided heart failure (cor pulmonale) may result. Most interstitial lung disease diagnoses are rendered by multidisciplinary teams of clinicians, radiologists, and pathologists.

Fibrosing Diseases

Idiopathic Pulmonary Fibrosis

Idiopathic pulmonary fibrosis (IPF) refers to a clinicopathologic syndrome marked by progressive interstitial pulmonary fibrosis and respiratory failure. IPF has characteristic radiologic, pathologic, and clinical features. The histologic pattern of fibrosis characteristic of IPF is referred to as usual interstitial pneumonia (UIP), which can often be diagnosed based on its unique appearance on computed tomography scans. The same pattern of fibrosis can also be seen in other disorders, notably connective tissue diseases, chronic hypersensitivity pneumonia, and asbestosis. These must be distinguished from IPF based on clinical, laboratory, and additional histologic features.

Pathogenesis

IPF appears to arise in genetically predisposed individuals who are prone to aberrant repair of recurrent alveolar epithelial cell injury caused by environmental exposures (Fig. 15.13). The implicated factors are:

- **Environmental factors.** Most important among these is cigarette smoking, which increases the risk of IPF several-fold. The incidence of IPF is also increased in individuals who are exposed to air pollution, microaspiration, metal fumes, and wood dust, or who work in certain occupations, including farming, hairdressing, and stone polishing. It is hypothesized that exposure to environmental irritants or toxins in each of these contexts causes recurrent alveolar epithelial cell damage.
- **Genetic factors.** The vast majority of individuals who smoke or who have other environmental exposures linked to IPF do not develop the disorder, indicating that additional factors are required for its development. Mutations in the *TERT*, *TERC*, *PARN*, and *RTEL1* genes, all of which are involved with the maintenance of telomeres, are associated with increased risk of IPF. You will recall that maintenance of telomeres (the ends of chromosomes) is necessary to prevent cellular senescence. Up to 15% of familial IPF is associated with inherited defects in genes that maintain telomeres, while up to 25% of sporadic IPF cases are associated with abnormal telomere shortening in peripheral blood lymphocytes, a finding that also points to a problem with telomere maintenance. Other, rare familial forms of IPF are associated with mutations in genes encoding components of surfactant; these mutations create folding defects in the affected proteins, leading to activation of the unfolded protein response in type II pneumocytes. This in turn appears to make pneumocytes more sensitive to environmental insults, leading to cellular dysfunction and injury. Finally, roughly one-third of IPF cases are associated with a single-nucleotide polymorphism in the promoter of the *MUC5B* gene that greatly increases the secretion of MUC5B, a member of the mucin family. This may in turn alter mucociliary clearance, but precisely how this alteration relates to IPF risk is uncertain.
- **Age.** IPF is a disease of older individuals, rarely appearing before the age of 50 years. Whether this association stems from aging-related telomere shortening or from other acquired changes associated with aging is unknown.

TABLE 15.5 Major Categories of Chronic Interstitial Lung Disease

Fibrosing
Usual interstitial pneumonia (idiopathic pulmonary fibrosis)
Nonspecific interstitial pneumonia
Cryptogenic organizing pneumonia
Connective tissue disease–associated
Pneumoconiosis
Drug reactions
Radiation pneumonitis
Granulomatous
Sarcoidosis
Hypersensitivity pneumonitis
Eosinophilic
Smoking-Related
Desquamative interstitial pneumonia
Respiratory bronchiolitis–associated interstitial lung disease
Other
Langerhans cell histiocytosis
Pulmonary alveolar proteinosis
Lymphoid interstitial pneumonia

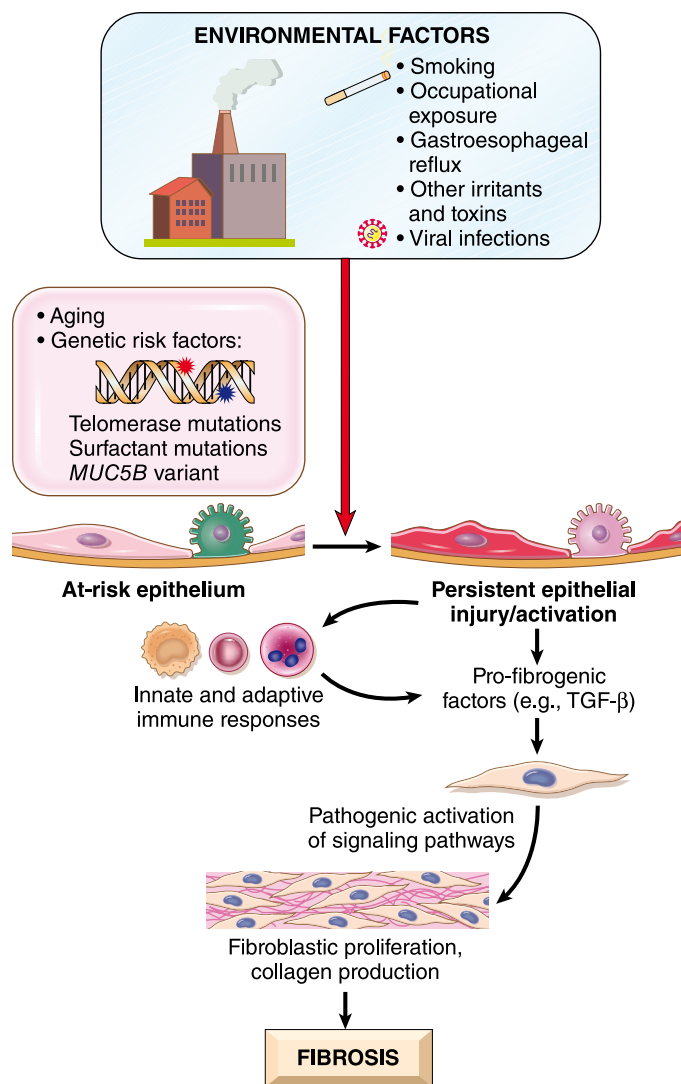


Fig. 15.13 Proposed pathogenic mechanisms in idiopathic pulmonary fibrosis. Environmental factors that are potentially injurious to alveolar epithelium interact with genetic or aging-related factors that place epithelium at risk, creating a persistent epithelial injury. Factors secreted from injured/activated epithelium, possibly augmented by factors released from innate and adaptive immune cells responding to “danger” signals produced by damaged epithelium, activate interstitial fibroblasts. There is some evidence that these activated fibroblasts exhibit abnormalities such as increased signaling through the PI3K/AKT pathway. The activated fibroblasts synthesize and deposit collagen, leading to interstitial fibrosis and eventual respiratory failure.

The pathogenesis of IPF remains poorly understood. Epithelial injury seems to be the initiating event but it is unknown precisely how alveolar epithelial cell damage translates into interstitial fibrosis. One model holds that the injured epithelial cells are the source of profibrotic factors such as TGF- β , whereas a second, non-mutually exclusive model proposes that innate and adaptive immune cells (such as alternatively activated macrophages, see Chapter 3) produce such factors as part of the host response to epithelial cell damage. Other work has described abnormalities in the fibroblasts themselves that include changes in PI3 kinase signaling (Chapter 7), but the causal links between these alterations and fibrosis has not been established.

MORPHOLOGY

Grossly, the pleural surfaces of the lung are cobblestoned due to the retraction of scars along the interlobular septa. The cut surface shows firm, rubbery white areas of fibrosis, which occurs preferentially in the lower lobes, the **subpleural regions**, and along the **interlobular septa** (Fig. 15.14). Microscopically, the hallmark is **patchy interstitial fibrosis** that varies in location, severity, and chronicity. At low power magnification, this is seen as relatively normal segments of lung juxtaposed to areas that are architecturally effaced (*spatial heterogeneity*). The earliest lesions contain an exuberant proliferation of fibroblasts (**fibroblastic foci**). With time these areas become more fibrotic and less cellular. The coexistence of both early and late lesions is typical, a feature referred to as *temporal heterogeneity*. The dense fibrosis causes the destruction of alveolar architecture and the formation of cystic spaces lined by hyperplastic type II pneumocytes or bronchiolar epithelium (**honeycomb fibrosis**). With adequate sampling, these diagnostic histologic changes (i.e., areas of dense fibrosis and fibroblastic foci) can be identified even in advanced IPF. There is usually a mild to moderate amount of inflammation within the fibrotic areas, consisting of mostly lymphocytes admixed with a few plasma cells, neutrophils, eosinophils, and mast cells. Foci of squamous metaplasia and smooth muscle hyperplasia may be present, along with pulmonary arterial hypertensive changes (intimal fibrosis and medial thickening). In acute exacerbations, DAD may be superimposed on these chronic changes.

Clinical Features

IPF begins insidiously with gradually increasing dyspnea on exertion and dry cough. Most patients are 55 to 75 years old at presentation. Hypoxemia, cyanosis, and clubbing occur late in the course. The course in individual patients is unpredictable. Usually there is slowly progressive respiratory failure, but some patients have acute exacerbations and follow a rapid downhill clinical course. The median survival is about 3.8 years after diagnosis. Lung transplantation is the only definitive therapy; however, two drugs, a tyrosine kinase inhibitor and a TGF- β antagonist, have both been shown to slow disease progression and represent the first effective (albeit modestly so) targeted therapies for IPF.

Nonspecific Interstitial Pneumonia

Despite its “nonspecific” name, this entity is important to recognize since these patients have a much better prognosis than patients with UIP. Nonspecific interstitial pneumonia is most often associated with connective tissue disease or another underlying etiology but may also be idiopathic.

MORPHOLOGY

Nonspecific interstitial pneumonia is divided into cellular and fibrosing histologic patterns. The **cellular pattern** consists primarily of mild to moderate chronic interstitial inflammation consisting of lymphocytes and a few plasma cells in a uniform or patchy distribution. The **fibrosing pattern** consists of diffuse or patchy interstitial fibrotic lesions of roughly the same appearance and age (*temporal homogeneity*), an important distinction from UIP. Fibroblastic foci, honeycombing, hyaline membranes, and granulomas are absent.

Clinical Features

Patients present with dyspnea and cough of several months' duration. They are often female nonsmokers in their sixth decade of life. On imaging, the lesions have the appearance of bilateral, symmetric, predominantly lower lobe reticular opacities. Patients having the cellular pattern are somewhat younger than those with the fibrosing pattern and have a better prognosis. Unlike UIP, immunosuppressive drugs

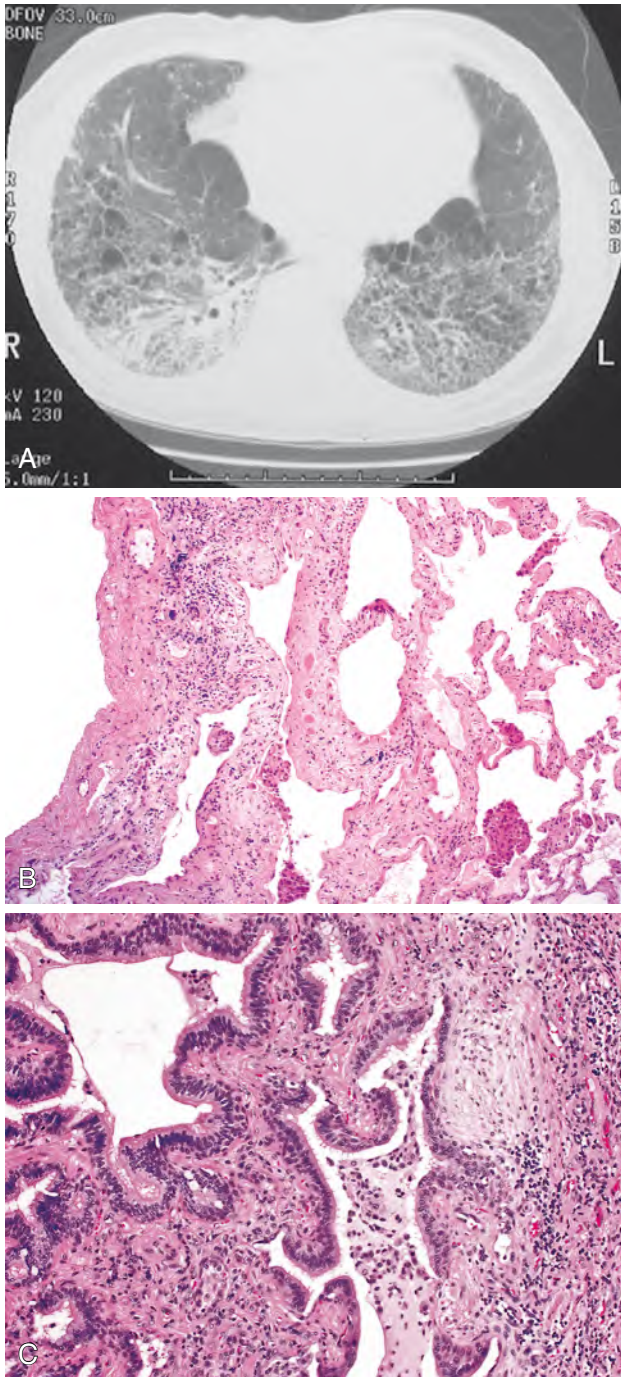


Fig. 15.14 Usual interstitial pneumonia. (A) Computed tomography of lung showing honeycombing and interstitial thickening. (B) Microscopic correlate to (A) showing variable pleural and alveolar septal fibrosis. (C) Lung, with an area of honeycombing on the left and a fibroblastic focus with bluish myxoid extracellular matrix on the right. (Courtesy Dr. Nicole Cipriani, Department of Pathology, University of Chicago.)

may be beneficial in patients with NSIP, suggesting that NSIP has an immune basis.

Cryptogenic Organizing Pneumonia

Organizing pneumonia is most often a response to infection or inflammatory injury of the lungs. This pattern of injury is nonspecific and has been associated with viral and bacterial pneumonias, inhaled

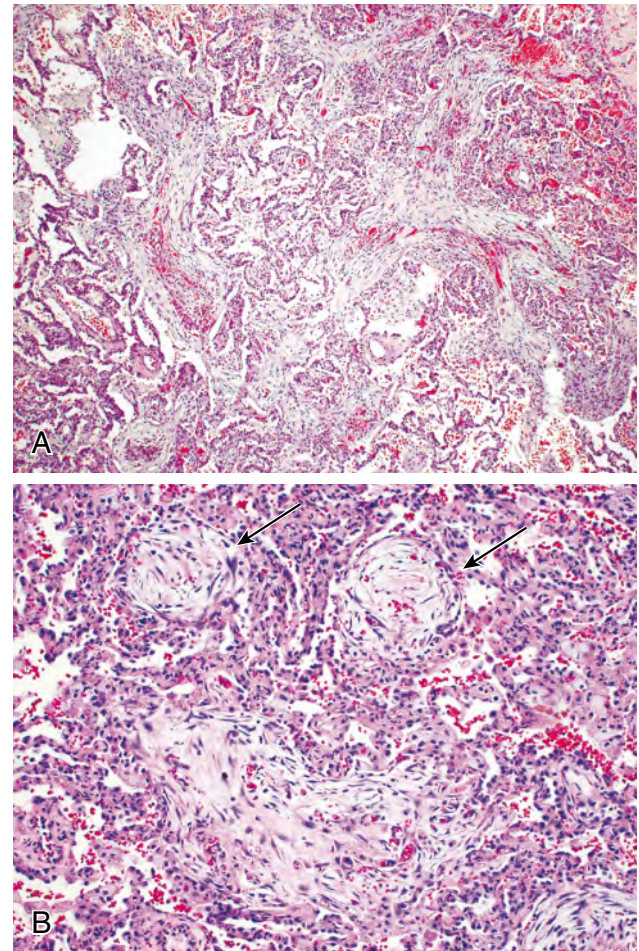


Fig. 15.15 Cryptogenic organizing pneumonia. Some alveolar spaces are filled with balls of fibroblasts (Masson bodies, arrows), while the alveolar walls are relatively normal. (A) Low power. (B) High power.

toxins, drugs, connective tissue disease, and graft-versus-host disease in hematopoietic stem cell transplant recipients. In cases where no etiology is identified, a diagnosis of *cryptogenic organizing pneumonia* may be rendered. Patients present with cough and dyspnea and have patchy subpleural or peribronchial areas of airspace consolidation radiographically. Histologically, it is characterized by the presence of polypoid plugs of loose organizing connective tissue (*Masson bodies*) within alveolar ducts, alveoli, and often bronchioles (Fig. 15.15). The connective tissue is all the same age, and the underlying lung architecture is normal. There is no interstitial fibrosis or honeycomb lung. Some patients recover spontaneously, but most need treatment with oral steroids for 6 months or longer for complete recovery. The long-term prognosis is dependent on the underlying disorder. Of note, some patients with organizing pneumonia histologic pattern present with a clinical picture resembling acute ARDS.

Pulmonary Involvement in Autoimmune Diseases

Many autoimmune diseases (also referred to as connective tissue diseases because of their frequent association with arthritis) can involve the lung at some point in their course. Those that are well recognized for producing pulmonary disease include systemic lupus erythematosus, rheumatoid arthritis, progressive systemic sclerosis (scleroderma), and dermatomyositis-polymyositis. Pulmonary involvement can take different histologic patterns; nonspecific interstitial pneumonia, usual

interstitial pneumonia, organizing pneumonia, and bronchiolitis are the most common.

- *Rheumatoid arthritis* is associated with pulmonary involvement in 30% to 40% of patients as (1) chronic pleuritis, with or without effusion; (2) diffuse interstitial pneumonitis and fibrosis; (3) intrapulmonary rheumatoid nodules; (4) follicular bronchiolitis; or (5) pulmonary hypertension. When lung disease occurs in the setting of rheumatoid arthritis and pneumoconiosis (described next), it is referred to as *Caplan syndrome*.
- *Systemic sclerosis* (scleroderma) is associated with diffuse interstitial fibrosis (nonspecific interstitial pattern more common than usual interstitial pattern) and pleural involvement.
- *Lupus erythematosus* may cause patchy, transient parenchymal infiltrates or occasionally severe lupus pneumonitis, as well as pleuritis and pleural effusions.

Pulmonary involvement in these diseases has a variable prognosis that is determined by the extent and histologic pattern of involvement.

KEY CONCEPTS

Chronic Interstitial Lung Diseases

- Diffuse interstitial fibrosis of the lung gives rise to restrictive lung diseases characterized by reduced lung compliance and reduced FVC. The ratio of FEV₁ to FVC is normal.

- Idiopathic pulmonary fibrosis (IPF) is prototypic of restrictive lung diseases. It is characterized by patchy interstitial fibrosis, fibroblastic foci, and formation of cystic spaces (honeycomb lung). This histologic pattern is known as usual interstitial pneumonia (UIP).
- The cause of idiopathic pulmonary fibrosis is unknown, but genetic analyses point to roles for senescence of alveolar epithelium (due to telomere shortening), altered mucin production, and abnormal signaling in alveolar fibroblasts. Injury to alveolar epithelial cells sets in motion events that lead to increased local production of fibrogenic cytokines, such as TGF- β .
- Unlike UIP, nonspecific interstitial pneumonia (NSIP) has significantly better prognosis and may respond to immunosuppressive drugs.
- The exact etiology of cryptogenic organizing pneumonia (COP) is not known; most patients with COP are effectively treated with corticosteroids.

Pneumoconioses

The term *pneumoconiosis* was originally coined to describe lung disease caused by inhalation of mineral dusts encountered in the workplace, but now also includes disease induced by chemical fumes and vapors. A simplified classification is presented in Table 15.6. Where implemented, regulations limiting worker exposure have resulted in a marked decrease in dust-associated diseases.

TABLE 15.6 Lung Diseases Caused by Air Pollutants

Agent	Disease	Exposure
Mineral Dusts		
Coal dust	Anthracosis Macules Progressive massive fibrosis Caplan syndrome	Coal mining (particularly hard coal)
Silica	Silicosis Caplan syndrome	Metal casting work, sandblasting, hard rock mining, stone cutting, others
Asbestos	Asbestosis Pleural plaques Caplan syndrome Mesothelioma Carcinoma of the lung, larynx, stomach, colon	Mining, milling, manufacturing, and installation and removal of insulation
Beryllium	Acute berylliosis Beryllium granulomatosis Lung carcinoma (?)	Mining, manufacturing
Iron oxide	Siderosis	Welding
Barium sulfate	Baritosis	Mining
Tin oxide	Stannosis	Mining
Organic Dusts That Induce Hypersensitivity Pneumonitis		
Moldy hay	Farmer's lung	Farming
Bagasse	Bagassosis	Manufacturing wallboard, paper
Bird droppings	Bird breeder's lung	Bird handling
Organic Dusts That Induce Asthma		
Cotton, flax, hemp	Byssinosis	Textile manufacturing
Red cedar dust	Asthma	Lumbering, carpentry
Chemical Fumes and Vapors		
Nitrous oxide, sulfur dioxide, ammonia, benzene, insecticides	Bronchitis, asthma Pulmonary edema ARDS Mucosal injury Fulminant poisoning	Occupational and accidental exposure

ARDS, Acute respiratory distress syndrome.

Pathogenesis

Specific factors that influence the development of dust-borne pneumoconiosis include:

- **Dust retention**, which is determined by the dust concentration in ambient air, the duration of exposure, and the effectiveness of clearance mechanisms. Any influence that impairs mucociliary clearance significantly (such as cigarette smoking) increases the accumulation of dust in the lungs.
- **Particle size**. The most dangerous particles are from 1 to 5 μm in diameter because they are able to reach the terminal small airways and air sacs and deposit in their linings.
- **Particle solubility and cytotoxicity**, which are influenced by particle size. In general, small particles composed of injurious substances of high solubility are more likely to produce rapid-onset acute lung injury. By contrast, larger particles are more likely to resist dissolution and may persist within the lung parenchyma for years. These tend to evoke fibrosing collagenous pneumoconioses, such as is characteristic of *silicosis*.
- **Particle uptake by epithelial cells or egress across epithelial linings**, which allows direct interactions with fibroblasts and interstitial macrophages to occur. Some particles may be carried to lymph nodes through lymphatic drainage directly or within migrating macrophages and thereby initiate an adaptive immune response to components of the particulates, to self-proteins modified by the particles, or to both.
- **Activation of the inflammasome** (Chapter 3), which occurs following the phagocytosis of certain particles by macrophages. This innate immune response amplifies the intensity and the duration of the local inflammatory reaction.
- **Tobacco smoking**, which exacerbates the injurious effects of all inhaled mineral dusts, particularly those caused by asbestos.

In general, only a small percentage of exposed people develop occupational respiratory diseases, implying a genetic predisposition to their development. Many of the diseases listed in Table 15.6 are quite uncommon; hence only a select few that cause pulmonary fibrosis are presented next.

Coal Workers' Pneumoconiosis. Coal workers' pneumoconiosis (CWP) is a lung disease caused by inhalation of coal particles and other admixed forms of dust. Dust reduction measures in coal mines around the globe have drastically reduced its incidence. The spectrum of lung findings in coal workers is wide, varying from asymptomatic anthracosis, to simple coal workers' pneumoconiosis with little to no pulmonary dysfunction, to complicated coal workers' pneumoconiosis, or *progressive massive fibrosis*, in which lung function is compromised. Contaminating silica in the coal dust favors the development of progressive disease. In most cases, carbon dust itself is the major culprit, and studies have shown that complicated lesions contain much more dust than simple lesions. Coal workers may also develop emphysema and chronic bronchitis independent of smoking.

MORPHOLOGY

Carbon deposits are dark black in color and are readily visible grossly and microscopically. **Anthracosis** is the most innocuous coal-induced pulmonary lesion in coal miners and is also seen to some degree in urban dwellers and tobacco smokers. Inhaled carbon pigment is engulfed by alveolar or interstitial macrophages, which accumulate in the connective tissue adjacent to the lymphatics and in organized lymphoid tissue adjacent to the bronchi or in the lung hilus.

Simple coal workers' pneumoconiosis is characterized by **coal macules** (1 to 2 mm in diameter) and somewhat larger **coal nodules**. Coal macules

consist of carbon-laden macrophages; nodules also contain a delicate network of collagen fibers. Although these lesions are scattered throughout the lung, the upper lobes and upper zones of the lower lobes are more heavily involved. They are located primarily adjacent to respiratory bronchioles, the site of initial dust accumulation. In due course dilation of adjacent alveoli occurs, sometimes giving rise to **centrilobular emphysema**.

Complicated coal workers' pneumoconiosis (progressive massive fibrosis) occurs on a background of simple disease and generally requires many years to develop. It is characterized by intensely blackened scars 1 cm or larger, sometimes up to 10 cm in greatest diameter. They are usually multiple. Microscopically, the lesions consist of dense collagen and pigment (Fig. 15.16). The center of the lesion is often necrotic, most likely due to local ischemia.

Clinical Features

Coal workers' pneumoconiosis is usually benign, causing little decrement in lung function. Even mild forms of complicated coal workers' pneumoconiosis do not affect lung function significantly. In a minority of cases (fewer than 10%), progressive massive fibrosis develops, leading to increasing pulmonary dysfunction, pulmonary hypertension, and cor pulmonale. Once progressive massive fibrosis develops, it may continue to worsen even if further exposure to dust is prevented. Unlike silicosis (discussed next), there is no convincing evidence that coal workers' pneumoconiosis increases susceptibility to tuberculosis, nor does it predispose to cancer in the absence of smoking. However, domestic indoor use of "smoky coal" (bituminous) for cooking and heating, a common practice in low-resource parts of the world, is associated with an increased risk of lung cancer death, even in those who do not smoke.

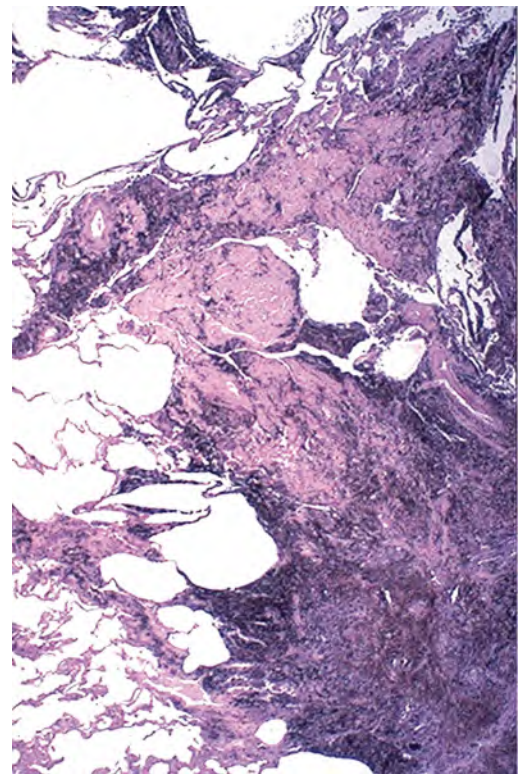


Fig. 15.16 Progressive massive fibrosis in a coal worker. A large amount of black pigment is associated with dense interstitial fibrosis. (From Klatt EC. *Robbins and Cotran Atlas of Pathology*. 2nd ed. Saunders; 2010:121.)

Silicosis. Silicosis is a common lung disease caused by inhalation of proinflammatory crystalline silicon dioxide (silica). It usually presents after decades of exposure as slowly progressing, nodular, fibrosing pneumoconiosis. Currently, silicosis is the most prevalent chronic occupational disease in the world. As shown in Table 15.6, workers in many occupations are at risk, including individuals involved with the repair, rehabilitation, or demolition of concrete structures such as buildings and roads. The disease also occurs in workers producing stressed denim by sandblasting, stone carvers, and jewelers using chalk molds. Occasionally, heavy exposure over months to a few years can result in acute silicosis, a disorder characterized by the accumulation of abundant lipoproteinaceous material within alveoli (identical morphologically to alveolar proteinosis, discussed later).

Pathogenesis

Phagocytosis of inhaled silica crystals by macrophages activates the inflammasome and stimulates the release of inflammatory mediators, particularly IL-1 and IL-18. This in turn leads to the recruitment of additional inflammatory cells and activates interstitial fibroblasts, leading to collagen deposition. Silica occurs in both crystalline and amorphous forms, but crystalline forms (including quartz, cristobalite, and tridymite) are much more fibrogenic. Of these, quartz is most commonly implicated. The lack of severe responses to silica in coal and hematite miners is thought to be due to coating of silica with other minerals, especially clay components, which render the silica less toxic. Although amorphous silicates are biologically less active than crystalline silica, heavy lung burdens of these minerals may also produce lesions.

MORPHOLOGY

Silicosis is characterized grossly in its early stages by tiny, barely palpable, discrete pale to blackened (if coal dust is also present) nodules in the hilar

lymph nodes and upper zones of the lungs. As the disease progresses, these nodules coalesce into **hard, collagenous scars** (Fig. 15.17A). Some nodules may undergo central softening and cavitation due to superimposed tuberculosis or to ischemia. Fibrotic lesions may also occur in the hilar lymph nodes and pleura. Sometimes, thin sheets of calcification occur in the lymph nodes and are seen radiographically as **eggshell calcification** (i.e., calcium surrounding a zone lacking calcification). If the disease continues to progress, expansion and coalescence of lesions may produce progressive massive fibrosis. Histologic examination reveals the hallmark lesion characterized by a central area of whorled collagen fibers with a more peripheral zone of dust-laden macrophages (Fig. 15.17B). Examination of the nodules by polarized microscopy reveals weakly birefringent silicate particles.

Clinical Features

The onset of silicosis may be slow and insidious (10 to 30 years after exposure; this is most common), accelerated (within 10 years of exposure), or rapid (weeks or months after intense exposure to fine dust high in silica; this is rare). Chest radiographs typically show a fine nodularity in the upper zones of the lung. Pulmonary function is either normal or only moderately affected early in the course, and most patients do not develop shortness of breath until progressive massive fibrosis supervenes. The disease may continue to worsen even if the patient is no longer exposed. It is slow to kill, but impaired pulmonary function may severely limit activity.

Silicosis also is associated with an increased susceptibility to *tuberculosis* and a two-fold increased risk of lung cancer. The former may be because crystalline silica inhibits the ability of pulmonary macrophages to kill phagocytosed mycobacteria. The link to cancer is not fully understood, but it is but one of many chronic inflammatory conditions that increase the risk of carcinoma in involved tissues (Chapter 7).

Asbestos-Related Diseases. Asbestos is a family of proinflammatory crystalline hydrated silicates that are associated with pulmonary fibrosis and various forms of cancer. Use of asbestos is

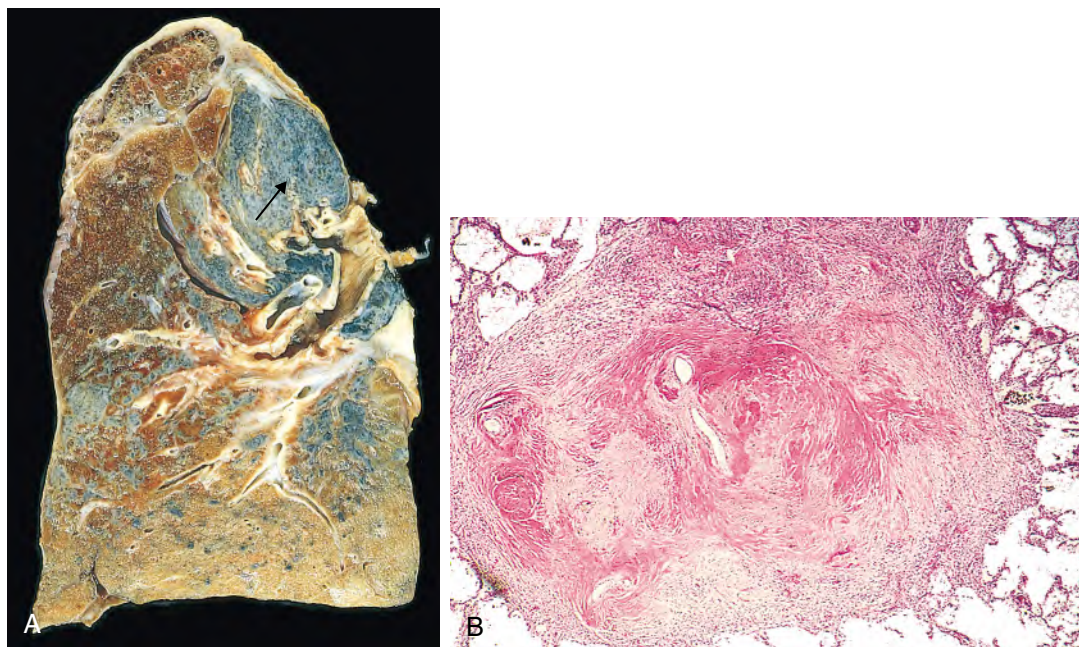


Fig. 15.17 Advanced silicosis. (A) Scarring has contracted the upper lobe into a small dark mass (arrow). Note the dense pleural thickening. (B) Several coalescent collagenous silicotic nodules. (Courtesy Dr. John Godleski, Brigham and Women's Hospital, Boston.)

tightly restricted in many high-resource countries; however, there is little, if any, control of its use in low-resource parts of the world. Asbestos-related diseases include:

- Localized fibrous plaques or, rarely, diffuse pleural fibrosis
- Pleural effusions, recurrent
- Parenchymal interstitial fibrosis (*asbestosis*)
- Lung carcinoma
- Mesothelioma
- Laryngeal, ovarian, and perhaps other extrapulmonary neoplasms, including colon carcinoma

The increased incidence of asbestos-related cancers in family members of asbestos workers has alerted the general public to the potential hazards of even low-level exposure to asbestos. However, the necessity of expensive asbestos abatement programs for environments, such as schools with low but measurable airborne asbestos fiber counts, remains a matter of contention.

Pathogenesis

The disease-causing capabilities of the different forms of asbestos depend on concentration, size, shape, and solubility. Asbestos occurs in two distinct geometric forms, serpentine and amphibole. The serpentine chrysotile form accounts for 90% of the asbestos used in industry. Amphiboles, even though less prevalent, are more pathogenic than chrysotiles, particularly with respect to induction of mesothelioma, a malignant tumor derived from the lining cells of pleural surfaces (described later).

The greater pathogenicity of amphiboles is apparently related to their aerodynamic properties and solubility. Chrysotiles, with their more flexible, curled structure, are likely to become impacted in the upper respiratory passages and removed by the mucociliary elevator. Furthermore, once trapped in the lungs, chrysotiles are gradually leached from the tissues because they are more soluble than amphiboles. In contrast, the straight, stiff amphiboles may align themselves with the airstream and thus be delivered deeper into the lungs, where they can penetrate epithelial cells and reach the interstitium. Both amphiboles and serpentines are fibrogenic, and increasing doses are associated with a higher incidence of asbestos-related diseases.

In contrast to other inorganic dusts, asbestos acts as a tumor initiator and a tumor promoter (Chapter 7). Some of its oncogenic effects are mediated by reactive free radicals generated by asbestos fibers, which preferentially localize in the distal lung, close to the mesothelial cells of the pleura. Oxygen and nitrogen free radicals may form via reactions catalyzed by iron molecules within asbestos fibers or because of the activation of inflammatory cells. Toxic chemicals adsorbed onto the asbestos fibers also likely contribute to the oncogenicity of the fibers. For example, the adsorption of carcinogens in tobacco smoke onto asbestos fibers may be the basis for the remarkable synergy between tobacco smoking and the development of lung carcinoma in asbestos workers. Smoking also enhances the effect of asbestos by interfering with the mucociliary clearance of fibers. One study of asbestos workers found a five-fold increase of lung carcinoma with asbestos exposure alone, while asbestos exposure and smoking together led to a 55-fold increase in the risk.

Once phagocytosed by macrophages, asbestos fibers activate the inflammasome and stimulate the release of proinflammatory factors and fibrogenic mediators. The initial injury occurs at bifurcations of small airways and ducts, where asbestos fibers land, penetrate, and are directly toxic to pulmonary parenchymal cells. Macrophages, both alveolar and interstitial, attempt to ingest and clear the fibers. Long-term deposition of fibers and persistent release of mediators (e.g., reactive oxygen species, proteases, cytokines, and growth factors) eventually lead to generalized interstitial pulmonary inflammation and fibrosis.

MORPHOLOGY

Asbestosis is marked by **diffuse pulmonary interstitial fibrosis**, which is distinguished from diffuse interstitial fibrosis resulting from other causes only by the presence of **asbestos bodies**. Asbestos bodies are golden brown, fusiform or beaded rods with a translucent center that consist of asbestos fibers coated with an iron-containing proteinaceous material (Fig. 15.18A). They arise when macrophages phagocytose asbestos fibers; the iron is presumably derived from phagocyte ferritin. Other inorganic particulates may become coated with similar iron-protein complexes and are called **ferruginous bodies**.

Asbestosis begins as fibrosis around respiratory bronchioles and alveolar ducts and extends to involve adjacent alveolar sacs and alveoli. The fibrosis distorts the architecture, creating enlarged airspaces enclosed by thick fibrous walls; eventually the affected regions become honeycombed. The pattern of fibrosis is histologically similar to that seen in usual interstitial fibrosis, with fibroblastic foci and varying degrees of fibrosis. In contrast to coal workers' pneumoconiosis and silicosis, asbestosis begins in the lower lobes and sub-pleurally, with the middle and upper lobes becoming affected as fibrosis progresses. The scarring may trap and narrow pulmonary arteries and arterioles, causing pulmonary hypertension and cor pulmonale.

Pleural plaques, the most common manifestation of asbestos exposure, are well-circumscribed plaques of dense collagen that are often calcified (Fig. 15.18B). They develop most frequently on the anterior and posterolateral aspects of the parietal pleura and over the domes of the diaphragm. The size and number of pleural plaques do not correlate with the level of exposure to

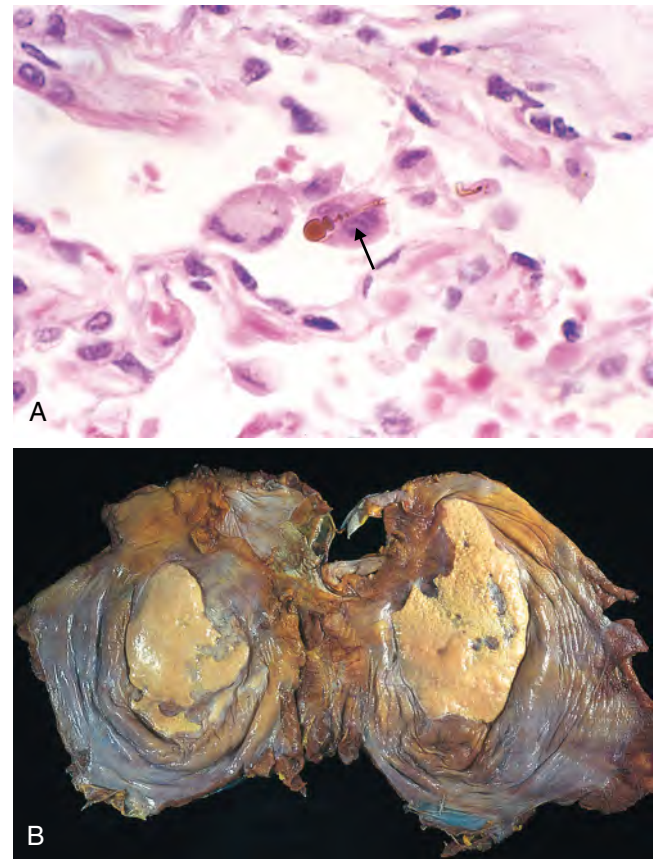


Fig. 15.18 Asbestosis. (A) High-power detail of an asbestos body, revealing the typical beading and knobbed ends (arrow). (B) Asbestos-related pleural plaques. Large, discrete fibrocalcific plaques are seen on the pleural surface of the diaphragm. (Courtesy Dr. John Godleski, Brigham and Women's Hospital, Boston.)

asbestos or the time since exposure. They also do not contain identifiable asbestos bodies; however, only rarely do they occur in individuals without a history or evidence of asbestos exposure. Uncommonly, asbestos exposure induces pleural effusions, which are usually serous but may be bloody. Rarely, diffuse visceral pleural fibrosis may occur and, in advanced cases, bind the lung to the thoracic wall.

Both lung carcinomas and mesotheliomas (pleural and peritoneal) develop in workers exposed to asbestos (see sections on Carcinomas and Pleural Tumors).

Clinical Features

The clinical findings in asbestosis are very similar to those caused by other diffuse interstitial lung diseases (discussed earlier). They rarely appear earlier than 10 years after first exposure and are more common after 20 to 30 years. Dyspnea is usually the first manifestation; at first, it is provoked by exertion, but later is present even at rest. Cough associated with production of sputum, when present, is likely to be due to smoking rather than asbestosis. Chest x-ray studies reveal irregular linear densities, particularly in both lower lobes. With advancement of the pneumoconiosis, a honeycomb pattern develops. The disease may remain static or progress to respiratory failure, cor pulmonale, and death. Pleural plaques are usually asymptomatic and are detected on radiographs as circumscribed densities. Asbestosis complicated by lung or pleural cancer is associated with a particularly grim prognosis.

KEY CONCEPTS

Pneumoconioses

- Pneumoconioses encompass a group of chronic fibrosing diseases of the lung resulting from exposure to organic and inorganic particulates, most commonly mineral dust.
- Pulmonary alveolar macrophages play a central role in the pathogenesis of lung injury by promoting inflammation and producing fibrogenic cytokines.
- Coal dust–induced disease varies from asymptomatic anthracosis to simple coal workers' pneumoconiosis (coal macules or nodules, and centrilobular emphysema), to progressive massive fibrosis, manifested by increasing pulmonary dysfunction, pulmonary hypertension, and cor pulmonale.
- Silicosis is the most common pneumoconiosis in the world, and crystalline silica (e.g., quartz) is the usual culprit. The lung disease is progressive even after exposure stops.
- The manifestations of silicosis range from asymptomatic silicotic nodules to large areas of dense fibrosis; persons with silicosis also have an increased susceptibility to tuberculosis. There is a two-fold increased risk of lung cancer.
- Asbestos fibers come in two forms; the stiff amphiboles have a greater fibrogenic and carcinogenic potential than the serpentine chrysotiles.
- Asbestos exposure is linked with six disease processes: (1) parenchymal interstitial fibrosis (asbestosis); (2) localized pleural plaques (asymptomatic) or rarely diffuse pleural fibrosis; (3) recurrent pleural effusions; (4) lung carcinoma; (5) malignant pleural and peritoneal mesotheliomas; and (6) laryngeal cancer.
- Cigarette smoking increases the risk of lung cancer in the setting of asbestos exposure; even family members of workers exposed to asbestos are at increased risk for lung carcinoma and mesothelioma.

Complications of Therapies

Drug-Induced Lung Diseases. An increasing number of prescription drugs have been found to cause a variety of acute and chronic alterations in lung structure and function, interstitial fibrosis, bronchiolitis obliterans, and eosinophilic pneumonia. For example, cytotoxic drugs used in cancer therapy (e.g., bleomycin) cause pulmonary damage and fibrosis

as a result of direct toxicity and by stimulating the influx of inflammatory cells into the alveoli. Amiodarone, a drug used to treat cardiac arrhythmias, is preferentially concentrated in the lung, and causes significant pneumonitis in 5% to 15% of patients receiving it. Cough induced by angiotensin-converting enzyme inhibitors is very common.

Intravenous drug use disorder most often causes lung infections. In addition, particulate matter used to cut drugs may lodge in the lung microvasculature, producing granulomatous inflammation and fibrosis. Nitrofurantoin, an antibiotic used to treat urinary tract infection, has also been linked to pulmonary fibrosis.

Radiation-Induced Lung Diseases. Radiation pneumonitis is a well-known complication of radiotherapy for thoracic tumors (lung, esophageal, breast, mediastinal). It most often involves the lung within the radiation field and occurs in acute and chronic forms. *Acute radiation pneumonitis* (lymphocytic alveolitis or hypersensitivity pneumonitis) occurs 1 to 6 months after irradiation in 3% to 44% of patients, depending on dose and age. It manifests with fever, dyspnea out of proportion to the volume of lung irradiated, pleural effusion, and pulmonary infiltrates. Morphologic changes are those of diffuse alveolar damage associated with atypia of hyperplastic type II pneumocytes and fibroblasts. Epithelial cell atypia and foam cells within vessel walls are also characteristic of radiation damage. With steroid therapy, these symptoms may resolve completely, but other cases progress to *chronic radiation pneumonitis* (pulmonary fibrosis), which also may occur without antecedent, clinically apparent, acute radiation pneumonitis. In its most severe form, chronic radiation pneumonitis and associated progressive fibrosis can lead to cyanosis, pulmonary hypertension, and cor pulmonale.

Granulomatous Diseases

Sarcoidosis

Sarcoidosis is a systemic granulomatous disease of unknown cause that may involve many tissues and organs. Its various clinical presentations are protean, but the most common are bilateral hilar lymphadenopathy or parenchymal lung involvement, occurring in 90% of cases. Eye and skin lesions are next in frequency. Since other diseases, including mycobacterial and fungal infections and berylliosis, can also produce noncaseating granulomas, the diagnosis is one of exclusion.

Sarcoidosis usually occurs in adults younger than 40 years of age but can affect any age group. The prevalence is higher in females but varies widely in different countries and populations. In the United States the rates are highest in the southeast and are 10 times higher in African American than in White Americans. In contrast, the disease is rare among Chinese and Southeast Asian populations. Patterns of organ involvement also vary with race.

Pathogenesis

Although several lines of evidence suggest that sarcoidosis is a disease of disordered immune regulation in genetically predisposed individuals, its etiology remains unknown. There are several immunologic abnormalities in the local milieu of sarcoid granulomas that suggest a cell-mediated immune response to an unidentified antigen. These abnormalities include:

- Intraalveolar and interstitial accumulation of CD4+ T cells, resulting in CD4/CD8 T-cell ratios ranging from 5:1 to 15:1, suggesting a pathogenic role for CD4+ helper T cells. There is oligoclonal expansion of T-cell subsets as determined by analysis of T-cell receptor rearrangement, consistent with an antigen-driven proliferation.
- Increased levels of T cell–derived Th1 cytokines such as IL-2 and interferon (IFN)- γ , which may be responsible for T-cell expansion and macrophage activation, respectively.

- Increased levels of several cytokines in the local environment (IL-8, TNF, macrophage inflammatory protein 1 α) that favor recruitment of additional T cells and monocytes and contribute to the formation of granulomas. TNF is released in particularly large amounts by activated alveolar macrophages, and the TNF concentration in the bronchoalveolar fluid is a marker of disease activity.
- Impaired dendritic cell function has been suggested but not proven.

Additionally, there are systemic immunologic abnormalities in individuals with sarcoidosis. Both anergy to common skin test antigens, such as *Candida* or tuberculosis purified protein derivative (PPD), and polyclonal hypergammaglobulinemia, another manifestation of helper T-cell dysregulation, are frequently observed. Evidence of genetic influences includes familial and racial clustering of cases and the association with certain human leukocyte antigen (HLA) genotypes (e.g., HLA-A1 and HLA-B8).

MORPHOLOGY

Virtually every organ in the body may be affected by sarcoidosis, at least on rare occasions. Involved tissues contain well-formed **nonnecrotizing granulomas** (Fig. 15.19) composed of aggregates of tightly clustered epithelioid macrophages, often with giant cells. Central necrosis is unusual. With chronicity, the granulomas may become enclosed within fibrous rims or may eventually be replaced by hyaline fibrous scars. Laminated concretions composed of calcium and proteins known as **Schaumann bodies** and stellate inclusions known as **asteroid bodies** are found within giant cells in approximately 60% of the granulomas. Though characteristic, these microscopic features are not pathognomonic of sarcoidosis because asteroid and Schaumann bodies may be encountered in other granulomatous diseases (e.g., tuberculosis).

The **lung** is a common site of involvement. Macroscopically, there is usually no demonstrable alteration, although in advanced cases coalescence of granulomas produces small nodules that are palpable or visible as 1 to 2 cm,

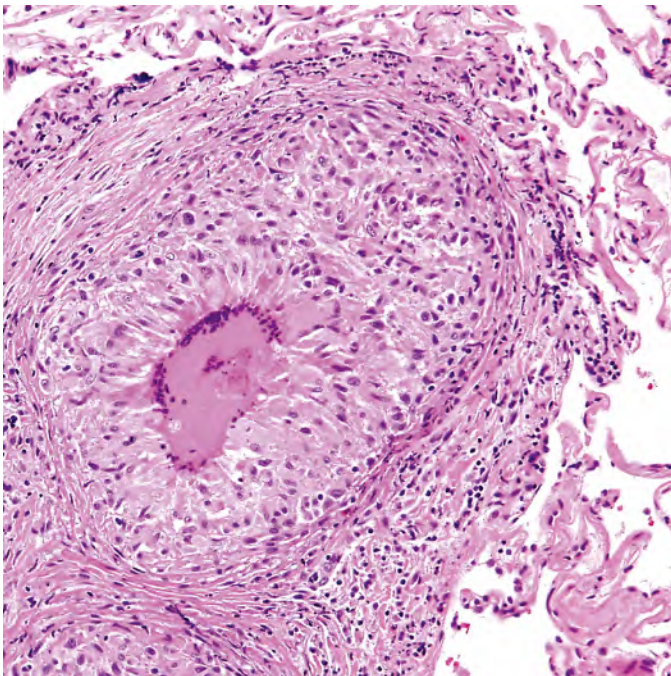


Fig. 15.19 Sarcoidosis. A large noncaseating granuloma with many epithelioid cells and several multinucleated cells is present. Note the surrounding fibrosis and sparse lymphocytic infiltrate.

noncaseating, noncavitated consolidations. The lesions are distributed primarily along lymphatics around bronchi and blood vessels, although alveolar lesions and pleural involvement are also seen. The relatively high frequency of granulomas in the bronchial submucosa accounts for the high diagnostic yield of bronchoscopic biopsies. There seems to be a strong tendency for lesions to heal in the lungs, so varying stages of fibrosis and hyalinization are often found.

Lymph nodes are involved in almost all cases, particularly the hilar and mediastinal nodes, but any node in the body may be affected. Nodes are characteristically enlarged, discrete, and sometimes calcified. Tonsillar granulomas are seen in about one-quarter to one-third of cases. The **spleen** is involved in about 75% of cases, but overt splenomegaly is seen in only 20% of cases. On occasion, granulomas may coalesce to form small nodules that are visible macroscopically. The **liver** is affected slightly less often than the spleen. It may be moderately enlarged and typically contains scattered granulomas, more in portal triads than in the lobular parenchyma.

The **bone marrow** is involved in about 20% of cases. Radiologically, visible bone lesions have a particular tendency to involve phalangeal bones of the hands and feet, creating small, circumscribed areas of bone resorption within the marrow cavity and a diffuse reticulated pattern throughout the cavity, with widening of the bony shafts or new bone formation on the outer surfaces.

Skin lesions, encountered in 25% of cases, assume a variety of appearances, including discrete subcutaneous nodules; focal, slightly elevated, erythematous plaques; or flat lesions that are slightly reddened and scaling, resembling those of systemic lupus erythematosus. Lesions may also appear on the mucous membranes of the oral cavity, larynx, and upper respiratory tract. Other patients present with **erythema nodosum**, painful erythematous nodules on the shins that stem from septal panniculitis.

Ocular involvement, seen in 25% of cases, takes the form of iritis or iridocyclitis and may be bilateral or unilateral. Consequently, corneal opacities, glaucoma, and total loss of vision may occur. These ocular lesions are frequently accompanied by inflammation of the lacrimal glands and suppression of lacrimation (**sicca syndrome**). Bilateral sarcoidosis of the parotid, submaxillary, and sublingual glands (combined uveoparotid involvement) is referred to as **Mikulicz syndrome** (Chapter 6).

Muscle involvement is underdiagnosed since it may be asymptomatic. Muscle weakness, aches, tenderness, and fatigue should prompt consideration of occult sarcoid myositis, which can be diagnosed by muscle biopsy. Sarcoid granulomas occasionally occur in the heart, kidneys, central nervous system (neurosarcoidosis, seen in 5% to 15% of cases), and endocrine glands, particularly in the pituitary, as well as in other body tissues.

Clinical Features

Because of its varying severity and inconstant tissue distribution, sarcoidosis may present with diverse features. It may be discovered unexpectedly on routine chest films as bilateral hilar adenopathy or may present with peripheral lymphadenopathy, cutaneous lesions, eye involvement, splenomegaly, or hepatomegaly. In the great majority of cases, however, individuals seek medical attention because of the insidious onset of respiratory abnormalities (shortness of breath, cough, chest pain, hemoptysis) or of constitutional signs and symptoms (fever, fatigue, weight loss, anorexia, night sweats).

Sarcoidosis follows an unpredictable course. It may be inexorably progressive or marked by periods of activity interspersed with remissions, sometimes permanent, that may be spontaneous or induced by steroid therapy. Overall, 65% to 70% of affected patients recover with minimal or no residual manifestations. Twenty percent have permanent loss of some lung function or some permanent visual impairment. Of the remaining 10% to 15%, some die of cardiac or central nervous system damage, but most succumb to progressive pulmonary fibrosis and cor pulmonale.

KEY CONCEPTS

Sarcoidosis

- Sarcoidosis is a multisystem disease of unknown etiology; the diagnostic histopathologic feature is the presence of noncaseating granulomas in various tissues.
- Immunologic abnormalities include high levels of CD4+ Th1 T cells in the lung that secrete cytokines such as IFN- γ and IL-2 locally.
- Clinical manifestations include lymph node enlargement, eye involvement (sicca syndrome [dry eyes], iritis, or iridocyclitis), skin lesions (e.g., erythema nodosum), and visceral (liver, skin, marrow) involvement. Lung involvement occurs in 90% of cases, with formation of granulomas and interstitial fibrosis.

Hypersensitivity Pneumonitis

The term *hypersensitivity pneumonitis* describes a spectrum of immunologically mediated, predominantly interstitial lung disorders caused by intense, often prolonged exposure to inhaled organic antigens. Affected individuals have an abnormal sensitivity or heightened reactivity to the causative antigen, which, in contrast to asthma, leads to pathologic changes that primarily involve the alveolar walls (thus the synonym *extrinsic allergic alveolitis*). It is important to recognize these diseases early in their course because progression to serious chronic fibrotic lung disease can be prevented by removal of the environmental agent.

Most commonly, hypersensitivity results from the inhalation of organic dust containing antigens made up of the spores of thermophilic bacteria, fungi, animal proteins, or bacterial products. Numerous syndromes are described, depending on the occupation or exposure of the individual, but all share the same stereotypic features. *Farmer's lung* results from exposure to dusts generated from humid, warm, newly harvested hay that permits the rapid proliferation of the spores of thermophilic actinomycetes. *Pigeon breeder's lung* (bird fancier's disease) is provoked by proteins from serum, excreta, or feathers of birds. *Humidifier or air-conditioner lung* is caused by thermophilic bacteria in heated water reservoirs. Pet birds and moldy basements are easily missed unless asked about specifically.

Several lines of evidence suggest that hypersensitivity pneumonitis is an immunologically mediated disease:

- Bronchoalveolar lavage specimens from the acute phase show increased levels of proinflammatory chemokines such as macrophage inflammatory protein 1 α (also known as CCL3) and IL-8.
- Bronchoalveolar lavage specimens also consistently demonstrate increased numbers of both CD4+ and CD8+ T lymphocytes.
- Most patients have specific antibodies against the causative antigen in their serum.
- Complement and immunoglobulins have been demonstrated within vessel walls by immunofluorescence.
- The presence of nonnecrotizing granulomas in two-thirds of the patients suggests that T cell-mediated (type IV) hypersensitivity reactions against the implicated antigens have a pathogenic role.

MORPHOLOGY

Histologic changes depend on the phase of the disease; acute alveolar damage is seen in the first hours and few days after antigen exposure, while subacute changes are characteristically centered on bronchioles. They include interstitial pneumonitis, consisting primarily of lymphocytes, plasma cells, and macrophages (eosinophils are rare), as well as nonnecrotizing granulomas (Fig. 15.20). Interstitial fibrosis with fibroblastic foci, honeycombing, and obliterative bronchiolitis together with granulomas is seen in the chronic phase. The patterns of fibrosis present may vary. The subacute phase classically shows interstitial

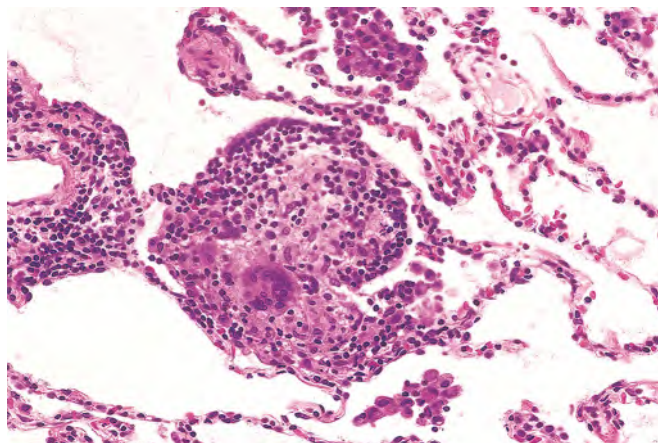


Fig. 15.20 Hypersensitivity pneumonitis. Loosely formed interstitial granulomas and chronic inflammation are characteristic.

fibrosis with features that overlap with nonspecific interstitial pneumonia and organizing pneumonia, bronchocentric injury (such as peribronchiolar metaplasia of the alveolar duct/respiratory bronchiole), and poorly formed granulomas. Some chronic cases progress to a usual interstitial pneumonia-type pattern of fibrosis that lacks characteristic features.

Clinical Features

The clinical manifestations are varied. Episodic acute attacks following inhalation of antigenic dust in sensitized patients are marked by fever, dyspnea, cough, and leukocytosis. Micronodular interstitial infiltrates may appear in the chest radiograph and pulmonary function tests show an acute restrictive disorder. Symptoms usually appear 4 to 6 hours after exposure and may last for 12 hours to several days. They recur with re-exposure. If exposure is continuous and protracted, a chronic form of the disease supervenes, leading to progressive fibrosis, dyspnea, and cyanosis—a picture similar to that seen in other forms of chronic interstitial disease.

Pulmonary Eosinophilia

Although relatively rare, there are several clinical and pathologic pulmonary entities that are characterized by infiltrates of eosinophils, recruited in part by elevated alveolar levels of eosinophil attractants such as IL-5. Pulmonary eosinophilia is divided into the following categories:

- *Acute eosinophilic pneumonia with respiratory failure.* This is an acute illness of unknown cause. It is marked by rapid onset of fever, dyspnea, and hypoxemic respiratory failure. The chest radiograph shows diffuse infiltrates, and bronchoalveolar lavage fluid contains more than 25% eosinophils. Histology shows diffuse alveolar damage and many eosinophils. There is a prompt response to corticosteroids.
- *Secondary eosinophilia*, which occurs in several parasitic, fungal, and bacterial infections; in hypersensitivity pneumonitis; in drug allergies; and in association with asthma, allergic bronchopulmonary aspergillosis, or *Churg-Strauss syndrome*, a form of vasculitis. *Löffler syndrome* is a form of secondary eosinophilia caused by invasive infection of alveoli by the larvae of *Ascaris*, hookworm, or *Strongyloides* helminth species (Chapter 17).
- *Idiopathic chronic eosinophilic pneumonia*, typically characterized by focal areas of consolidation in the periphery of the lung due to infiltrates of lymphocytes and eosinophils within the septal walls and the alveolar spaces. Interstitial fibrosis and organizing pneumonia are often present. These patients have cough, fever, night sweats, dyspnea, and weight loss, all of which respond to corticosteroid therapy.

Chronic eosinophilic pneumonia is diagnosed when other causes of pulmonary eosinophilia are excluded.

Smoking-Related Interstitial Diseases

Smoking-related diseases can be grouped into obstructive diseases (emphysema and chronic bronchitis, already discussed) and restrictive or interstitial diseases. Most individuals with idiopathic pulmonary fibrosis are smokers; however, the role of cigarette smoking in its pathogenesis has not been clarified yet. Desquamative interstitial pneumonia and respiratory bronchiolitis–associated interstitial lung disease are two other smoking-associated interstitial lung diseases worthy of brief mention.

Desquamative Interstitial Pneumonia

Desquamative interstitial pneumonia is characterized by large collections of macrophages in the airspaces of a current or former smoker. The macrophages were originally thought to be desquamated pneumocytes—thus the misnomer “desquamative interstitial pneumonia.”

MORPHOLOGY

The most striking finding is the accumulation of numerous macrophages with abundant cytoplasm containing dusty brown pigment (**smokers’ macrophages**) in the airspaces (Fig. 15.21). Some of the macrophages contain lamellar bodies (composed of surfactant) within phagocytic vacuoles, presumably derived from necrotic type II pneumocytes. The alveolar septa are thickened by a sparse inflammatory infiltrate of lymphocytes, plasma cells, and occasional eosinophils. The septa are lined by plump, cuboidal pneumocytes. Interstitial fibrosis, when present, is mild. Emphysema and airway changes suggestive of obstructive lung disease are often present.

Desquamative interstitial pneumonia usually presents in the fourth or fifth decade of life and is now equally common in males and females. Virtually all patients are cigarette smokers. Presenting symptoms include an insidious onset of dyspnea and dry cough over weeks or months, often associated with clubbing of digits. Pulmonary function tests usually show a mild restrictive abnormality and moderately decreased diffusing capacity. Patients with desquamative interstitial pneumonia typically have an excellent response to steroid therapy and cessation of smoking, but occasionally patients progress to interstitial fibrosis.

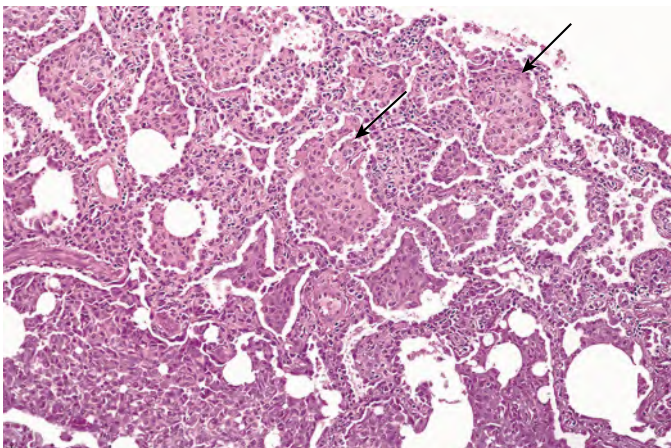


Fig. 15.21 Desquamative interstitial pneumonia. Medium-power detail of lung demonstrates the accumulation of large numbers of macrophages (arrows) within the alveolar spaces and only mild fibrous thickening of the alveolar walls.

Respiratory Bronchiolitis–Associated Interstitial Lung Disease

Respiratory bronchiolitis–associated interstitial lung disease is marked by chronic inflammation and peribronchiolar fibrosis. It is a common histologic lesion in cigarette smokers. It is characterized by the presence of pigmented intraluminal macrophages within first- and second-order respiratory bronchioles. In its mildest form, it is most often an incidental finding in the lungs of smokers or ex-smokers. The term *respiratory bronchiolitis–associated interstitial lung disease* is used for patients who develop significant pulmonary symptoms, abnormal pulmonary function, and imaging abnormalities.

MORPHOLOGY

The changes are patchy at low magnification and have a bronchiolocentric distribution. Respiratory bronchioles, alveolar ducts, and peribronchiolar spaces contain aggregates of dusty brown macrophages (**smokers’ macrophages**) similar to those seen in desquamative interstitial pneumonia. There is a patchy submucosal and peribronchiolar infiltrate of lymphocytes and histiocytes. Mild peribronchiolar fibrosis is also seen, which expands contiguous alveolar septa. Centrilobular emphysema is common but not severe. Desquamative interstitial pneumonia can sometimes be found in different parts of the same lung.

Symptoms are usually mild, consisting of gradual onset of dyspnea and cough in patients who are typically current cigarette smokers with exposures of over 30 pack-years in the fourth or fifth decade of life. Cessation of smoking usually results in improvement.

Pulmonary Langerhans Cell Histiocytosis

Pulmonary Langerhans cell histiocytosis is a rare disease characterized by focal collections of Langerhans cells (often accompanied by eosinophils). As these lesions progress, scarring occurs, leading to airway destruction and alveolar damage that result in the appearance of irregular cystic spaces. Imaging of the chest shows characteristic cystic and nodular abnormalities.

More than 90% of affected patients are relatively young adult smokers or ex-smokers; among smokers, about half improve after smoking cessation, suggesting that in some cases the lesions are a reactive inflammatory process. However, in other cases the Langerhans cells have activating mutations in the serine/threonine kinase BRAF, a feature shared with Langerhans cell histiocytosis involving other tissues that marks these proliferations as neoplasms (Chapter 13). While usually associated with an excellent prognosis, the disease is progressive in some patients, sometimes even necessitating lung transplantation.

Pulmonary Alveolar Proteinosis

Pulmonary alveolar proteinosis (PAP) is a rare disease caused by defects in pulmonary macrophage function due to deficient granulocyte-macrophage colony-stimulating factor (GM-CSF) signaling, which results in the accumulation of surfactant in the intraalveolar and bronchiolar spaces. PAP is characterized radiologically by bilateral patchy asymmetric pulmonary opacifications. There are three distinct classes of disease—autoimmune (formerly called acquired), secondary, and congenital—each with a similar spectrum of histologic changes.

- **Autoimmune PAP** is caused by autoantibodies that bind and neutralize the function of GM-CSF. It occurs primarily in adults, represents 90% of all cases of PAP, and lacks any familial predisposition. Knockout of the GM-CSF gene in mice induces PAP, and these mice are “cured” by treatment with GM-CSF. Loss of GM-CSF signaling

blocks the terminal differentiation of alveolar macrophages, impairing their ability to catabolize surfactant.

- **Secondary PAP** is uncommon and is associated with diverse diseases, including hematopoietic disorders, malignancies, immunodeficiency disorders, lysinuric protein intolerance (an inborn error of amino acid metabolism), and acute silicosis and other inhalational syndromes. It is speculated that these diseases somehow impair GM-CSF-dependent signaling or downstream events involved in macrophage maturation or function, again leading to inadequate clearance of surfactant from alveolar spaces.
- **Hereditary PAP** is extremely rare, occurs in neonates, and is caused by loss-of-function mutations in the genes that encode GM-CSF or the GM-CSF receptor.

MORPHOLOGY

The disease is characterized by the accumulation of intraalveolar precipitates containing surfactant proteins, causing focal-to-confluent consolidation of large areas of the lungs with minimal inflammatory reaction (Fig. 15.22). Consequently, there is a marked increase in the size and weight of the lung. The alveolar precipitate is pink, homogeneous, and periodic acid–Schiff–positive and contains cholesterol clefts and surfactant proteins (which can be demonstrated by immunohistochemical stains). Ultrastructurally, the surfactant lamellae in type II pneumocytes are normal, in contrast to surfactant dysfunction disorders (described below).

Clinical Features

Adult patients, for the most part, present with cough and abundant sputum that often contains chunks of gelatinous material. Some have symptoms lasting for years, often with intermittent febrile illnesses caused by secondary pulmonary infections with a variety of organisms. Progressive dyspnea, cyanosis, and respiratory insufficiency may occur, but other patients follow a benign course, with eventual resolution of the lesions. Whole-lung lavage is the standard of care and provides benefit regardless of the underlying defect. GM-CSF therapy is safe and effective in more than half of patients with autoimmune PAP,

and therapy directed at the underlying disorder may also be helpful. Primary disease is treated with GM-CSF replacement therapy, sometimes followed by allogeneic hematopoietic stem cell transplantation, which can be curative.

Surfactant Dysfunction Disorders

Surfactant dysfunction disorders are diseases caused by diverse mutations in genes encoding proteins involved in surfactant trafficking or secretion. Clinical manifestations range from neonatal respiratory failure to adult-onset interstitial lung disease. The most commonly mutated genes are:

- **ATP-binding cassette protein member 3 (ABCA3)** is the most frequently mutated gene in surfactant dysfunction disorders. The protein is a transmembrane lipid transporter that is involved in regulating surfactant homeostasis. Mutations in *ABCA3* are associated with an autosomal recessive disorder that usually presents in the first few months of life with rapidly progressive respiratory failure followed by death. Less commonly, it comes to attention in older children and in adults with chronic interstitial lung disease.
- The second most commonly mutated gene in surfactant dysfunction disorders encodes *surfactant protein C*. This form has an autosomal dominant mode of inheritance and has a highly variable course.
- The third most commonly mutated gene in surfactant dysfunction disorders encodes *surfactant protein B*. This form has an autosomal recessive mode of inheritance. Typically, the infant is full-term and develops progressive respiratory distress shortly after birth. Death ensues between 3 and 6 months of age unless lung transplantation is performed.

MORPHOLOGY

There is a variable amount of intraalveolar pink granular material, type II pneumocyte hyperplasia, interstitial fibrosis, and alveolar simplification. Immunohistochemical stains show the lack of surfactant proteins C and B in their respective deficiencies. Ultrastructurally, abnormalities in lamellar bodies in type II pneumocytes can be seen in all three; small lamellar bodies with electron dense cores are diagnostic for *ABCA3* mutation (Fig. 15.22B).

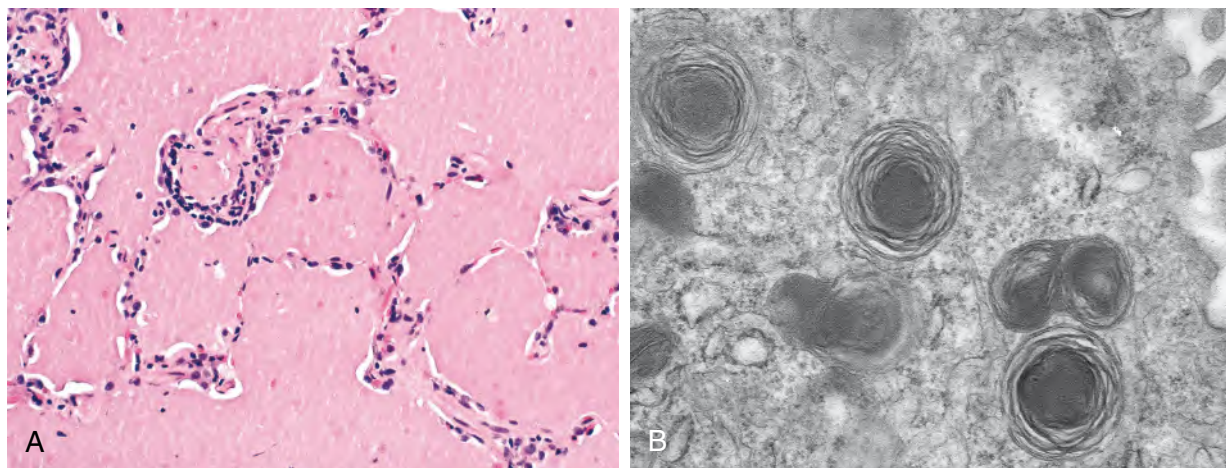


Fig. 15.22 Pulmonary alveolar proteinosis and surfactant dysfunction disorder. (A) The alveoli are filled with a dense, amorphous, protein-lipid granular precipitate, while the alveolar walls are normal in this case of pulmonary alveolar proteinosis. (B) An electron micrograph shows type II pneumocytes containing small surfactant lamellae with electron dense cores, an appearance that is characteristic of cases of surfactant dysfunction disorder associated with *ABCA3* mutations.

DISEASES OF VASCULAR ORIGIN

Pulmonary Embolism and Infarction

Pulmonary embolism is an important cause of morbidity and mortality, particularly in patients who are bedridden, but also in a wide range of conditions that are associated with hypercoagulability. Blood clots that occlude the large pulmonary arteries are almost always embolic in origin. The usual source—thrombi in the deep veins of the leg (>95% of cases)—and the magnitude of the clinical problem were discussed in Chapter 4. Pulmonary embolism causes more than 50,000 deaths in the United States each year. Its incidence at autopsy has varied from 1% in the general population of hospital patients to 30% in patients dying after severe burns, trauma, or fractures. It is the sole or major contributing cause of death in about 10% of adults who die acutely in hospitals. By contrast, large-vessel pulmonary thromboses are rare and develop only in the presence of pulmonary hypertension and heart failure.

Pathogenesis

Pulmonary embolism usually occurs in patients with a predisposing condition that increases the risk of thrombosis (Chapter 4). Patients often have cardiac disease or cancer or have been immobilized for several days or weeks prior to the appearance of a symptomatic embolism. Patients with hip fractures are at particularly high risk. Hypercoagulable states, either primary (e.g., factor V Leiden) or secondary (e.g., obesity, recent surgery, cancer, oral contraceptive use, pregnancy), are important risk factors. Indwelling central venous lines can be a nidus for formation of right atrial thrombi, which can embolize to the lungs. Rarely, pulmonary embolism may consist of fat, air, or tumor. Small bone marrow emboli are often seen in patients who die after chest compressions performed during resuscitative efforts.

The pathophysiologic response and clinical significance of pulmonary embolism depend on the extent to which pulmonary artery blood flow is obstructed, the size of the occluded vessels, the number of emboli, and the cardiovascular health of the patient. Emboli have two deleterious pathophysiologic consequences: *respiratory compromise* due to the non-perfused, although ventilated, segment; and *hemodynamic compromise* due to increased resistance to pulmonary blood flow caused by the embolic obstruction. Sudden death often ensues, largely as a result of the blockage of blood flow through the lungs. Death may also be caused by acute right-sided heart failure (*acute cor pulmonale*).

MORPHOLOGY

Large emboli lodge in the main pulmonary artery or its major branches or at the bifurcation as a **saddle embolus** (Fig. 15.23A). Smaller emboli travel out into the more peripheral vessels, where they may cause hemorrhage or infarction. In patients with adequate cardiovascular function, the bronchial arterial supply sustains the lung parenchyma; in this instance, hemorrhage may occur, but there is no infarction. In those in whom cardiovascular function is already compromised, such as patients with heart or lung disease, infarction is more likely. Overall, about 10% of emboli cause infarction. About 75% of infarcts affect the lower lobes, and in more than half, multiple lesions occur. They vary in size from barely visible to massive lesions involving large parts of a lobe. Typically, they extend to the periphery of the lung as a wedge with the apex pointing toward the hilus of the lung. In many cases, an occluded vessel is identified near the apex of the infarct. Pulmonary embolus can be distinguished from a postmortem clot by the presence of the lines of Zahn in the thrombus (Chapter 4).

The pulmonary infarct is classically hemorrhagic and appears as a raised, red-blue area in the early stages (Fig. 15.23B). Often, the apposed pleural surface is covered by a fibrinous exudate. The red cells begin to lyse within 48 hours, and the infarct becomes paler and eventually red-brown as hemosiderin is produced. With the passage of time, fibrous replacement begins at the margins as a gray-white peripheral zone and eventually converts the infarct into a contracted scar. Histologically, the hemorrhagic area shows ischemic necrosis of the alveolar walls, bronchioles, and vessels. If the infarct is caused by an infected embolus, the neutrophilic inflammatory reaction can be intense. Such lesions are referred to as **septic infarcts**, some of which turn into abscesses.

Clinical Features

A large pulmonary embolus is one of the few causes of virtually instantaneous death. During cardiopulmonary resuscitation in such instances, the patient frequently is said to have electromechanical dissociation, in which the electrocardiogram has a rhythm, but no pulses are palpated because no blood is entering the pulmonary arterial circulation. If the patient survives after a sizable pulmonary embolus, however, the clinical syndrome may mimic myocardial infarction, with severe chest pain, dyspnea, and shock. Small emboli are silent or induce only transient chest pain and cough. In the remaining group of patients with symptomatic pulmonary embolism, the most common presenting symptoms

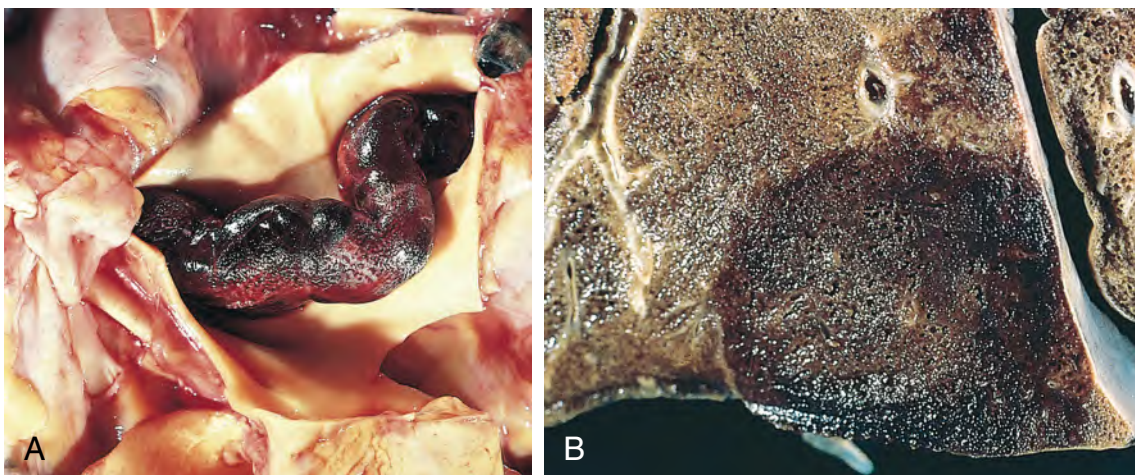


Fig. 15.23 Pulmonary embolism. (A) Large saddle embolus from the femoral vein lying astride the main left and right pulmonary arteries. (B) Acute hemorrhagic pulmonary infarct. (From the Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

(in descending order) are dyspnea, pleuritic pain, and cough, accompanied in about half of cases by calf or thigh swelling or pain. Emboli that lead to pulmonary infarction may additionally produce fever and hemoptysis. An overlying fibrinous pleuritis may produce a pleural friction rub. Recurrent small emboli can give rise to pulmonary hypertension (discussed later).

In hemodynamically stable patients with a low to moderate risk of pulmonary embolism, D-dimer measurement is a useful screening test, as a normal D-dimer level excludes pulmonary embolism. Definitive diagnosis is usually made by computed tomographic pulmonary angiogram, which identifies obstructed pulmonary arteries. Rarely, other diagnostic methods, such as ventilation-perfusion scanning, are required. Deep vein thrombosis can be diagnosed with duplex ultrasonography. Chest radiography may be normal or disclose a pulmonary infarct, usually 12 to 36 hours after it has occurred, as a wedge-shaped infiltrate.

After the initial acute insult, emboli often resolve via contraction and fibrinolysis, particularly in relatively young patients. If unresolved, with time multiple small emboli may lead to pulmonary hypertension and chronic cor pulmonale. Perhaps most important is that a small embolus may presage a larger one. In the presence of an underlying predisposing condition, patients with a pulmonary embolus have a 30% chance of experiencing a second embolus.

Prevention of pulmonary embolism is a major clinical challenge for which there is no easy solution. Prophylactic therapy includes early ambulation in postoperative and postpartum patients, elastic stockings and graduated compression stockings for bedridden patients, and anticoagulation in high-risk individuals. Treatment of pulmonary embolism includes anticoagulation and supportive measures; thrombolysis may have some benefit in patients with severe complications (e.g., shock), but carries a high risk of bleeding. Those at risk of recurrent pulmonary embolism in whom anticoagulation is contraindicated may be fitted with an inferior vena cava filter (an “umbrella”) that catches clots before they reach the lungs.

KEY CONCEPTS

Pulmonary Embolism

- Almost all large pulmonary artery thrombi are embolic in origin, usually arising from the deep veins of the lower leg.
- Risk factors include prolonged bed rest, leg surgery, severe trauma, congestive heart failure, use of oral contraceptives (especially those with high estrogen content), disseminated cancer, and inherited forms of hypercoagulability.
- The vast majority (60% to 80%) of emboli are clinically silent, a minority (5%) cause acute cor pulmonale, shock, or death (typically from large “saddle emboli”), and the remainder cause symptoms related to ventilation-perfusion mismatch and/or pulmonary infarction, particularly dyspnea and pleuritic chest pain.
- Risk of recurrence is high, and recurrent embolism may eventually lead to pulmonary hypertension and cor pulmonale.

Pulmonary Hypertension

Pulmonary hypertension is defined as a mean pulmonary artery pressure greater than or equal to 25 mm Hg at rest. Based on underlying mechanisms, the WHO has classified pulmonary hypertension into five groups: (1) pulmonary arterial hypertension, a diverse collection of disorders that all primarily impact small pulmonary muscular arteries; (2) pulmonary hypertension due to left heart failure; (3) pulmonary hypertension due to lung diseases and/or hypoxia; (4) chronic thromboembolic pulmonary hypertension and other obstructions; and

(5) pulmonary hypertension with unclear and/or multifactorial mechanisms.

Pathogenesis

As can be gathered from the WHO classification, pulmonary hypertension has diverse causes even within each group. It is most frequently associated with structural cardiopulmonary conditions that increase either pulmonary blood flow, pulmonary vascular resistance, or left heart resistance to blood flow. Some of the more common causes are:

- *Chronic obstructive or interstitial lung diseases* (group 3). These diseases obliterate alveolar capillaries, increasing pulmonary resistance to blood flow and, secondarily, pulmonary blood pressure.
- *Antecedent congenital or acquired heart disease* (group 2). Mitral stenosis, for example, causes an increase in left atrial pressure and pulmonary venous pressure that is eventually transmitted to the arterial side of the pulmonary vasculature, leading to hypertension.
- *Recurrent thromboemboli* (group 4). Recurrent pulmonary emboli may cause pulmonary hypertension by reducing the functional cross-sectional area of the pulmonary vascular bed, which in turn leads to an increase in pulmonary vascular resistance.
- *Autoimmune diseases* (group 1). Several of these diseases (most notably systemic sclerosis) involve the pulmonary vasculature and/or the interstitium, leading to increased vascular resistance and pulmonary hypertension.
- *Obstructive sleep apnea* (group 3) is a common disorder that is associated with obesity and hypoxemia. It is a significant contributor to the development of pulmonary hypertension and cor pulmonale.

Uncommonly, pulmonary hypertension is encountered in patients in whom all known causes are excluded; this is referred to as *idiopathic pulmonary arterial hypertension* and falls into group 1 disease. However, “idiopathic” is a bit of a misnomer, as up to 80% of “idiopathic” cases (sometimes referred to as primary pulmonary hypertension) have a genetic basis, sometimes being inherited in families as an autosomal dominant trait. Within these families, there is incomplete penetrance, and only 10% to 20% of the family members actually develop overt disease.

As is often the case, much has been learned about the pathogenesis of pulmonary hypertension by investigating the molecular basis of the uncommon familial form of the disease. The first mutation to be discovered in familial pulmonary arterial hypertension was in the gene encoding bone morphogenetic protein receptor type 2 (*BMPR2*). Inactivating germline mutations in the *BMPR2* gene are found in 75% of the familial cases of pulmonary hypertension and 25% of sporadic cases. Other mutations have been discovered that also converge on the *BMPR2* pathway and affect intracellular signaling. It has also been demonstrated that *BMPR2* is downregulated in lungs from some patients with idiopathic pulmonary arterial hypertension without *BMPR2* mutations.

BMPR2 is a cell surface protein belonging to the TGF- β receptor superfamily, which binds a variety of cytokines, including TGF- β , bone morphogenetic protein (BMP), activin, and inhibin. Originally described as a pathway that regulates bone growth, BMP-BMPR2 signaling is now known to be important for embryogenesis, apoptosis, and cell proliferation and differentiation. Details remain to be worked out, but it appears that haploinsufficiency for *BMPR2* leads to dysfunction and proliferation of endothelial cells and vascular smooth muscle cells. Because only 10% to 20% of individuals with *BMPR2* mutations develop disease, it is likely that modifier genes and/or environmental triggers also contribute to the pathogenesis of the disorder. A two-hit

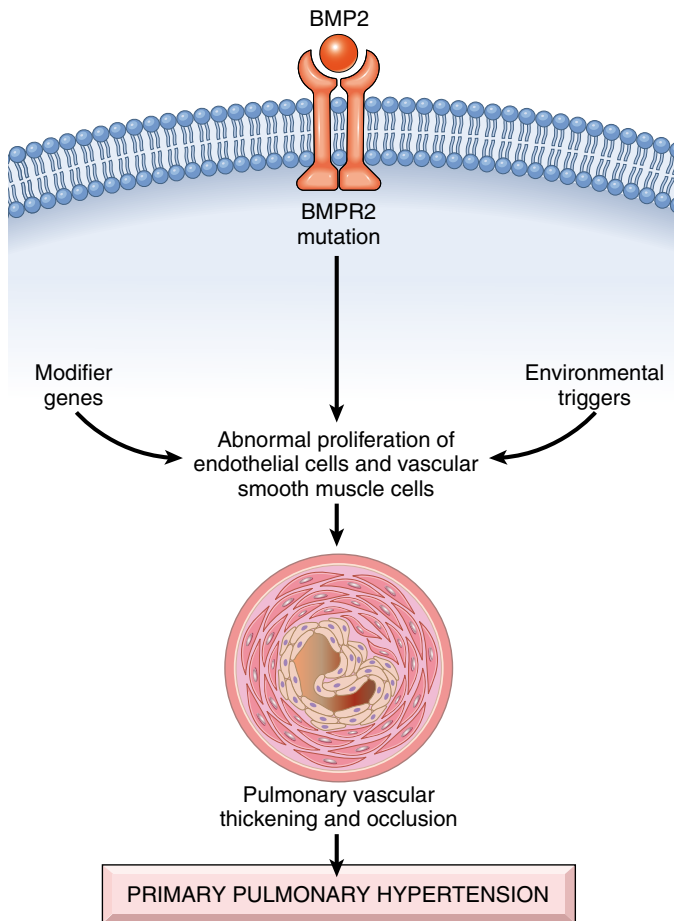


Fig. 15.24 Pathogenesis of primary (idiopathic) pulmonary hypertension. See text for details.

model has been proposed whereby a genetically susceptible individual with a *BMPRII* mutation requires additional genetic or environmental insults to develop the disease (Fig. 15.24).

MORPHOLOGY

All forms of pulmonary hypertension are associated with **medial hypertrophy of the pulmonary muscular and elastic arteries** and **right ventricular hypertrophy**. The presence of many organizing or recanalized thrombi favors recurrent pulmonary emboli as the cause, and the coexistence of diffuse pulmonary fibrosis, or severe emphysema and chronic bronchitis, points to chronic hypoxia and loss of capillary beds as initiating events. The vessel changes can involve the entire arterial tree, from the main pulmonary arteries down to the arterioles (Fig. 15.25A–C). In the most severe cases, the thickening of the walls of the pulmonary artery and its major branches take on some features of systemic atherosclerosis, but classic atherosclerotic changes are not seen (see Fig. 15.25A). Instead, the arterioles and small arteries (40 to 300 μm in diameter) are prominently affected by striking medial hypertrophy and intimal fibrosis, sometimes narrowing the lumens to pinpoint channels (see Fig. 15.25B). One extreme in the spectrum of pathologic changes is the **plexiform lesion**, so-called because a tuft of capillary formations is present, producing a network, or web, that spans the lumens of dilated thin-walled, small arteries and may extend outside the vessel (see Fig. 15.25C). Plexiform lesions are most prominent in idiopathic and familial pulmonary hypertension (group 1), unrepaired congenital heart disease with left-to-right shunts (group 2), and pulmonary hypertension associated with human immunodeficiency virus (HIV) infection and drugs (also group 1).

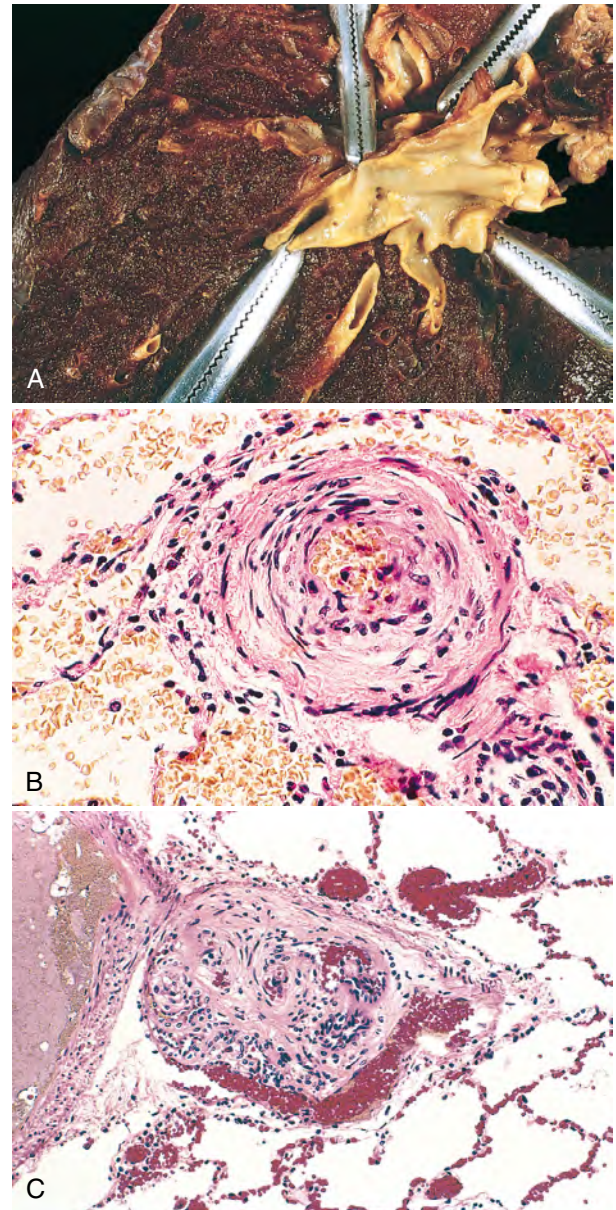


Fig. 15.25 Vascular changes in pulmonary arterial hypertension. (A) Atheroma-like changes, a finding usually limited to large vessels. (B) Marked medial hypertrophy. (C) Plexiform lesion of small arteries that is characteristic of advanced pulmonary hypertension.

Clinical Features

Idiopathic (often inherited) forms of pulmonary hypertension are most common in females who are 20 to 40 years of age but may occasionally present in childhood. Clinical signs and symptoms in all types become evident only in advanced disease. In cases of idiopathic disease, the presenting features are usually dyspnea and fatigue, but some patients have chest pain of the anginal type. Over time, severe respiratory distress, cyanosis, and right ventricular hypertrophy occur, and death from decompensated cor pulmonale, often with superimposed thromboembolism and pneumonia, ensues within 2 to 5 years in 80% of patients.

Treatment choices depend on the underlying cause. For patients with secondary disease, therapy is directed at the trigger (e.g., thromboembolic disease or hypoxemia). A variety of vasodilators have been used with varying success in those with group 1 or refractory disease

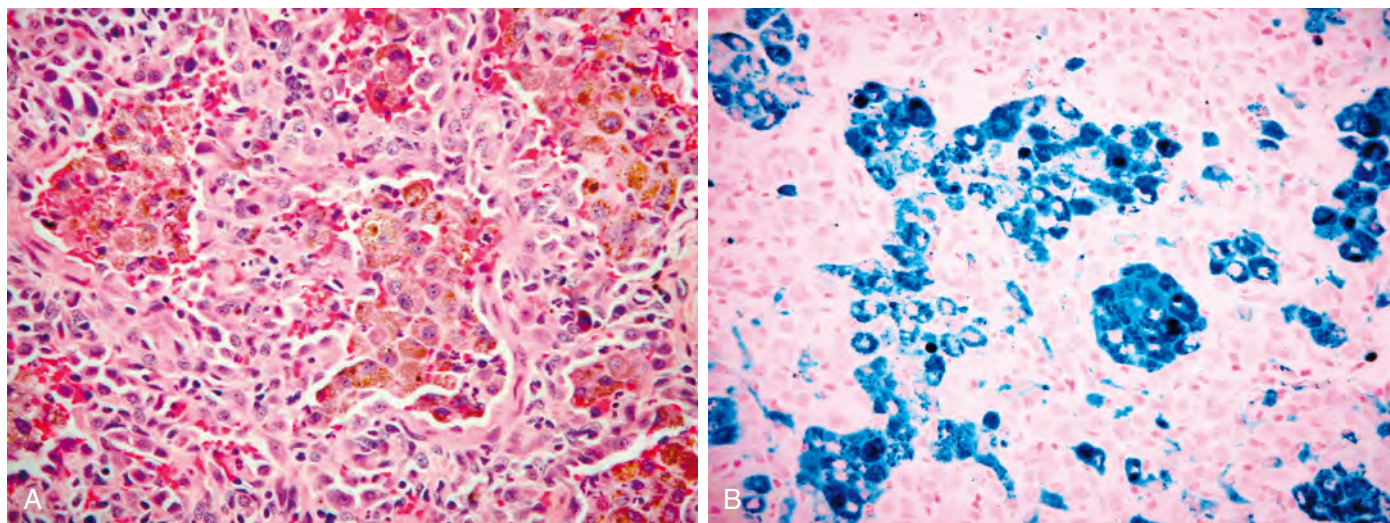


Fig. 15.26 Diffuse pulmonary hemorrhage. (A) Intraalveolar spaces contain large numbers of macrophages laden with phagocytosed red cells and the hemoglobin breakdown product hemosiderin. (B) A Prussian-blue stain identifies iron derived from hemoglobin in the macrophages.

belonging to other groups. Lung transplantation provides definitive treatment for selected patients.

Diffuse Pulmonary Hemorrhage Syndromes

Pulmonary hemorrhage is a dramatic complication of some interstitial lung disorders. *Pulmonary hemorrhage syndromes* (Fig. 15.26) include (1) Goodpasture syndrome, (2) idiopathic pulmonary hemosiderosis, and (3) vasculitis-associated hemorrhage. The latter encompasses conditions such as hypersensitivity angiitis, granulomatosis with polyangiitis, and systemic lupus erythematosus (Chapter 11).

Goodpasture Syndrome (Anti–Glomerular Basement Membrane Antibody Disease With Pulmonary Involvement)

Goodpasture syndrome is an uncommon autoimmune disease in which kidney and lung injury is caused by circulating autoantibodies against the noncollagenous domain of the $\alpha 3$ chain of collagen IV. When only renal disease is caused by this antibody, it is called anti-glomerular basement membrane disease. The term *Goodpasture syndrome* applies to the 40% to 60% of patients who develop pulmonary hemorrhage in addition to renal disease. Although any age can be affected, most cases occur in people in their teens or 20s, and in contrast to many other autoimmune diseases, there is a male preponderance. Most patients are active smokers.

Pathogenesis

The immunopathogenesis of the syndrome and the nature of Goodpasture syndrome antigens are described in Chapter 20. The pathogenic antibodies initiate inflammatory destruction of the basement membrane in renal glomeruli and pulmonary alveoli, giving rise to rapidly progressive glomerulonephritis and a necrotizing hemorrhagic interstitial pneumonitis. The trigger that initiates the production of anti-basement membrane antibodies is unknown. In addition to autoreactive B cells, some experimental evidence suggests that T cells also contribute, both by enhancing B-cell function and by participating directly in glomerular damage and crescent formation. As with other autoimmune disorders, there is an association with certain HLA subtypes (e.g., HLA-DRB1*1501 and HLA-DRB1*1502).

MORPHOLOGY

In the classic case, the lungs are heavy, with areas of red-brown consolidation. Histologically, there is focal necrosis of alveolar walls associated with intraalveolar hemorrhages. Often the alveoli contain hemosiderin-laden macrophages (see Fig. 15.26). In later stages there may be fibrous thickening of the septa, hyperplasia of type II pneumocytes, and organization of blood in alveolar spaces. In many cases, immunofluorescence studies reveal linear deposits of immunoglobulins along the basement membranes of the septal walls. The kidneys have characteristic findings of focal proliferative glomerulonephritis in early cases or crescentic glomerulonephritis in patients with rapidly progressive glomerulonephritis. Diagnostic linear deposits of immunoglobulins and complement are seen by immunofluorescence studies along the glomerular basement membranes even in the few patients without renal disease.

Clinical Features

Most cases begin with respiratory symptoms, principally hemoptysis, and radiographic evidence of focal pulmonary consolidations. Soon, manifestations of glomerulonephritis appear, leading to rapidly progressive renal failure. The most common cause of death is uremia. The once dismal prognosis for this disease has been markedly improved by intensive plasmapheresis. This procedure is thought to be beneficial by removing anti-basement membrane antibodies and possibly other mediators of immunologic injury. Simultaneous immunosuppressive therapy inhibits further antibody production, ameliorating both lung hemorrhage and glomerulonephritis.

Idiopathic Pulmonary Hemosiderosis

Idiopathic pulmonary hemosiderosis is a rare disorder characterized by intermittent, diffuse alveolar hemorrhage. Most cases occur in young children, although the disease has been reported in adults as well. It usually presents with an insidious onset of productive cough, hemoptysis, and anemia associated with diffuse pulmonary infiltrates.

The cause and pathogenesis are unknown, and anti-basement membrane antibodies (the cause of Goodpasture syndrome) are undetectable. However, favorable response to long-term immunosuppression with prednisone and/or azathioprine indicates that an immunologic

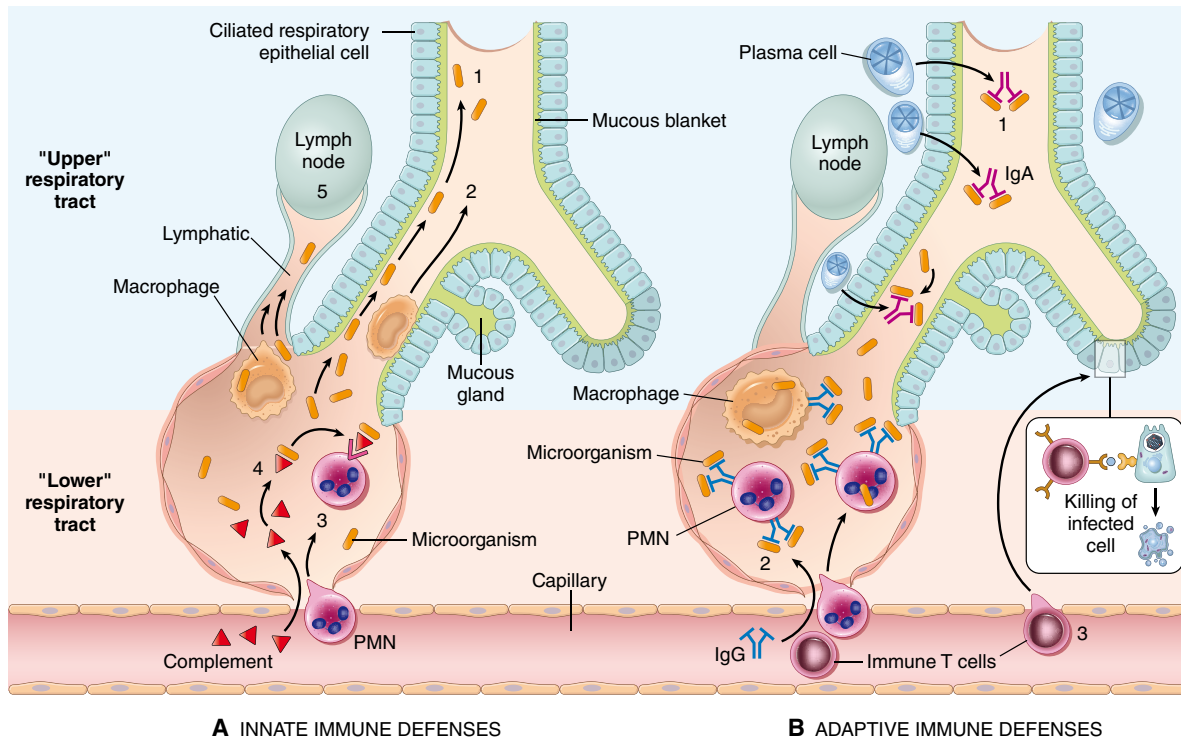


Fig. 15.27 Lung defense mechanisms. (A) Innate defenses against infection: 1, In the healthy lung, removal of microbial organisms depends on entrapment in the mucous blanket and removal by means of the mucociliary elevator; 2, phagocytosis by alveolar macrophages can kill and degrade organisms and remove them from the air spaces by migrating onto the mucociliary elevator; or 3, phagocytosis and killing by neutrophils recruited by macrophage factors; 4, complement may enter the alveoli and be activated by the alternative pathway to produce the opsonin C3b, which enhances phagocytosis; 5, organisms, including those ingested by phagocytes, may reach the draining lymph nodes to initiate adaptive immune responses. (B) Additional mechanisms operate after development of adaptive immunity: 1, Secreted IgA can block attachment of the microorganism to epithelium in the upper respiratory tract; 2, in the lower respiratory tract, serum antibodies (IgM, IgG) are present in the alveolar lining fluid and activate complement more efficiently by the classical pathway, yielding C3b (not shown); in addition, IgG is an opsonin; 3, the accumulation of immune T cells is important for controlling infections by viruses and other intracellular microorganisms. *Ig*, Immunoglobulin; *PMN*, neutrophil.

mechanism may be involved in the pulmonary capillary damage underlying alveolar bleeding. In addition, long-term follow-up shows that some affected patients develop other immune disorders.

Granulomatosis With Polyangiitis

Previously called Wegener granulomatosis, this autoimmune disease most often involves the upper respiratory tract and/or the lungs, with hemoptysis being the common presenting symptom. Its features are discussed in Chapter 11. Here, it suffices to emphasize that a transbronchial lung biopsy might provide the only tissue available for diagnosis. Since the amount of tissue is small, necrosis and granulomatous vasculitis might not be present. Rather, the diagnostically important histologic features are capillaritis and scattered, poorly formed granulomas (unlike those of sarcoidosis, which are rounded and well-defined).

PULMONARY INFECTIONS

Respiratory tract infections are more frequent than infections of any other organ and account for the largest number of workdays lost in the general population. The vast majority consist of upper respiratory tract infections caused by viruses (common cold, pharyngitis), but bacterial, viral, mycoplasmal, and fungal infections of the lung (pneumonia) account for an enormous amount of morbidity and are responsible for

2.3% of all deaths in the United States. Pneumonia can be very broadly defined as any infection of the lung parenchyma.

Pulmonary antimicrobial defense mechanisms are described in Chapter 8 and are summarized in Fig. 15.27. Pneumonia can result whenever these local defense mechanisms are impaired or the systemic resistance of the host is lowered. Factors that impair resistance include chronic diseases, immunologic deficiencies, treatment with immunosuppressive agents, and leukopenia. Local pulmonary defense mechanisms may also be compromised by many factors, including:

- *Loss or suppression of the cough reflex*, because of altered sensorium (e.g., coma), anesthesia, neuromuscular disorders, drugs, or chest pain, any of which may lead to *aspiration* of gastric contents.
- *Dysfunction of the mucociliary apparatus*, which can be caused by cigarette smoke, inhalation of hot or corrosive gases, viral diseases, or genetic defects of ciliary function (e.g., immotile cilia syndrome).
- *Accumulation of secretions* in conditions such as cystic fibrosis and bronchial obstruction.
- *Interference with the phagocytic and bactericidal activities of alveolar macrophages* by alcohol, tobacco smoke, anoxia, or oxygen intoxication.
- *Pulmonary congestion and edema*.

Defects in innate immunity (including neutrophil and complement defects) and humoral immunodeficiency typically lead to an increased

incidence of infections with pyogenic bacteria. Germline mutations in *MYD88* (which encodes an adaptor for several Toll-like receptors [TLRs] that is important for activation of the transcription factor NF- κ B) are also associated with destructive bacterial (pneumococcal) pneumonias. On the other hand, cell-mediated immune defects (congenital and acquired) lead to increased infections with intracellular microbes such as mycobacteria and herpesviruses as well as with microorganisms of very low virulence, such as the fungus *Pneumocystis jirovecii*.

Several other points should be emphasized. First, to paraphrase the French physician Louis Cruveilhier in 1919 (during the Spanish flu epidemic), “flu condemns, and additional infection executes.” The most common cause of death in viral influenza epidemics is superimposed bacterial pneumonia. Second, although the portal of entry for most bacterial pneumonias is the respiratory tract, hematogenous seeding of the lungs from another organ may occur and may be difficult to distinguish from primary pneumonia. Finally, many patients with chronic diseases acquire terminal pneumonia while hospitalized (*nosocomial infection*) because of several factors: bacteria common to the hospital environment may have acquired resistance to antibiotics; opportunities for spread are increased; invasive procedures that allow entry of microbes, such as intubations and injections, are common; and bacteria may contaminate equipment used in respiratory care units.

Pneumonia is classified based on the etiologic agent or, if no pathogen can be isolated (which occurs in about 50% of cases), by the clinical setting in which the infection occurs. The latter considerably narrows the list of suspected pathogens, providing a guide for empirical antimicrobial therapy. As Table 15.7 indicates, pneumonia can arise in seven distinct clinical settings (“pneumonia syndromes”), and the implicated pathogens are fairly specific to each category.

Community-Acquired Bacterial Pneumonias

Community-acquired acute pneumonia refers to lung infection in otherwise healthy individuals that is acquired from the normal environment (in contrast to hospital-acquired pneumonia). It may be bacterial or viral. Clinical and radiologic features usually do not clearly distinguish between viral and bacterial infections, and culture and other assays are often necessary to establish the diagnosis.

Often, a bacterial infection follows an upper respiratory tract viral infection. Bacterial invasion of the lung parenchyma causes the alveoli to be filled with an inflammatory exudate, thus causing consolidation (“solidification”) of the pulmonary tissue. Many variables, such as the specific etiologic agent, the host reaction, and the extent of involvement, determine the precise form of pneumonia. Predisposing conditions include extremes of age, chronic diseases (congestive heart failure, COPD, and diabetes), congenital or acquired immune deficiencies, and decreased or absent splenic function. The latter puts the patient at risk for infection with encapsulated bacteria such as pneumococcus.

Streptococcus pneumoniae

Streptococcus pneumoniae (pneumococcus) is the most common cause of community-acquired acute pneumonia. Examination of Gram-stained sputum is an important step in the diagnosis of acute pneumonia. The presence of numerous neutrophils containing the typical gram-positive, lancet-shaped diplococci supports the diagnosis of pneumococcal pneumonia, but it must be remembered that *S. pneumoniae* is a part of the endogenous flora in 20% of adults, and therefore false-positive results may be obtained. Isolation of pneumococci from blood cultures is more specific but less sensitive (in the early phase of illness, only 20% to 30% of patients have positive blood cultures). Pneumococcal vaccines containing capsular polysaccharides from the

TABLE 15.7 Pneumonia Syndromes

Community-Acquired Acute Pneumonia
<i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> <i>Moraxella catarrhalis</i> <i>Staphylococcus aureus</i> <i>Legionella pneumophila</i> Enterobacteriaceae (<i>Klebsiella pneumoniae</i>) and <i>Pseudomonas</i> spp. <i>Mycoplasma pneumoniae</i> Chlamydia spp. (<i>C. pneumoniae</i> , <i>C. psittaci</i> , <i>C. trachomatis</i>) <i>Coxiella burnetii</i> (Q fever) Viruses: respiratory syncytial virus, parainfluenza virus, and human metapneumovirus (children); influenza A and B (adults); adenovirus (military recruits)
Healthcare-Associated Pneumonia
<i>Staphylococcus aureus</i> , methicillin-sensitive <i>Staphylococcus aureus</i> , methicillin-resistant <i>Pseudomonas aeruginosa</i> <i>Streptococcus pneumoniae</i>
Hospital-Acquired Pneumonia
Gram-negative rods, Enterobacteriaceae (<i>Klebsiella</i> spp., <i>Serratia marcescens</i> , <i>Escherichia coli</i>) and <i>Pseudomonas</i> spp. <i>Staphylococcus aureus</i> (usually methicillin-resistant)
Aspiration Pneumonia
Anaerobic oral flora (<i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>), admixed with aerobic bacteria (<i>Streptococcus pneumoniae</i> , <i>Staphylococcus aureus</i> , <i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i>)
Chronic Pneumonia
Nocardia Actinomyces Granulomatous: <i>Mycobacterium tuberculosis</i> and atypical mycobacteria, <i>Histoplasma capsulatum</i> , <i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i>
Necrotizing Pneumonia and Lung Abscess
Anaerobic bacteria (extremely common), with or without mixed aerobic infection <i>Staphylococcus aureus</i> , <i>Klebsiella pneumoniae</i> , <i>Streptococcus pyogenes</i> , and type 3 pneumococcus (uncommon)
Pneumonia in the Immunocompromised Host
Cytomegalovirus <i>Pneumocystis jirovecii</i> <i>Mycobacterium avium-intracellulare</i> complex Invasive aspergillosis Invasive candidiasis “Usual” bacterial, viral, and fungal organisms (listed herein)

common serotypes are used in individuals at high risk for pneumococcal sepsis.

Haemophilus influenzae

Haemophilus influenzae is a pleomorphic, gram-negative organism that occurs in encapsulated and nonencapsulated forms. There are six serotypes of the encapsulated form (types a to f), of which type b is the most virulent. Antibodies against the capsule protect the host from *H. influenzae* infection; hence the capsular polysaccharide b is incorporated in the widely used vaccine against *H. influenzae*. With routine use of *H. influenzae* vaccines, the incidence of disease caused by the b serotype has declined significantly. By contrast, infections with non-encapsulated forms, also called *nontypeable forms*, are increasing. These are less virulent and tend to spread along the surface of the upper respiratory tract, producing otitis media (infection of the middle

ear), sinusitis, and bronchopneumonia. Neonates and children with comorbidities such as prematurity, malignancy, and immunodeficiency are at high risk for development of invasive infection.

H. influenzae pneumonia, which may follow a viral respiratory infection, is a pediatric emergency and has a high mortality rate. Descending laryngotracheobronchitis results in airway obstruction as the smaller bronchi are plugged by dense, fibrin-rich exudates containing neutrophils, similar to that seen in pneumococcal pneumonias. Pulmonary consolidation is usually lobular and patchy but may be confluent and involve the entire lung lobe. Before a vaccine became widely available, *H. influenzae* was a common cause of suppurative meningitis in children up to 5 years of age. *H. influenzae* also causes an acute, purulent conjunctivitis (pink eye) in children and, in predisposed older patients, may cause septicemia, endocarditis, pyelonephritis, cholecystitis, and suppurative arthritis. Finally, *H. influenzae* is the most common bacterial cause of acute exacerbations of COPD.

Moraxella catarrhalis

Moraxella catarrhalis is recognized as a cause of bacterial pneumonia, especially in the elderly. It is the second most common bacterial cause of acute exacerbation of COPD. Along with *S. pneumoniae* and *H. influenzae*, *M. catarrhalis* is one of the three most common causes of otitis media in children.

Staphylococcus aureus

Staphylococcus aureus is an important cause of secondary bacterial pneumonia in children and healthy adults following viral respiratory illnesses (e.g., measles in children and influenza in both children and adults). Staphylococcal pneumonia is associated with a high incidence of complications, such as lung abscess and empyema. *Intravenous drug users* are at high risk for development of staphylococcal pneumonia in association with endocarditis. It is also an important cause of hospital-acquired pneumonia.

Klebsiella pneumoniae

Klebsiella pneumoniae is the most frequent cause of gram-negative bacterial pneumonia. It commonly afflicts debilitated and malnourished people, particularly *chronic alcoholics*. Thick, mucoid (often blood-tinged) sputum is characteristic because the organism produces an abundant viscid capsular polysaccharide, which the patient may have difficulty expectorating.

Pseudomonas aeruginosa

Although *Pseudomonas aeruginosa* most commonly causes hospital-acquired infections, it is mentioned here because of its occurrence in cystic fibrosis and immunocompromised patients. It is common in patients who are neutropenic, and it has a propensity to invade blood vessels with consequent extrapulmonary spread. *Pseudomonas* septicemia is a very fulminant disease.

Legionella pneumophila

Legionella pneumophila is the agent of Legionnaires' disease, the form of pneumonia caused by this organism. It also causes Pontiac fever, a related self-limited upper respiratory tract infection. This organism flourishes in artificial aquatic environments, such as water-cooling towers and the tubing systems of domestic (potable) water supplies. It is transmitted by either inhalation of aerosolized organisms or aspiration of contaminated drinking water. *Legionella* pneumonia is common in individuals with predisposing conditions such as cardiac, renal, immunologic, or hematologic disease. Organ transplant recipients are particularly susceptible. It can be quite severe, frequently requiring hospitalization, and immunosuppressed patients have fatality rates of

up to 50%. The diagnosis can be made rapidly by detecting *Legionella* DNA in sputum using a polymerase chain reaction (PCR)-based test or by identification of *Legionella* antigens in the urine; culture remains the diagnostic gold standard but takes 3 to 5 days.

Mycoplasma pneumoniae

Mycoplasma infections are particularly common among children and young adults. They occur sporadically or as local epidemics in closed communities (schools, military camps, and prisons).

MORPHOLOGY

Bacterial pneumonia has two patterns of anatomic distribution: lobular bronchopneumonia and lobar pneumonia (Figs. 15.28 and 15.29). Patchy consolidation of the lung is the dominant characteristic of **bronchopneumonia**, while consolidation of a large portion of a lobe or of an entire lobe defines **lobar pneumonia**. These anatomic categorizations may be difficult to apply in individual cases because patterns overlap. The patchy involvement may become confluent, producing lobar consolidation. Moreover, the same organisms may produce either pattern depending on patient susceptibility. **Most important from the clinical standpoint are identification of the causative agent and determination of the extent of disease.**

In lobar pneumonia, four stages of the inflammatory response have classically been described: congestion, red hepatization, gray hepatization, and resolution. In the first stage of **congestion**, the lung is heavy, boggy, and red. It is characterized by vascular engorgement, intraalveolar edema fluid containing a few neutrophils, and the presence of bacteria, which may be numerous. In the next stage of **red hepatization**, there is massive confluent exudation, as neutrophils, red cells, and fibrin fill the alveolar spaces (Fig. 15.30A). On gross examination, the lobe is red, firm, and airless, with a liver-like consistency, hence the name hepatization. The third stage of **gray hepatization** is marked by progressive disintegration of red cells and the persistence of a fibrinosuppurative exudate (Fig. 15.30B), resulting in a color change to grayish-brown. In the final stage of **resolution**, the exudate within the alveolar spaces is broken down by enzymatic digestion to produce granular, semifluid debris that is resorbed, ingested by macrophages, expectorated, or organized by fibroblasts growing into it (Fig. 15.30C). Pleural fibrinous reaction to the underlying inflammation, often present in the early stages if the consolidation extends to the lung surface (**pleuritis**), may similarly resolve. More often it undergoes organization, leaving fibrous thickening or permanent adhesions.

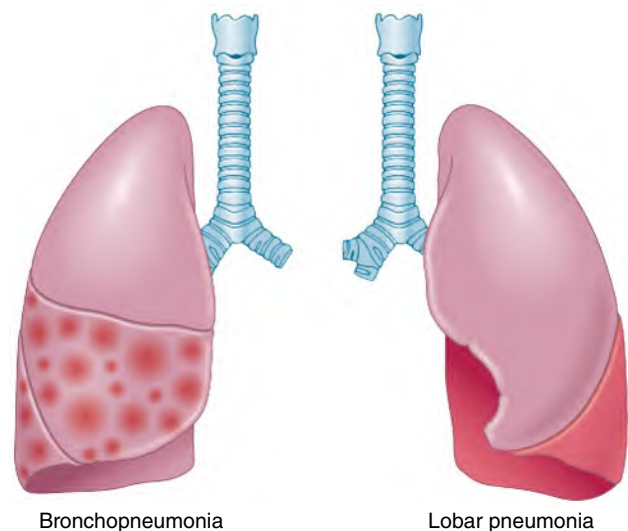


Fig. 15.28 Comparison of bronchopneumonia and lobar pneumonia.

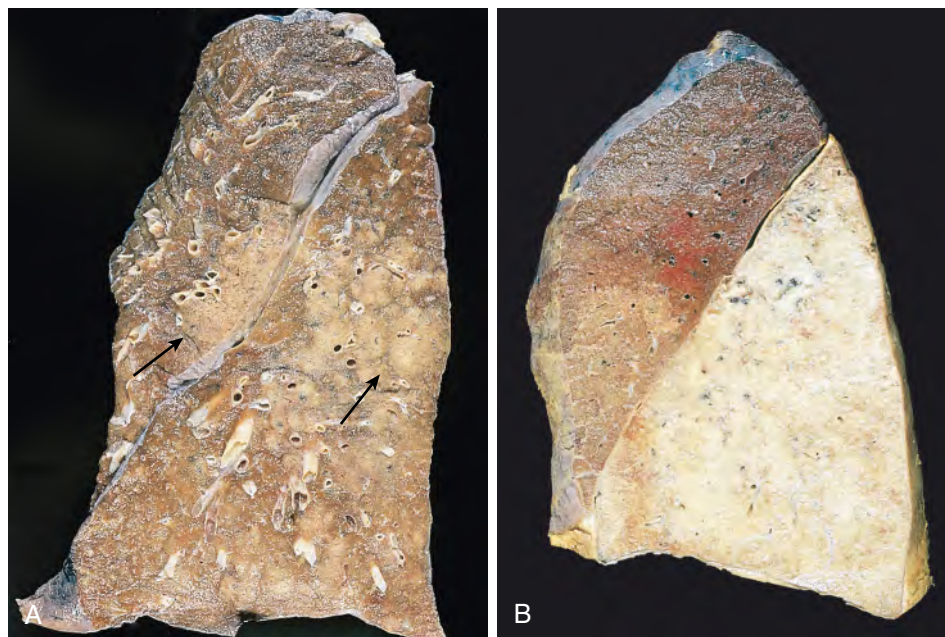


Fig. 15.29 Pneumonia. (A) Bronchopneumonia. Section of lung showing patches of consolidation (arrows). (B) Lobar pneumonia—gray hepatization. The lower lobe is uniformly consolidated.

Foci of **bronchopneumonia** are consolidated areas of acute suppurative inflammation. The consolidation may be confined to one lobe but is more often multilobar and frequently bilateral and basal because of the tendency of secretions to gravitate to the lower lobes. Well-developed lesions are slightly elevated, dry, granular, gray-red to yellow, and poorly delimited at their margins (see Fig. 15.29A). Histologically, the reaction usually elicits a neutrophil-rich exudate that fills the bronchi, bronchioles, and adjacent alveolar spaces (see Fig. 15.30A).

Complications of pneumonia include (1) tissue destruction and necrosis, causing **abscess formation** (particularly common with pneumococcal or *Klebsiella* infections) (Fig. 15.31); (2) spread of infection to the pleural cavity, causing an intrapleural fibrinosuppurative reaction known as **empyema**; and (3) **bacteremic dissemination** to the heart valves, pericardium, brain, kidneys, spleen, or joints, causing abscesses, endocarditis, meningitis, or suppurative arthritis.

Clinical Features

The major symptoms of community-acquired acute bacterial pneumonia are abrupt onset of high fever, shaking chills, and cough producing mucopurulent sputum and occasionally hemoptysis. When pleuritis is present it is accompanied by pleuritic pain and pleural friction rub. The whole lobe is radiopaque in lobar pneumonia, whereas there are focal opacities in bronchopneumonia.

The clinical picture is markedly modified by the administration of effective antibiotics. Appropriately treated patients may become afebrile with few clinical signs 48 to 72 hours after the initiation of antibiotics. The identification of the organism and the determination of its antibiotic sensitivity are the keystones of therapy. Fewer than 10% of patients with pneumonia severe enough to merit hospitalization now succumb, and in most instances death results from a complication, such as empyema, meningitis, endocarditis, or pericarditis, or is attributable to some predisposing influence, such as debility or chronic alcoholism.

Community-Acquired Viral Pneumonia

Common viral infections include influenza virus types A and B, respiratory syncytial viruses, human metapneumovirus, adenovirus, rhinoviruses, rubeola, and varicella viruses. Any of these agents can cause a relatively mild upper respiratory tract infection, recognized as the common cold, or a more severe lower respiratory tract infection. Factors that favor extension of the infection to the lung include extremes of age, malnutrition, alcoholism, and underlying debilitating illnesses.

Although the molecular details vary, all of the viruses that cause pneumonia produce disease through similar general mechanisms. These viruses have mechanisms that allow them to attach to and enter respiratory epithelial cells. Viral replication and gene expression lead to cytopathic changes, inducing cell death and secondary inflammation. The resulting damage and impairment of local pulmonary defenses, such as mucociliary clearance, may predispose to bacterial superinfections, which are often more serious than the viral infection itself.

Influenza

Influenza viruses of type A infect humans, pigs, horses, and birds and are the major cause of pandemic and epidemic influenza infections.

The influenza genome encodes several proteins, but the most important from the vantage point of viral virulence are the *hemagglutinin* and *neuraminidase* proteins. Hemagglutinin has three major subtypes (H1, H2, H3), while neuraminidase has two (N1, N2). Both proteins are embedded in a lipid bilayer, which constitutes the influenza virus envelope. Hemagglutinin is particularly important, as it serves to attach the virus to its cellular target via sialic acid residues on surface polysaccharides. Following uptake of the virus into endosomal vesicles, the acidic environment of the endosome triggers a conformation change in hemagglutinin that allows the viral envelope to fuse with the host cell membrane, releasing the viral genomic RNAs into the cytoplasm of the cell. Neuraminidase in turn facilitates the release of newly formed virions that are budding from infected cells by cleaving sialic acid residues. Neutralizing host antibodies against viral hemagglutinin and neuraminidase prevent and ameliorate, respectively, infection with influenza virus.

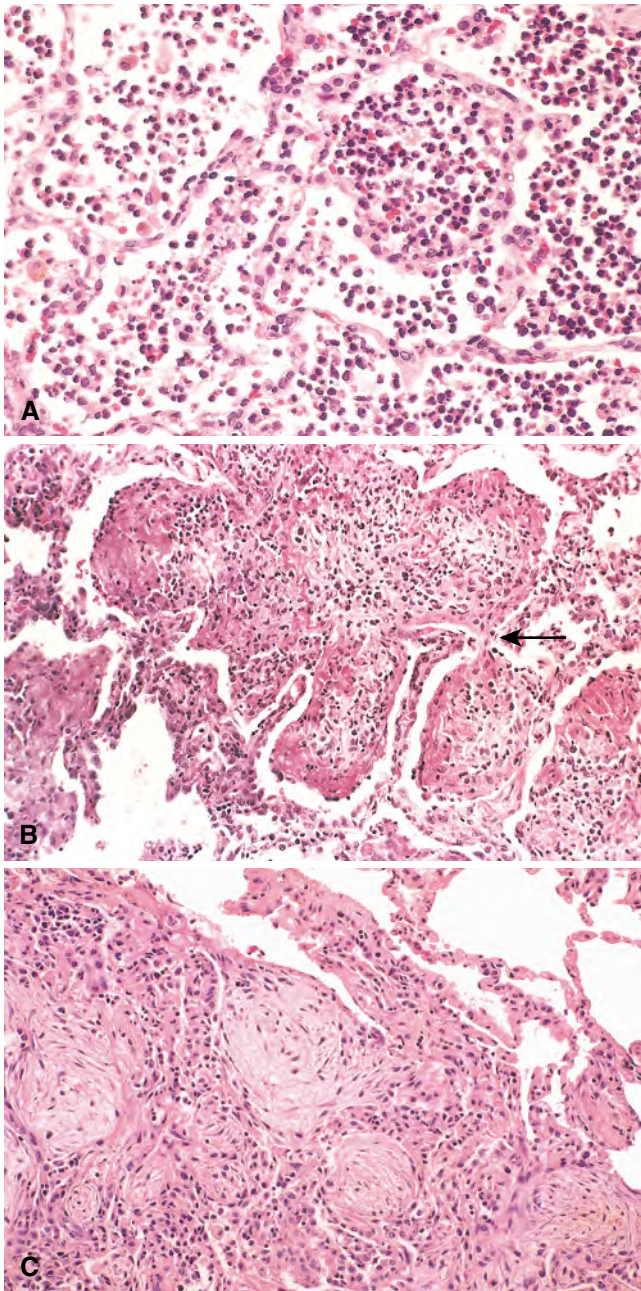


Fig. 15.30 Stages of bacterial pneumonia. (A) Acute pneumonia. The congested septal capillaries and numerous intraalveolar neutrophils are characteristic of early red hepatization. Fibrin nets have not yet formed. (B) Early organization of intraalveolar exudate, seen focally to be streaming through the pores of Kohn (arrow). (C) Advanced organizing pneumonia. The exudates have been converted to fibromyxoid masses rich in macrophages and fibroblasts.

The viral genome is composed of eight single-stranded RNAs, each encoding one or more proteins. The RNAs are packaged into helices by nucleoproteins that determine the influenza virus type (A, B, or C). A single subtype of influenza virus A predominates throughout the world at a given time. Epidemics of influenza are caused by spontaneous mutations that alter antigenic epitopes on the viral hemagglutinin and neuraminidase proteins. These antigenic changes (*antigenic drift*) result in new viral strains that are sufficiently different to elude, at least in part, antiinfluenza antibodies produced in members of the

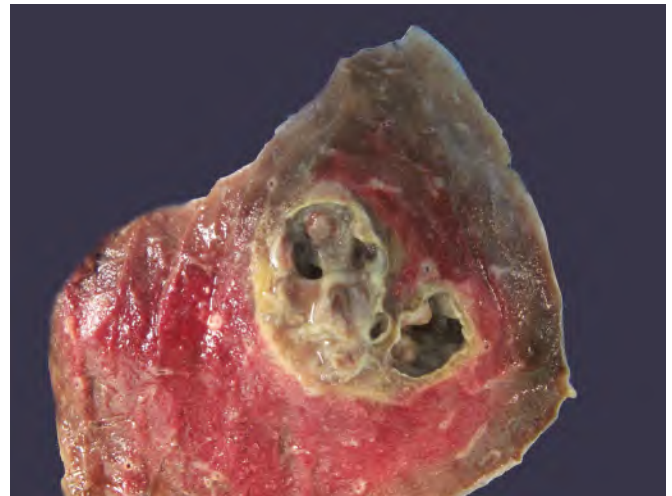


Fig. 15.31 Cut surface of lung showing two abscesses. (Courtesy Dr. M. Kamran Mirza, University of Chicago.)

population in response to prior exposures to other flu strains. Usually, however, these new strains bear sufficient resemblance to prior strains that some members of the population are at least partially resistant to infection. By contrast, pandemics, which are longer and more widespread than epidemics, occur when both the hemagglutinin and the neuraminidase genes are replaced through recombination with animal influenza viruses (*antigenic shift*). In this instance, essentially all individuals are susceptible to the new influenza virus.

If the host lacks protective antibodies, the virus infects pneumocytes and elicits several cytopathic changes. Shortly after entry into pneumocytes, the viral infection inhibits sodium channels, producing electrolyte and water shifts that lead to fluid accumulation in the alveolar lumen. This is followed by death of the infected cells through several mechanisms, including inhibition of host cell messenger RNA translation and activation of caspases leading to apoptosis. The death of epithelial cells exacerbates the fluid accumulation and releases “danger signals” that activate resident macrophages. In addition, prior to their death, infected epithelial cells release a variety of inflammatory mediators, including several chemokines and cytokines, adding fuel to the inflammatory fire. Mediators are also released from epithelial cells and macrophages that activate the nearby pulmonary endothelium and serve as chemoattractants for neutrophils, which migrate into the interstitium within the first day or two of infection. In some cases, viral infection may cause sufficient lung injury to produce ARDS, but more often severe pulmonary disease stems from a superimposed bacterial pneumonia. Of these, secondary pneumonias caused by *S. aureus* are particularly common and often life-threatening.

Control of the infection relies on several host mechanisms. The presence of viral products induces innate immune responses in infected cells, such as the production of type I interferon. These mediators upregulate the expression of the *MX1* gene, which encodes a guanosine triphosphatase that interferes with viral gene transcription and viral replication. As with other viral infections, natural killer cells and cytotoxic T cells can recognize and kill infected host cells, limiting viral replication and viral spread to adjacent pneumocytes. The cellular immune response is eventually augmented by development of antibody responses to the viral hemagglutinin and neuraminidase proteins.

Insight into future pandemics has come from studying past pandemics. DNA analysis of viral genomes retrieved from the lungs of a soldier who died in the great 1918 influenza pandemic that killed

between 20 million and 40 million people worldwide identified swine influenza sequences, consistent with this virus having its origin in an “antigenic shift.” The first flu pandemic of this century, in 2009, was also caused by an antigenic shift involving a virus of swine origin. It caused particularly severe infections in young adults, apparently because older adults had antibodies against past influenza strains that conveyed at least partial protection. Comorbidities such as diabetes, heart disease, lung disease, and immunosuppression were also associated with a higher risk of severe infection.

What then might be the source of the next great pandemic? There is no certainty, but one concern is centered on avian influenza, which normally infects birds. One such strain, type H5N1, has spread throughout the world in wild and domestic birds. Fortunately, the transmission of the current H5N1 avian virus is inefficient. However, if H5N1 influenza recombines with an influenza that is highly infectious for humans, a strain might result that is capable of sustained human-to-human transmission (and thus of causing the next great pandemic).

Human Metapneumovirus

Human metapneumovirus, a paramyxovirus discovered in 2001, is found worldwide and is associated with upper and lower respiratory tract infections. Infections can occur in any age group but are most common in young children, elderly adults, and immunocompromised patients. Some infections, such as bronchiolitis and pneumonia, are severe; overall, metapneumovirus is responsible for 5% to 10% of hospitalizations and 12% to 20% of outpatient visits of children with acute respiratory tract infections. Such infections are clinically indistinguishable from those caused by human respiratory syncytial virus and are often mistaken for influenza. The first human metapneumovirus infection occurs during early childhood, but reinfections are common throughout life, especially in older subjects. Diagnostic methods include PCR tests for viral RNA. Treatment generally focuses on supportive measures. Although work is ongoing, a clinically effective and safe vaccine has yet to be developed.

Human Coronaviruses

Coronaviruses are enveloped, positive-sense RNA viruses that infect humans and several other vertebrate species. Weakly pathogenic coronaviruses cause mild cold-like upper respiratory tract infections, while highly pathogenic ones may cause severe, often fatal pneumonia. An example of a highly pathogenic type is SARS-CoV-2, the coronavirus strain that emerged in China in late 2019 and caused the COVID-19 pandemic spanning from 2019 to the present (discussed in detail in Chapter 8). SARS-CoV-2 binds the ACE2 protein on the surface of respiratory epithelial cells, explaining the tropism of the respiratory viruses for the respiratory tract. With highly pathogenic forms in susceptible hosts, typically older individuals with comorbid conditions, the host immune response and locally released cytokines often produce acute lung injury and ARDS.

MORPHOLOGY

All viral infections produce similar morphologic changes. Upper respiratory infections are marked by mucosal hyperemia and swelling, infiltration of the submucosa by mononuclear cells (mainly lymphocytes and monocytes), and overproduction of mucus secretions. The swollen mucosa and viscous exudate may plug the nasal channels, sinuses, or the Eustachian tubes, leading to suppurative secondary bacterial infection. Virus-induced tonsillitis causing hyperplasia of the lymphoid tissue within the Waldeyer ring is frequent in children.

In viral **laryngotracheobronchitis** and **bronchiolitis** there is vocal cord swelling and abundant mucus production. Impairment of bronchociliary function

invites bacterial superinfection with more marked suppuration. Plugging of small airways may give rise to focal lung atelectasis. With more severe bronchiolar involvement, widespread plugging of secondary and terminal airways by cell debris, fibrin, and inflammatory exudate may, if prolonged, lead to organization and fibrosis, resulting in obliterative bronchiolitis and permanent lung damage.

Lung involvement may be quite patchy or may involve whole lobes bilaterally or unilaterally. The affected areas are red-blue and congested. Pleuritis or pleural effusions are infrequent. The histologic pattern depends on the severity of the disease. **Predominant is an interstitial inflammatory reaction involving the walls of the alveoli.** The alveolar septa are widened and edematous and usually contain a mononuclear inflammatory infiltrate of lymphocytes, macrophages, and occasionally plasma cells. In severe cases, neutrophils may also be present. The alveoli may be free of exudate, but in many patients, there is intraalveolar proteinaceous material and a cellular exudate. When complicated by ARDS, pink hyaline membranes line the alveolar walls (see Fig. 15.4B). Eradication of the infection is followed by reconstitution of normal lung architecture.

Superimposed bacterial infection modifies this picture by causing ulcerative bronchitis, bronchiolitis, and bacterial pneumonia. Some viruses, such as herpes simplex, varicella, and adenovirus, may be associated with necrosis of bronchial and alveolar epithelium and acute inflammation. Characteristic viral cytopathic changes are described in Chapter 8.

Clinical Features

The clinical course of viral pneumonia is extremely varied. Many cases masquerade as severe upper respiratory tract infections or as chest colds. Even individuals with well-developed atypical pneumonia have few localizing symptoms. Cough may be absent, and the major manifestations may consist only of fever, headache, and myalgia. The edema and exudation often cause ventilation-perfusion mismatch leading to hypoxemia and thus evoke symptoms out of proportion to the scant physical findings.

Viral pneumonias are usually mild and resolve spontaneously without any lasting sequelae. However, interstitial viral pneumonias may assume epidemic proportions, and in such instances even a low rate of complications can lead to significant morbidity and mortality, as is typically true of influenza epidemics.

Healthcare-Associated Pneumonia

Healthcare-associated pneumonia was recently described as a distinct clinical entity associated with several risk factors. These are hospitalization of at least 2 days within the last 90 days; presentation from a nursing home or long-term care facility; attending a hospital or hemodialysis clinic; and recent intravenous antibiotic therapy, chemotherapy, or wound care. The most common organisms isolated are methicillin-resistant *S. aureus* and *P. aeruginosa*. As there is a higher risk of infection with multidrug-resistant pathogens due to frequent healthcare contacts, healthcare-associated pneumonia requires broad spectrum empirical antimicrobial treatment. These patients have a higher mortality than those with community-acquired pneumonia.

Hospital-Acquired Pneumonia

Hospital-acquired pneumonias (or nosocomial pneumonias) are defined as pulmonary infections acquired in the course of a hospital stay. They are common in patients with severe underlying disease, immunosuppression, prolonged antibiotic therapy, or invasive access devices such as intravascular catheters. Patients on mechanical ventilation are at particularly high risk. Superimposed on an underlying disease (that caused hospitalization), hospital-acquired infections are serious and often life-threatening. Gram-positive cocci (mainly *S. aureus*) and gram-negative rods (Enterobacteriaceae and *Pseudomonas* species) are

the most common isolates. The same organisms predominate in ventilator-associated pneumonia, with gram-negative bacilli being somewhat more common in this setting.

KEY CONCEPTS

Acute Pneumonia

- *S. pneumoniae* (the pneumococcus) is the most common cause of community-acquired acute pneumonia; the distribution of inflammation is usually lobar.
- Lobar pneumonias evolve through four stages: congestion, red hepatization, gray hepatization, and resolution.
- Other common causes of acute bacterial pneumonias in the community include *H. influenzae* and *M. catarrhalis* (both associated with acute exacerbations of COPD), *S. aureus* (usually secondary to viral respiratory infections), *K. pneumoniae* (observed in patients who are chronic alcoholics), *P. aeruginosa* (seen in persons with cystic fibrosis and in those with neutropenia), and *L. pneumophila*, seen particularly in individuals with comorbid conditions (e.g., heart or lung disease) and in organ transplant recipients.
- Important causes of community-acquired viral pneumonia include influenza virus, metapneumovirus, and the coronavirus COVID-19, the latter a newly emergent pathogen.
- Bacterial pneumonias are characterized by predominantly intraalveolar neutrophilic inflammation, while viral pneumonia shows interstitial lymphocytic inflammation.

Aspiration Pneumonia

Aspiration pneumonia occurs in markedly debilitated patients or those who aspirate gastric contents either while unconscious (e.g., after a stroke) or during repeated vomiting. These patients have abnormal gag and swallowing reflexes that predispose to aspiration. The resultant pneumonia is partly chemical due to the irritating effects of gastric acid and partly bacterial (from the oral flora). Typically, more than one organism is recovered on culture, aerobes being more common than anaerobes. This type of pneumonia is often necrotizing, pursues a fulminant clinical course, and is a frequent cause of death. In patients who survive, lung abscess is a common complication.

Microaspiration, in contrast, occurs frequently in almost all people, especially those with gastroesophageal reflux disease. It usually results in small, poorly formed nonnecrotizing granulomas with multinucleated foreign body giant cell reaction. It is usually inconsequential, but may exacerbate other preexisting lung diseases such as asthma, interstitial fibrosis, and lung rejection.

Lung Abscess

The term *pulmonary abscess* describes a local suppurative process that produces necrosis of lung tissue. Oropharyngeal surgical or dental procedures, sinobronchial infections, and bronchiectasis play important roles in their development.

Etiology and Pathogenesis

Under appropriate circumstances any bacterial pathogen can produce an abscess; those that do so most commonly include aerobic and anaerobic streptococci, *S. aureus*, and a host of gram-negative organisms. Mixed infections often occur because of the important causal role played by inhalation of foreign material. Anaerobic organisms normally found in the oral cavity, including members of the *Bacteroides*, *Fusobacterium*, and *Peptococcus* genera, are the exclusive isolates in about 60% of cases. The causative organisms are introduced by the following mechanisms:

- **Aspiration of infective material** (the most frequent cause). Risk factors include suppressed cough reflexes (e.g., acute alcohol intoxication,

opioid use disorder, coma, anesthesia, seizure disorders), severe dysphagia (e.g., neurologic deficits, esophageal disease), protracted vomiting, and poor dental hygiene. Aspiration first causes pneumonia, which progresses to tissue necrosis and formation of lung abscess.

- **Antecedent primary lung infection.** Postpneumonic abscess formations are usually associated with *S. aureus*, *K. pneumoniae*, and pneumococcus. Posttransplant or otherwise immunosuppressed individuals are at special risk.
- **Septic embolism.** Infected emboli may arise from thrombophlebitis in any portion of the systemic venous circulation or from the vegetations of infective bacterial endocarditis on the right side of the heart and lodge in the lung.
- **Neoplasia.** Secondary infection is particularly common in bronchopulmonary segments obstructed by a primary or secondary malignancy (*postobstructive pneumonia*).
- **Miscellaneous.** Traumatic penetrations of the lungs; direct extension of suppurative infections from the esophagus, spine, subphrenic space, or pleural cavity; and hematogenous seeding of the lung by pyogenic organisms all may lead to lung abscess formation.

When all these causes are excluded, there are still cases in which no discernible basis for the abscess formation can be identified. These are referred to as *primary cryptogenic lung abscesses*.

MORPHOLOGY

Abscesses vary in diameter from a few millimeters to large cavities of 5 to 6 cm (see Fig. 15.31). They may affect any part of the lung and may be single or multiple. Pulmonary abscesses due to aspiration are more common on the right (because of the more vertical right main bronchus) and are most often single. Abscesses that develop due to pneumonia or bronchiectasis are usually multiple, basal, and diffusely scattered. Septic emboli and pyemic abscesses are multiple and may affect any region of the lungs.

The cardinal histologic change in all abscesses is suppurative destruction of lung parenchyma within a central area of cavitation.

The abscess cavity may be filled with suppurative debris or, if there is communication with an air passage, may partially drain to create an air-containing cavity. Superimposed saprophytic infections are prone to develop within the necrotic debris. Continued infection leads to large, poorly demarcated, fetid, green-black, multilocular cavities designated gangrene of the lung. In chronic cases considerable fibroblastic proliferation produces a fibrous wall.

Clinical Features

The manifestations of pulmonary abscesses are much like those of bronchiectasis and characteristically include cough, fever, and copious amounts of foul-smelling purulent or sanguineous sputum. Fever, chest pain, and weight loss are common. Clubbing of the fingers and toes may appear. The diagnosis can be only suspected from the clinical findings and must be confirmed radiologically. Whenever an abscess is discovered in older individuals, it is important to rule out an underlying carcinoma, which is present in 10% to 15% of cases.

The course of abscesses is variable. With antimicrobial therapy, most resolve, leaving behind a scar. Complications include extension of the infection into the pleural cavity, hemorrhage, the development of *brain abscesses* or *meningitis* from septic emboli, and (rarely) secondary amyloidosis (type AA).

Chronic Pneumonia

Chronic pneumonia is most often a localized lesion in the immunocompetent patient, with or without regional lymph node involvement. Typically, the inflammatory reaction is granulomatous and is caused by

bacteria (e.g., *Mycobacterium tuberculosis*) or fungi (e.g., *Histoplasma capsulatum*, *Blastomycosis dermatitidis*, or *Coccidioides immitis*). Tuberculosis and chronic pneumonias caused by fungi are discussed in Chapter 8.

Pneumonia in the Immunocompromised Host

The appearance of a pulmonary infiltrate, with or without signs of infection (e.g., fever), is one of the most common and serious complications in patients whose immune defenses are suppressed by disease, immunosuppressive therapy for organ or hematopoietic stem cell transplants, chemotherapy for tumors, or irradiation. In addition to the usual pathogens, a wide variety of so-called opportunistic infectious agents, many of which rarely cause infection in normal hosts, can cause pneumonia, and often more than one agent is involved. Mortality from these opportunistic infections is high. Table 15.8 lists some of the opportunistic agents according to their prevalence and whether they cause local or diffuse pulmonary infiltrates. The differential diagnosis of such infiltrates includes drug reactions and involvement of the lung by tumor. The specific infections are discussed in Chapter 8. Of these, the ones that commonly involve the lung can be classified according to the etiologic agent: (1) bacteria (*P. aeruginosa*, *Mycobacterium* species, *L. pneumophila*, and *Listeria monocytogenes*), (2) viruses (cytomegalovirus [CMV] and herpesvirus), and (3) fungi (*P. jirovecii*, *Candida* species, *Aspergillus* species, the Phycomycetes, and *Cryptococcus neoformans*).

Pulmonary Disease in Human Immunodeficiency Virus Infection

Pulmonary disease accounts for 30% to 40% of hospitalizations in HIV-infected individuals. Although the use of potent antiretroviral agents and effective chemoprophylaxis has markedly altered the incidence and outcome of pulmonary disease in HIV-infected persons, the plethora of infectious agents and other pulmonary lesions make diagnosis and treatment a distinct challenge. Some of the individual microbial agents afflicting HIV-infected individuals have already been discussed; this section focuses only on the general principles of HIV-associated pulmonary disease.

- *Despite the emphasis on opportunistic infections, it must be remembered that bacterial lower respiratory tract infections caused by the “usual” pathogens are among the most serious pulmonary disorders in HIV infection.* The implicated organisms include *S. pneumoniae*, *S. aureus*, *H. influenzae*, and gram-negative rods. Bacterial pneumonias in HIV-infected persons are more common, more severe,

and more often associated with bacteremia than in those without HIV infection.

- *Not all pulmonary infiltrates in HIV-infected individuals are infectious in etiology.* A host of noninfectious diseases, including Kaposi sarcoma (Chapters 6 and 11), non-Hodgkin lymphoma (Chapter 13), and lung cancer, occur with increased frequency and must be excluded.
- *The CD4+ T-cell count determines the risk of infection with specific organisms.* Bacterial and tubercular infections are more likely at higher CD4+ counts (>200 cells per mm³). *Pneumocystis* pneumonia usually strikes at CD4+ counts less than 200 cells per mm³, while CMV, fungal, and *Mycobacterium avium-intracellulare* complex infections are uncommon until the disease is very advanced (CD4+ counts less than 50 cells per mm³).

Finally, pulmonary disease in HIV-infected persons may result from more than one cause, and even common pathogens may present with atypical manifestations. Therefore, the diagnostic work-up of these patients may be more extensive (and expensive) than would be necessary in an immunocompetent individual.

LUNG TRANSPLANTATION

Indications for transplantation may include almost all nonneoplastic terminal lung diseases, provided that the patient does not have any other serious disease that would preclude lifelong immunosuppressive therapy. The most common indications are end-stage COPD, idiopathic pulmonary fibrosis, cystic fibrosis, and idiopathic/familial pulmonary arterial hypertension. Although bilateral lung transplantation is associated with better outcomes, single lung transplantation has the advantage of benefitting two recipients using organs harvested from a single donor. When bilateral chronic infection is present (e.g., cystic fibrosis, bronchiectasis), both lungs of the recipient must be replaced to remove the reservoir of infection.

With improving surgical and organ preservation techniques, postoperative complications (e.g., anastomotic dehiscence, vascular thrombosis, primary graft dysfunction) are becoming rare. The transplanted lung is subject to two major complications: infection and rejection.

- *Pulmonary infections* in lung transplant patients are essentially those of any immunocompromised host, discussed earlier. In the early posttransplant period (the first few weeks), bacterial infections are most common. With ganciclovir prophylaxis and matching of donor-recipient CMV status, CMV pneumonia occurs less frequently and is less severe, although some resistant strains are emerging. Most infections occur in the third to twelfth month after transplantation. *P. jirovecii* pneumonia is rare, since almost all patients receive adequate prophylaxis, usually with trimethoprim-sulfamethoxazole (Bactrim). Fungal infections are mostly due to *Aspergillus* and *Candida* species, and they may involve the bronchial anastomotic site and/or the lung.
- *Acute lung allograft rejection* occurs to some degree in all patients despite routine immunosuppression. It most often appears several weeks to months after surgery but also may present years later or whenever immunosuppression is decreased. Patients present with fever, dyspnea, cough, and radiologic infiltrates. Since these are similar to the picture of infections, diagnosis often relies on transbronchial biopsy.
- *Chronic lung allograft rejection* is a significant problem in at least half of all patients by 3 to 5 years posttransplant. It is manifested by cough, dyspnea, and an irreversible decrease in lung function due to pulmonary fibrosis.

TABLE 15.8 Causes of Pulmonary Infiltrates in Immunocompromised Hosts

Diffuse Infiltrates	Focal Infiltrates
Common	
Cytomegalovirus	Gram-negative bacterial infections
Pneumocystis jirovecii	Staphylococcus aureus
Drug reaction	Aspergillus
	Candida
	Malignancy
Uncommon	
Bacterial pneumonia	Cryptococcus
Aspergillus	Mucor
Cryptococcus	Pneumocystis jirovecii
Malignancy	Legionella pneumophila

MORPHOLOGY

The morphologic features of acute rejection are primarily those of inflammatory infiltrates (lymphocytes, plasma cells, and few neutrophils and eosinophils) around small vessels, in the submucosa of airways, or both. The major morphologic correlate of chronic rejection is **bronchiolitis obliterans**, the partial or complete occlusion of small airways by fibrosis, with or without active inflammation (Fig. 15.32). Bronchiolitis obliterans is patchy and therefore difficult to diagnose via transbronchial biopsy. Bronchiectasis and pulmonary fibrosis may also develop with long-standing chronic rejection.

Clinical Features

Acute cellular airway rejection (the presumed forerunner of later, fibrous obliteration of these airways) is generally responsive to therapy, but the treatment of established bronchiolitis obliterans has been disappointing. Its progress may be slowed or even halted for some time, but it cannot be reversed. Infrequent complications of lung transplantation include Epstein-Barr virus (EBV)-associated B-cell lymphoma, which most often arises within the lung allograft. With continuing improvement in surgical, immunosuppressive, and antimicrobial therapies, the outcome of lung transplantation has improved considerably. The overall median survival is 6 years, with younger patients and those undergoing bilateral lung transplantation having better outcomes.

TUMORS

Of the wide variety of benign and malignant tumors that may arise in the lung, 90% to 95% are carcinomas, about 5% are carcinoid tumors, and 2% to 5% are mesenchymal and other miscellaneous neoplasms.

Carcinomas

Lung cancer is currently the most frequently diagnosed major cancer and the most common cause of cancer mortality worldwide. Globally, in 2022 there were an estimated 2.5 million new cases and 1.8 million lung cancer deaths. The number of new cases of lung cancer in 2023 in the United States is expected to number approximately 238,000 (note that in 1950 it was 18,000), accounting for about 12% of cancer diagnoses and taking more than 125,000 lives, which amounts to roughly 21% of all cancer-related deaths. Each year, lung cancer kills more people than colon, breast, and prostate cancer combined. It is generally a disease of older adults, occurring most often between ages

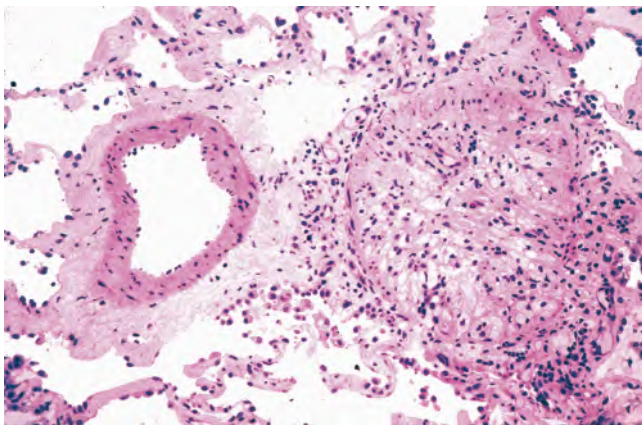


Fig. 15.32 Chronic rejection of lung allograft associated with bronchiolitis obliterans. An adjacent pulmonary artery is normal. (Courtesy Dr. Thomas Krausz, Department of Pathology, The University of Chicago, Pritzker School of Medicine, Chicago.)

55 and 84 years, with a peak incidence between 65 and 74 years. Only 2% of all cases occur before the age of 40.

Because lung cancer is strongly linked to cigarette smoking, changes in smoking habits greatly influence lung cancer incidence and mortality as well as the prevalence of the various histologic types of lung cancer. Since the early 1990s, lung cancer incidence and mortality rates have been decreasing in males due to a decrease in male smoking over the past 35 years. However, the decrease in smoking among females has lagged behind that of males, and since 1987 more females have died each year of lung cancer than of breast cancer.

Etiology and Pathogenesis

Most (but not all) lung cancers are associated with a well-known carcinogen—cigarette smoke. In addition, there are other genetic and environmental factors. Lung cancers are broadly classified into small cell and non-small cell types, with the latter group including adenocarcinoma and squamous cell carcinoma. The driver mutations that cause lung cancer vary among these histologic subtypes, as will be described later.

Tobacco Smoking. About 80% of lung cancers occur in active smokers or those who stopped recently, and there is a nearly linear correlation between the frequency of lung cancer and pack-years of cigarette smoking. The increased risk is 60 times greater in habitual heavy smokers (two packs a day for 20 years) than in nonsmokers. However, since lung cancer develops in only 10% to 15% of smokers, there are likely to be other factors that interact with smoking to predispose individuals to this deadly disease. For unclear reasons, it appears that females are more susceptible to carcinogens in tobacco than males. Although cessation of smoking decreases the risk for lung cancer over time, it may never return to baseline levels. In fact, genetic changes that predate lung cancer can persist for many years in the bronchial epithelium of former smokers. Pipe and cigar smokers also incur an elevated risk, albeit only modestly. Chewing tobacco is not a safe substitute for smoking cigarettes or cigars, as these products spare the lung but cause oral cancers and can lead to nicotine addiction. The long-term effects of electronic cigarette aerosols are not known, as “vaping” is a relatively recent phenomenon (Chapter 9).

Unfortunately, the carcinogenic effects of tobacco smoke extend to those who live and work with smokers. *Secondhand smoke*, or environmental tobacco smoke, contains numerous human carcinogens for which there is no safe level of exposure. It is estimated that each year about 3000 nonsmoking adults die of lung cancer due to breathing secondhand smoke.

What of heavy smokers who never develop cancer? While some of this may be a matter of chance, the mutagenic effect of carcinogens in smoke is modified by genetic variants. Recall that many chemicals (procarcinogens) are converted into carcinogens via activation by the highly polymorphic P-450 monooxygenase enzyme system (Chapter 9). Specific P-450 variants have an increased capacity to activate procarcinogens in cigarette smoke, and smokers with these variants incur a greater risk of lung cancer. Similarly, individuals whose peripheral blood lymphocytes show more numerous chromosomal breakages after exposure to tobacco-related carcinogens (mutagen sensitivity genotype) have a greater than 10-fold higher risk of developing lung cancer as compared with controls, presumably because of genetic variation in genes involved in DNA repair.

The histologic changes that correlate with steps along the path to neoplastic transformation are best documented for squamous cell carcinoma and are described in more detail later. There is a linear correlation between the intensity of exposure to cigarette smoke and the appearance of ever more worrisome epithelial changes. These begin with rather innocuous-appearing basal cell hyperplasia and squamous metaplasia

and progress to squamous dysplasia and carcinoma in situ, the last stage before progression to invasive cancer.

Industrial Hazards. Certain industrial exposures, such as asbestos, arsenic, chromium, uranium, nickel, vinyl chloride, and mustard gas, increase the risk of developing lung cancer. High-dose ionizing radiation is carcinogenic. There was an increased incidence of lung cancer among survivors of the Hiroshima and Nagasaki atomic bomb blasts, as well as in workers heavily involved in clean-up after the Chernobyl disaster. Uranium is weakly radioactive, but lung cancer rates among nonsmoking uranium miners are four times higher than those in the general population, and among smoking miners they are about 10 times higher. Asbestos exposure also increases the risk for lung cancer development. The latent period before the development of lung cancer is 10 to 30 years. Lung cancer is the most frequent malignancy in individuals exposed to asbestos, particularly when coupled with smoking. Asbestos workers who do not smoke have a five-fold greater risk of developing lung cancer than do nonsmoking control subjects, whereas those who smoke have a 55-fold greater risk.

Air Pollution. It is uncertain whether air pollution, by itself, increases the risk of lung cancer, but it likely adds to the risk in those who smoke or are exposed to secondhand smoke. It may do so through several different mechanisms. Chronic exposure to air particulates in smog may cause lung irritation, inflammation, and repair, and you will recall that chronic inflammation and repair increases the risk of a variety of cancers (Chapter 7). A specific form of air pollution that may contribute to an increased risk of lung cancer is radon gas. Radon is a ubiquitous radioactive gas that has been linked epidemiologically to increased lung cancer in uranium miners. Other underground miners and workers in locations below ground, such as subways, tunnels, and basements, are at increased risk for radon exposure. This has generated concern that low-level exposure (e.g., in well-insulated homes in areas with naturally high levels of radon in soil) may also increase the incidence of lung cancer.

Acquired Mutations. As with other cancers (Chapter 7), smoking-related carcinomas of the lung arise by a stepwise accumulation of oncogenic “driver” mutations that result in the neoplastic transformation of pulmonary epithelial cells. Some of the genetic changes associated with cancers can be found in the “benign” bronchial epithelium of smokers without lung cancers, suggesting that large areas of the respiratory mucosa are mutagenized by exposure to carcinogens in tobacco smoke (“field effect”). On this fertile soil, cells that accumulate just the “wrong” panoply of complementary driver mutations to acquire all the hallmarks of cancer develop into carcinomas.

Three of the major histologic subtypes of lung cancer each have distinctive molecular features:

- **Adenocarcinoma** is associated with tobacco smoking, but less so than other histologic subtypes; as a result, it is the most common subtype in never-smokers (described below). About one-third of adenocarcinomas have oncogenic gain-of-function mutations involving components of growth factor receptor signaling pathways. These are important to recognize because they often can be targeted with specific inhibitors (discussed later). These include gain-of-function mutations in genes encoding several different receptor tyrosine kinases, such as *EGFR* (in 10% to 15% of tumors in White people and a higher percentage of nonsmoking Asian females); *ALK* (in 3% to 5% of tumors); *ROS1* (in 1% of tumors); *MET* (in 2% to 5% of tumors); and *RET* (in 1% to 2% of tumors). Other tumors have gain-of-function mutations in serine/threonine kinases (*BRAF*, 2% of tumors; and *PI3K*, 2% of tumors) or in the *KRAS* gene (roughly 30% of tumors), all of which encode signaling

molecules that lie downstream of receptor tyrosine kinases in growth factor signaling pathways.

- **Squamous cell carcinoma** is highly associated with exposure to tobacco smoke and harbors diverse genetic aberrations, many of which are chromosome deletions involving tumor suppressor loci. These losses, especially those involving 3p, 9p (site of the *CDKN2A* gene), and 17p (site of the *TP53* gene), are early events in tumor evolution, being detected at an appreciable frequency in the histologically normal respiratory mucosal cells of smokers. Most tumors have mutations in *TP53*, and p53 protein overexpression (as seen by immunohistochemical staining), a marker of *TP53* mutations, is an early event, being reported in 10% to 50% of squamous dysplasias and 60% to 90% of squamous cell carcinoma in situ. The *CDKN2A* tumor suppressor gene, which encodes the cyclin-dependent kinase inhibitor p16, is mutated in 65% of tumors. Many squamous cell carcinomas also have amplification of *FGFR1*, a gene encoding the fibroblast growth factor receptor tyrosine kinase.
- **Small cell carcinoma** is virtually always smoking related and has the highest mutational burden among lung cancers. There is almost universal inactivation of both *TP53* and *RB*, and unusual transformations of non-small cell carcinoma to small cell carcinoma are often associated with acquisition of *RB* loss-of-function mutations, emphasizing the importance of *RB* inactivation in this lung cancer subtype. Loss of chromosome 3p also occurs in nearly all these tumors and is seen even in histologically normal lung epithelium, suggesting that this also is a critical early event. This subtype is also commonly associated with amplification of genes of the *MYC* family.

Lung Cancer in Never-Smokers. The WHO estimates that 25% of lung cancer worldwide occurs in never-smokers. This percentage is probably closer to 10% to 15% in Western countries. These cancers occur more commonly in females, and most are adenocarcinomas, often with targetable mutations. Cancers in nonsmokers are more likely to have *EGFR* mutations and almost never have *KRAS* mutations; *TP53* mutations are not uncommon but occur less frequently than in smoking-related cancers.

Precursor (Preinvasive) Lesions. Four types of morphologic precursor epithelial lesions are recognized: (1) atypical adenomatous hyperplasia (Fig. 15.33A), (2) adenocarcinoma in situ (Fig. 15.33B), (3) squamous metaplasia, dysplasia, and carcinoma in situ (Fig. 15.33C–E), and (4) diffuse idiopathic pulmonary neuroendocrine cell hyperplasia. It should be remembered that the term *precursor* does not imply that progression to cancer is inevitable. Currently it is not possible to distinguish between precursor lesions that progress and those that remain localized or regress.

Classification

Tumor classification is important for consistency in patient treatment and provides a uniform basis for epidemiologic and biologic studies. The most recent classification of lung cancer is given in Table 15.9. Several histologic variants of each type of lung cancer are described. The relative proportions of the major categories are:

- Adenocarcinoma (50%)
- Squamous cell carcinoma (20%)
- Small cell carcinoma (15%)
- Large cell carcinoma (2%)
- Other (13%)

There may be mixtures of histologic patterns, even in the same cancer. Thus, combinations of squamous cell carcinoma and adenocarcinoma or small cell and squamous cell carcinoma occur in about 14% and 5% of patients, respectively.

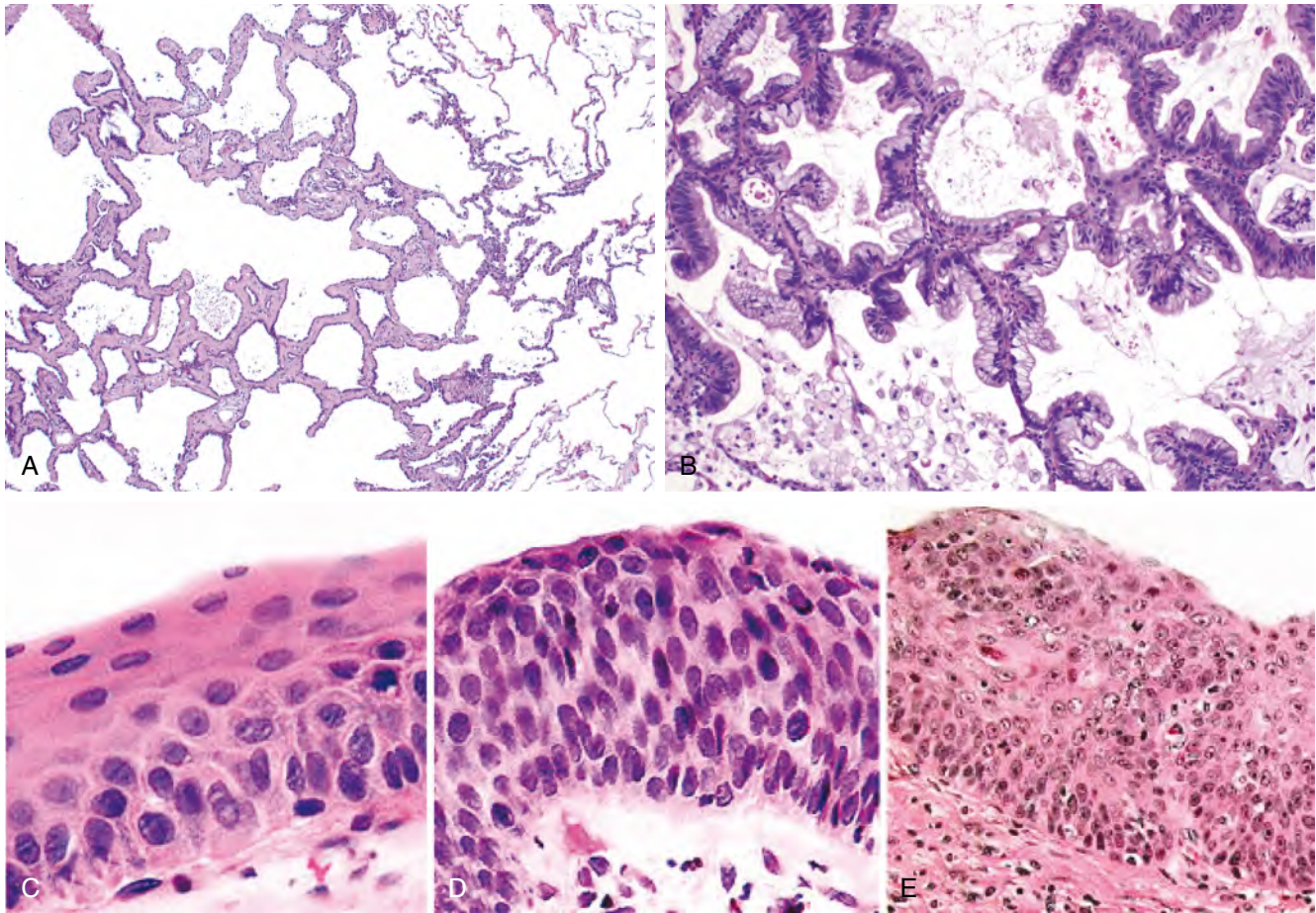


Fig. 15.33 Precursor lesions of lung cancer. (A) Atypical adenomatous hyperplasia. Note the septal thickening due to interstitial fibrosis. (B) Adenocarcinoma in situ, mucinous type. The epithelium is tall, columnar mucinous cells growing along existing alveolar septations. (C–E) Squamous cell precursors. Basal cell (or reserve cell) hyperplasia and squamous metaplasia (C). More ominous changes include the appearance of squamous dysplasia (D), characterized by the presence of disordered squamous epithelium, with loss of nuclear polarity, nuclear hyperchromasia, pleomorphism, and mitotic figures. Squamous dysplasia may progress through the stages of mild, moderate, and severe dysplasia. Carcinoma in situ (E), the stage immediately preceding invasive squamous carcinoma by definition has not penetrated the basement membrane and has cytologic features similar to those of carcinoma. (Courtesy of the late Dr. Adi Gazdar, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

The incidence of adenocarcinoma has increased significantly in the last 2 decades, and it is now the most common form of lung cancer in adult females and males. The basis for this change is unclear. One possible factor is the increase in adult female smokers, but this still calls into question why females develop adenocarcinoma more frequently than other forms of lung cancer. Another possibility is that changes in cigarettes (altered filter tips and decreased tar and nicotine) may have caused smokers to inhale more deeply, increasing the exposure of peripheral airways and cells with a predilection to give rise to adenocarcinoma in response to carcinogens.

MORPHOLOGY

Lung carcinomas may arise in the peripheral lung (more often adenocarcinomas) or in the central/hilar region (more often squamous cell carcinomas), sometimes in association with recognizable precursor lesions. In all types of lung cancer, the neoplastic tissue is gray-white and firm to hard. Especially when the tumors are bulky, focal areas of hemorrhage or necrosis may appear

to produce red or yellow-white mottling and softening. Sometimes these necrotic foci cavitate.

Atypical adenomatous hyperplasia is a small precursor lesion (≤ 5 mm) characterized by dysplastic pneumocytes lining alveolar walls that may be mildly fibrotic (see Fig. 15.33A). It can be single or multiple and can be close to or distant from an invasive tumor.

Adenocarcinoma in situ (formerly called bronchioloalveolar carcinoma) is a lesion that is less than 3 cm in size and is composed entirely of dysplastic cells growing along pre-existing alveolar septa in a so-called **lepidic pattern** (see later). The cells may or may not have intracellular mucin (see Fig. 15.33B).

Adenocarcinoma is an invasive malignant epithelial tumor with glandular differentiation or mucin production by the tumor cells. Adenocarcinomas grow in various patterns, including lepidic (noninvasive), acinar, papillary, micropapillary, and solid. Compared with squamous cell cancers, these lesions are usually more peripherally located and tend to be smaller (Fig. 15.34A). They vary histologically from well-differentiated tumors with obvious glandular elements (Fig. 15.34B), to papillary lesions resembling other papillary carcinomas, to solid masses with only occasional mucin-producing glands and cells. The

TABLE 15.9 Histologic Classification of Malignant Epithelial Lung Tumors

Tumor Classification
Adenocarcinoma
Lepidic, acinar, micropapillary, papillary, or solid (growth pattern)
Invasive mucinous adenocarcinoma
Minimally invasive adenocarcinoma (nonmucinous, mucinous)
Colloid adenocarcinoma
Enteric-type adenocarcinoma
Fetal adenocarcinoma
Squamous Cell Carcinoma
Keratinizing, Nonkeratinizing, Basaloid
Neuroendocrine Tumors
Small cell carcinoma
Combined small cell carcinoma
Large cell neuroendocrine carcinoma
Combined large cell neuroendocrine carcinoma
Carcinoid (neuroendocrine) tumor
Typical, atypical
Other Uncommon Types
Large cell carcinoma
Adenosquamous carcinoma
Sarcomatoid carcinoma
Pleomorphic carcinoma, carcinosarcoma, pulmonary blastoma
Others such as lymphoepithelioma-like carcinoma and NUT carcinoma
Salivary gland–type tumors
Adenoid cystic carcinoma, epithelial-myoepithelial carcinoma, mucoepidermoid carcinoma

majority express thyroid transcription factor-1 (TTF-1) (Fig. 15.34B inset), a protein first identified in the thyroid that is required for normal lung development. At the periphery of the tumor there is often a lepidic pattern of spread, in which the tumor cells “crawl” along normal-appearing alveolar septa. Smaller tumors (≤ 3 cm) with limited invasion (≤ 5 mm) associated with scarring and a lepidic-predominant growth pattern are called **microinvasive adenocarcinoma**. These have a far better prognosis than invasive carcinomas of the same size. **Mucinous adenocarcinomas** (see Fig. 15.33B) tend to spread aerogenously, forming satellite tumors; thus, these are less likely to be cured by surgery. They may present as a solitary nodule or as multiple nodules, or an entire lobe may be consolidated by tumor, mimicking lobar pneumonia.

Squamous cell carcinoma is more common in adult males and is strongly associated with smoking. Precursor lesions that give rise to invasive squamous cell carcinoma are well characterized. Squamous cell carcinomas are often antedated by **squamous metaplasia** or **dysplasia** in the bronchial epithelium, which then transforms to **carcinoma in situ**, a phase that may last for years (see Fig. 15.33E). By this time, atypical cells may be identified in cytologic smears of sputum or in bronchial lavage fluids or brushings (Fig. 15.35), but the lesion is asymptomatic and undetectable on radiographs. Eventually, an invasive squamous cell carcinoma appears. The tumor may then follow a variety of paths. It may grow into the bronchial lumen, producing an intraluminal mass. With further enlargement the bronchus may become obstructed, leading to distal atelectasis and infection. The tumor may also penetrate the wall of the bronchus and infiltrate along the peribronchial tissue (see Fig. 15.34C) into the adjacent carina or mediastinum. In other instances, the tumor grows along a broad front to produce a cauliflower-like intraparenchymal mass that compresses the surrounding lung.

Histologically, squamous cell carcinoma is characterized by the presence of keratinization and/or intercellular bridges. Keratinization may take the form of squamous pearls or individual cells with markedly eosinophilic cytoplasm (see Fig. 15.34D). These features are prominent in well-differentiated tumors, are

easily seen but not extensive in moderately differentiated tumors, and are seen focally in poorly differentiated tumors. Mitotic activity is higher in poorly differentiated tumors, and poorly differentiated tumors may lack keratinization and intracellular bridges altogether. Such cases require immunohistochemistry for squamous markers (i.e., p40) to prove squamous differentiation. In the past, most squamous cell carcinomas arose centrally from the segmental or subsegmental bronchi, but the incidence of squamous cell carcinoma of the peripheral lung is increasing. Squamous metaplasia, epithelial dysplasia, and foci of frank carcinoma in situ may be seen in bronchial epithelium adjacent to the tumor mass.

Small cell carcinoma is a highly malignant tumor with a strong relationship to cigarette smoking; only about 1% occurs in nonsmokers. Tumors may arise in major bronchi or in the periphery of the lung. There is no known preinvasive phase. They are the most aggressive of lung tumors, metastasizing widely and virtually always proving to be fatal.

Small cell carcinoma is composed of relatively small cells with scant cytoplasm, ill-defined cell borders, finely granular nuclear chromatin (salt and pepper pattern), and absent or inconspicuous nucleoli (see Fig. 15.34F). The cells are round, oval, or spindle-shaped, and nuclear molding is prominent. There is no absolute size for the tumor cells, but in general they are smaller than three times the diameter of a small resting lymphocyte (a size of about 25 μ m). The mitotic count is invariably high. Necrosis is common and often extensive. Basophilic staining of vascular walls due to encrustation by DNA from necrotic tumor cells (so-called **Azzopardi effect**, seen also in Fig. 15.34F) is frequently present. Combined small cell carcinoma is a variant in which typical small cell carcinoma is mixed with non-small cell histologies such as large cell neuroendocrine carcinoma or even spindled cell morphologies resembling sarcoma.

Electron microscopy is rarely used to assist in the diagnosis but classically shows dense-core neurosecretory granules, about 100 nm in diameter, in two-thirds of cases of small cell carcinoma. Other evidence of neuroendocrine differentiation includes the expression of neuroendocrine markers such as chromogranin, synaptophysin, and CD56, and the ability of some of these tumors to secrete hormones (e.g., parathormone-related protein, a cause of paraneoplastic hypercalcemia). Among the various types of lung cancer, small cell carcinoma is most commonly associated with ectopic hormone production (discussed later).

Large cell carcinoma is an undifferentiated malignant epithelial tumor that lacks the cytologic and immunophenotypic features of other forms of lung cancer. The cells typically have large nuclei, prominent nucleoli, and a moderate amount of cytoplasm (see Fig. 15.34E). It is important to note that large cell carcinoma and large cell neuroendocrine carcinoma are not variants of the same entity but represent two discrete tumor histologies. Large cell carcinoma shows no morphologic or immunophenotypic features of differentiation (i.e., is truly undifferentiated), whereas large cell neuroendocrine carcinoma shows neuroendocrine morphology and immunophenotype.

Combined Carcinoma. Approximately 4% to 5% of all lung carcinomas have two or more histologies consisting of the aforementioned types.

Spread of Carcinoma. Any type of lung carcinoma may extend to the pleural surface and then spread within the pleural cavity or into the pericardium. Metastases to bronchial, tracheal, and mediastinal nodes can be found in most cases. The frequency of nodal involvement varies slightly with the histologic pattern but averages greater than 50%.

Distant spread of lung carcinoma occurs through both lymphatic and hematogenous pathways. These tumors often spread early in their course throughout the body except for squamous cell carcinoma, which metastasizes outside the thorax later in its course. Metastasis may be the first manifestation of an underlying occult pulmonary lesion. No organ or tissue is spared but the adrenal glands, for obscure reasons, are involved in more than half of the cases. The liver (30% to 50%), brain (20%), and bone (20%) are other favored sites of metastases.

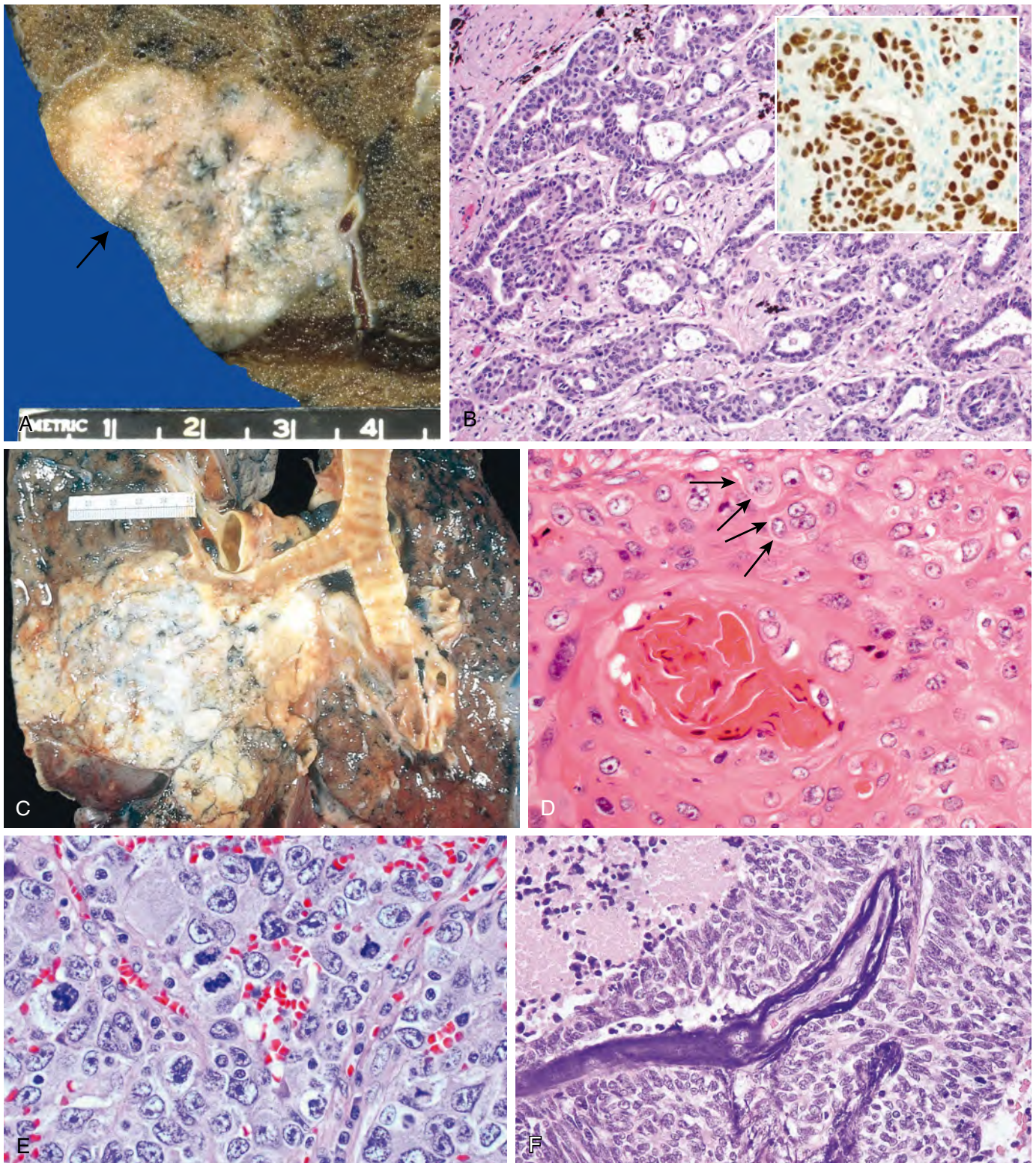


Fig. 15.34 Pathology of lung carcinomas. (A) Lung adenocarcinoma. Note the central scarring associated with anthracotic pigments and pleural puckering (arrow). (B) Gland-forming adenocarcinoma; inset shows staining for thyroid transcription factor 1 (TTF-1), which is characteristic. (C) Squamous cell carcinoma appearing as a central (hilar) mass that is invading contiguous parenchyma. (D) Well-differentiated squamous cell carcinoma, showing keratinization, pearls, and intercellular bridges (arrows). (E) Large cell carcinoma, consisting of sheets of large cells without gland formation or squamous differentiation. (F) Small cell carcinoma with small deeply basophilic cells and areas of necrosis (top left). Note basophilic staining of vascular walls due to encrustation by DNA from necrotic tumor cells (Azzopardi effect). (A, From Nose V. *Diagnostic Pathology: Familial Cancer Syndromes*, ed 2. Elsevier; 2020.)

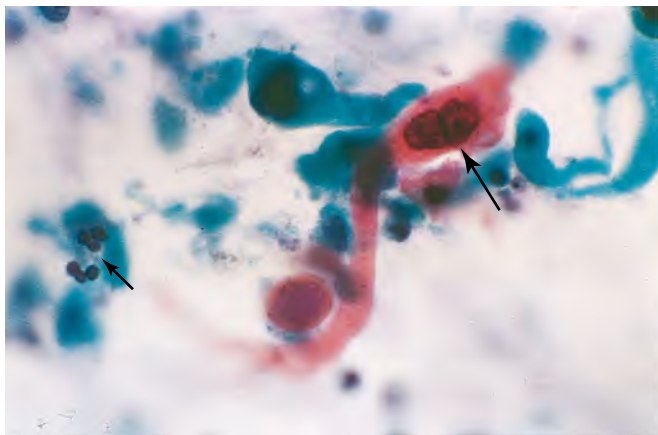


Fig. 15.35 Cytologic diagnosis of lung cancer. A sputum specimen shows an orange-staining, keratinized squamous carcinoma cell with a prominent hyperchromatic nucleus (*large arrow*). Note the size of the tumor cells compared with normal neutrophils (*small arrow*).

Secondary Pathology. Lung carcinomas have local effects that may cause several pathologic changes in the lung distal to the point of bronchial involvement. Partial obstruction may cause marked **focal emphysema**; total obstruction may lead to **atelectasis**. The impaired drainage of the airways is a common cause for **severe suppurative or ulcerative bronchitis** or **bronchiectasis**. **Pulmonary abscesses** sometimes call attention to an otherwise silent carcinoma. Compression or invasion of the superior vena cava can cause venous congestion and edema of the head and arm and, ultimately, circulatory compromise—the **superior vena cava syndrome**. Extension to the pericardial or pleural sacs may cause **pericarditis** (Chapter 12) or **pleuritis** with significant effusions. Apical lung cancers in the superior pulmonary sulcus tend to invade the neural structures around the trachea, including the cervical sympathetic plexus, and produce a group of clinical findings that includes severe pain in the distribution of the ulnar nerve and **Horner syndrome** (enophthalmos, ptosis, miosis, and anhidrosis) on the same side as the lesion. Such tumors are also referred to as *Pancoast tumors*.

Staging

A uniform TNM (tumor-node-metastasis) system for staging cancer according to its anatomic extent at the time of diagnosis is useful, particularly for comparing treatment results from different centers (**Table 15.10**). Lung tumors are staged primarily according to size, but involvement of certain thoracic structures (such as the mainstem bronchus, pleura, and chest wall) upgrades the tumor stage (pT) of relatively small tumors.

Clinical Features

Lung cancer is one of the most insidious and aggressive neoplasms in the realm of oncology. In the usual case it is discovered in patients in their 50s or older whose symptoms are of several months' duration. The major presenting complaints are cough (75%), weight loss (40%), chest pain (40%), and dyspnea (20%). Some of the more common local manifestations of lung cancer and their pathologic bases are listed in **Table 15.11**.

Not infrequently, lung cancer is recognized though biopsy of tissues involved by metastatic disease. Symptoms of metastases depend on the site, for example, back pain due to bone metastases, or headache, hemiparesis, cranial nerve damage, and seizures due to brain metastases.

TABLE 15.10 TNM Staging System for Lung Cancer

TNM Staging			
Tis	Carcinoma in situ Adenocarcinoma in situ: adenocarcinoma with pure lepidic pattern, ≤3 cm Squamous cell carcinoma in situ		
T1	Tumor ≤3 cm without pleural or mainstem bronchus involvement (T1mi, minimally invasive adenocarcinoma; T1a, <1 cm; T1b, 1–2 cm; T1c, 2–3 cm)		
T2	Tumor 3–5 cm or involvement of mainstem bronchus but not of carina, visceral pleural involvement, or lobar atelectasis (T2a, 3–4 cm, or <3 cm with pleural involvement; T2b, 4–5 cm)		
T3	Tumor >5–7 cm or one with involvement of parietal pleura, chest wall (including superior sulcus tumors), diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium, or separate tumor nodules in the same lobe		
T4	Tumor >7 cm or any tumor with invasion of mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, or carina, or separate tumor nodules in a different ipsilateral lobe		
N0	No metastasis to regional lymph nodes		
N1	Ipsilateral intraparenchymal or peribronchial or hilar nodal involvement		
N2	Metastasis to ipsilateral mediastinal or subcarinal lymph nodes		
N3	Metastasis to contralateral mediastinal or hilar lymph nodes, ipsilateral or contralateral scalene, or supraclavicular lymph nodes		
M0	No distant metastasis		
M1	Distant metastasis (M1a, separate tumor nodule in contralateral lobe or pleural nodules or malignant pleural or pericardial effusion; M1b, single extrathoracic metastasis in a single organ; M1c, multiple extrathoracic metastases)		
Stage Grouping			
Stage 0	Tis	N0	M0
Stage IA	IA1, T1mi or T1a; IA2, T1b; IA3, T1c	N0	M0
Stage IB	T2a	N0	M0
Stage IIA	T2b	N0 N1	M0 M0
Stage IIB	T2b T1a, T1b, T1c, T2a, T2b T3	N0 N1 N0	M0 M0 M0
Stage IIIA	T1a, T1b, T1c, T2a, T2b T3 T4	N2 N1 N0, N1	M0 M0 M0
Stage IIIB	T1a, T1b, T1c, T2a, T2b T3, T4	N3 N2	M0 M0
Stage IIIC	T3, T4	N3	M0
Stage IVA	T any	N any	M1a, M1b
Stage IVB	T any	N any	M1c

The best “treatment” for lung cancer is smoking prevention, which has lowered lung cancer incidence in the United States among adult males. However, approximately 15% of adults still smoke, and even those who quit remain at elevated risk for an extended period of time. This reality has led to early detection trials in high-risk individuals using low-dose computed tomography, which can detect some early

TABLE 15.11 Local Effects of Lung Tumor Spread

Clinical Feature	Pathologic Basis
Cough (50%–75%)	Involvement of central airways
Hemoptysis (25%–50%)	Hemorrhage from tumor in airway
Chest pain (20%)	Extension of tumor into mediastinum, pleura, or chest wall
Pneumonia, abscess, lobar collapse	Airway obstruction by tumor
Lipoid pneumonia	Tumor obstruction; accumulation of cellular lipid in foamy macrophages
Pleural effusion	Tumor spread into pleura
Hoarseness	Recurrent laryngeal nerve invasion
Dysphagia	Esophageal invasion
Diaphragm paralysis	Phrenic nerve invasion
Rib destruction	Chest wall invasion
SVC syndrome	SVC compression by tumor
Horner syndrome	Sympathetic ganglia invasion
Pericarditis, tamponade	Pericardial involvement

SVC, Superior vena cava.

(resectable) cancers but has a high incidence of false-positive (noncancer) findings. Overall, the outlook is poor for most patients. Even with incremental improvements in thoracic surgery, radiation therapy, and chemotherapy, the overall 5-year survival rate is still only 18.7%. The 5-year survival rate is 52% for cases detected when the disease is still localized, 22% when there is regional metastasis, and only 4% with distant metastases. In general, adenocarcinoma and squamous cell carcinoma tend to remain localized longer and have a slightly better prognosis than small cell carcinoma, which is usually advanced by the time it is discovered.

Treatment of patients with adenocarcinoma and activating mutations in *EGFR* (present in about 15% of all cases) or in other tyrosine kinases with specific kinase inhibitors prolongs survival. Many tumors that recur carry new mutations that generate resistance to these inhibitors, proving that these drugs are “hitting” their target. In contrast, activating *KRAS* mutations (present in approximately 30% of cases of adenocarcinoma) appear to be associated with a worse prognosis, regardless of treatment, in an already grim disease. This picture has brightened a bit with the development of cysteine residue-reactive *KRAS* inhibitors that target particular mutated forms of *KRAS*. Because of the mutagenic effects of carcinogens in tobacco smoke, lung cancers have a high burden of potentially antigenic neoantigens. Accordingly, both adenocarcinoma and squamous cell carcinoma respond in subsets of cases to checkpoint inhibitor therapy, which has produced improvements in survival and is now approved for use.

Small cell carcinoma is quite sensitive to radiation therapy and chemotherapy, and approximately 10% of patients with limited disease survive for 5 years and may be cured. Unfortunately, however, most patients present with advanced stage disease; for these patients, despite excellent initial responses to chemotherapy, the median survival is approximately 10 months, and the cure rate is close to zero. With the advent of lung cancer screening, some small cell carcinomas are detected at an earlier stage and can therefore be treated more promptly (even surgically in some cases). While preliminary studies show improved survival for early-stage small cell carcinoma relative to widely metastatic cases, such cases still represent <5% of all small cell carcinomas. New approaches involving use of antibody-drug conjugates that

deliver chemotherapy selectively to tumor cells and immune checkpoint inhibitors are being tested.

Paraneoplastic Syndromes

Lung carcinoma can be associated with several paraneoplastic syndromes (Chapter 7), some of which may antedate the development of a detectable pulmonary lesion. The hormones or hormone-like factors elaborated by lung cancer cells and associated syndromes include:

- *Antidiuretic hormone* (ADH), inducing hyponatremia as part of the syndrome of inappropriate ADH secretion
- *Adrenocorticotrophic hormone* (ACTH), producing Cushing syndrome
- *Parathormone, parathyroid hormone-related peptide, prostaglandin E, and some cytokines*, all implicated in the hypercalcemia often seen with lung cancer
- *Calcitonin*, causing hypocalcemia
- *Gonadotropins*, causing gynecomastia
- *Serotonin and bradykinin*, associated with the carcinoid syndrome

The incidence of clinically significant paraneoplastic syndromes related to these factors in lung cancer patients ranges from 1% to 10%. Any histologic type of tumor may occasionally produce any one of the hormones, but tumors that produce ACTH and ADH are predominantly small cell carcinomas, whereas those that produce hypercalcemia are mostly squamous cell carcinomas.

Other systemic manifestations of lung carcinoma include the *Lambert-Eaton myasthenic syndrome* (Chapter 27), in which muscle weakness is caused by autoantibodies against the neuronal calcium channel; *peripheral neuropathy*, usually purely sensory; dermatologic abnormalities, including *acanthosis nigricans* (Chapter 25); hematologic abnormalities, such as *leukemoid reactions*; hypercoagulable states, such as *Trousseau syndrome* (deep vein thrombosis and thromboembolism); and finally, a peculiar abnormality of connective tissue called *hypertrophic pulmonary osteoarthropathy*, associated with clubbing of the fingers.

KEY CONCEPTS

Carcinomas of the Lung

- The three major histologic subtypes are adenocarcinoma (most common), squamous cell carcinoma, and small cell carcinoma.
- Each of these is clinically and genetically distinct. Most small cell lung carcinomas are best treated by chemotherapy because almost all are metastatic at presentation. Those which are found at an early stage may be treated with a combination of surgery and chemo/radiotherapy. The other carcinomas may be curable by surgery if limited to the lung. Combination chemotherapy also is available along with tyrosine kinase inhibitors for tumors with *EGFR*, *ALK*, *ROS*, and *MET* mutations and *KRAS* inhibitors for specific mutated *KRAS* variants.
- Smoking is the most important risk factor for lung cancer; the most common subtype related to smoking in adults is adenocarcinoma. Adenocarcinoma is also the most common subtype in nonsmokers. Almost all small cell carcinomas are related to smoking.
- Precursor lesions include atypical adenomatous hyperplasia and adenocarcinoma in situ (formerly bronchioloalveolar carcinoma) for adenocarcinomas and squamous dysplasia for squamous cell carcinoma.
- Tumors 3 cm or less in diameter characterized by pure growth along pre-existing structures (lepidic pattern) without stromal invasion are now called adenocarcinoma in situ. Lepidic growth is the only noninvasive pattern of growth for adenocarcinomas.
- Lung cancers, particularly small cell lung carcinomas, often cause paraneoplastic syndromes.

Neuroendocrine Proliferations and Tumors

The normal lung contains neuroendocrine cells within the epithelium as single cells or as clusters called neuroepithelial bodies. Virtually all pulmonary neuroendocrine cell hyperplasias are secondary to airway fibrosis and/or inflammation. The exception is a rare disorder called *diffuse idiopathic pulmonary neuroendocrine cell hyperplasia*, in which hyperplasia occurs in the absence of an inflammatory stimulus.

Neoplasms of neuroendocrine cells in the lung include benign *tumors*, small, inconsequential, hyperplastic nests of neuroendocrine cells seen in areas of scarring or chronic inflammation; *carcinoids*; and the (already discussed) highly aggressive neuroendocrine carcinomas (small cell carcinoma and large cell neuroendocrine carcinoma). Carcinoid tumors are classified separately from neuroendocrine carcinomas and carcinomas in general since they differ significantly in terms of incidence and clinical, epidemiologic, histologic, and molecular characteristics.

Carcinoid Tumors

Carcinoid tumors represent 1% to 5% of all lung tumors. Most patients with these tumors are younger than 60 years of age, and the incidence is equal for both sexes. Approximately 20% to 40% of patients are nonsmokers. Carcinoid tumors are low-grade malignant epithelial neoplasms that are subclassified into *typical* and *atypical carcinoids*.

MORPHOLOGY

Carcinoids may arise centrally or may be peripheral. On gross examination, the central tumors grow as finger-like or spherical polypoid masses that commonly project into the lumen of the bronchus and are usually covered by an intact mucosa (Fig. 15.36A). They rarely exceed 3 to 4 cm in diameter. Most are confined to the mainstem bronchi. Others, however, penetrate the bronchial wall to fan out in the peribronchial tissue, producing the so-called **collar-button lesion**. Peripheral tumors are solid and nodular.

Histologically, the tumor is composed of organoid, trabecular, palisading, ribbon, or rosette-like arrangements of cells separated by a delicate fibrovascular stroma, similar to neuroendocrine tumors at other sites (Fig. 15.36B). Grading of lung carcinoid tumors is based on morphologic criteria and does not currently require the use of proliferation markers (such as Ki67), in contrast to neuroendocrine tumors arising in other organ systems. Typical carcinoids (corresponding to low-grade neuroendocrine tumors) have less than 2 mitoses

per 10 high-power fields and lack necrosis, while atypical carcinoids (corresponding to intermediate grade neuroendocrine tumors) have between 2 and 10 mitoses per 10 high-power fields and/or foci of necrosis. Atypical carcinoids also show increased pleomorphism, have more prominent nucleoli, and are more likely to grow in a disorganized fashion and invade lymphatics. On electron microscopy the cells exhibit the dense-core granules characteristic of other neuroendocrine tumors and, by immunohistochemistry, are found to contain serotonin, neuron-specific enolase, bombesin, calcitonin, or other peptides.

Clinical Features

The clinical manifestations of bronchial carcinoids result from their intraluminal growth, their capacity to metastasize, and the ability of some of the lesions to elaborate vasoactive amines. Persistent cough, hemoptysis, impairment of drainage of respiratory passages with secondary infections, bronchiectasis, emphysema, and atelectasis are all by-products of the intraluminal growth of these lesions.

Lesions composed of functional cells are capable of producing the classic *carcinoid syndrome*, characterized by intermittent attacks of diarrhea, flushing, and cyanosis. Approximately, 10% of bronchial carcinoids give rise to this syndrome. Overall, most bronchial carcinoids do not have secretory activity and do not metastasize to distant sites but instead follow a relatively benign course for long periods. These lesions are therefore mostly amenable to resection. The reported 5-year survival rates are 95% for typical carcinoids and 70% for atypical carcinoids, compared to 30% for large cell neuroendocrine carcinoma and 5% for small cell carcinoma, respectively.

Miscellaneous Tumors

Benign and malignant mesenchymal tumors, such as inflammatory myofibroblastic tumor, fibroma, fibrosarcoma, lymphangioleiomyomatosis, leiomyoma, leiomyosarcoma, lipoma, hemangioma, and chondroma, may occur in the lung but are rare. Hematolymphoid tumors similar to those described in other organs may also affect the lung, either as isolated lesions or, more commonly, as part of a generalized disorder. These include Langerhans cell histiocytosis, Hodgkin lymphomas, lymphomatoid granulomatosis, an unusual EBV-positive B-cell lymphoma, and low-grade extranodal marginal zone B-cell lymphoma (Chapter 13).

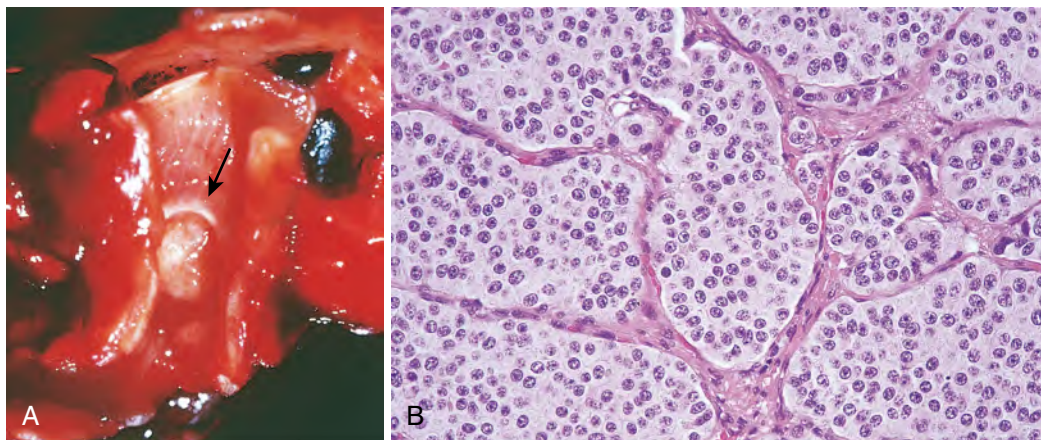


Fig. 15.36 Bronchial carcinoid. (A) Carcinoid growing as a spherical mass (arrow) protruding into the lumen of the bronchus. (B) The tumor cells have small, rounded, uniform nuclei and moderate amounts of cytoplasm. (Courtesy Dr. Thomas Krausz, Department of Pathology, The University of Chicago, Pritzker School of Medicine.)

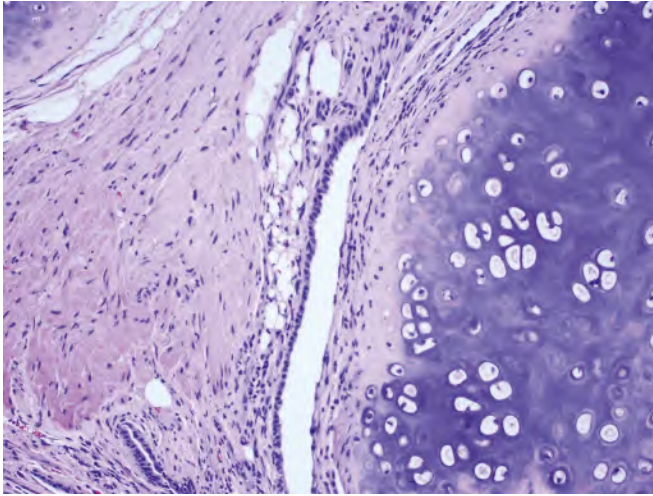


Fig. 15.37 Pulmonary hamartoma. There are islands of cartilage, fat, smooth muscle, and entrapped respiratory epithelium.

Pulmonary *hamartoma* is a relatively common benign lesion that is usually discovered as an incidental, rounded radio-opacity (*coin lesion*) on a routine chest x-ray. Most are solitary, less than 3 to 4 cm in diameter, and well circumscribed. Pulmonary hamartoma consists of nodules of connective tissue intersected by epithelial clefts. Cartilage is the most common connective tissue, but there may also be fibrous tissue and fat. The clefts are lined by ciliated or nonciliated epithelium (Fig. 15.37). The traditional term *hamartoma* is retained for this lesion, but it is in fact a clonal neoplasm associated with chromosomal aberrations involving either 6p21 or 12q14-q15. These aberrations are found in the mesenchymal component, while the epithelial component appears to represent entrapped respiratory epithelium. The presence of entrapped epithelial elements distinguishes hamartoma from chondroma, a benign cartilaginous neoplasm that can arise in the lung as well as other sites.

Lymphangioliomyomatosis is a pulmonary disorder that primarily affects young females of childbearing age. It is characterized by a proliferation of perivascular epithelioid cells that express markers of both melanocytes and smooth muscle cells. The proliferation distorts the involved lung, leading to cystic, emphysema-like dilation of terminal airspaces, thickening of the interstitium, and obstruction of lymphatic vessels. The lesional epithelioid cells frequently harbor loss-of-function mutations in the tumor suppressor *TSC2*, one of the loci linked to tuberous sclerosis (Chapter 28). The protein encoded by *TSC2*, tuberin, is a negative regulator of mechanistic target of rapamycin (mTOR), a key regulator of cellular metabolism. While *TSC2* mutations point to increased mTOR activity as a pathogenic factor, the disorder remains poorly understood. The strong tendency to affect young females suggests that estrogen contributes to the proliferation of perivascular epithelioid cells, which often express estrogen receptors. Patients most commonly present with dyspnea or spontaneous pneumothorax, the latter related to emphysematous changes. The disease tends to be slowly progressive over a period of several decades. mTOR inhibitors slow or prevent the deterioration of lung function but must be continued indefinitely. Only lung transplantation is curative.

Inflammatory myofibroblastic tumor is a rare tumor that is more common in children, with an equal male-to-female ratio. Presenting symptoms include fever, cough, chest pain, and hemoptysis. It may also be asymptomatic. Imaging studies usually show a round, well-defined, peripheral mass. Calcification is present in about a quarter of cases. Grossly, the lesion is firm, 3 to 10 cm in diameter, and grayish

TABLE 15.12 Mediastinal Neoplasms and Other Masses

Anterior Mediastinum
Thymoma
Teratoma
Lymphoma
Thyroid lesions
Parathyroid tumors
Metastatic carcinoma
Posterior Mediastinum
Neurogenic tumors (schwannoma, neurofibroma)
Lymphoma
Metastatic tumor (most are from the lung)
Bronchogenic cyst
Gastroenteric hernia
Middle Mediastinum
Bronchogenic cyst
Pericardial cyst
Lymphoma

white. Microscopically, there is proliferation of spindle-shaped fibroblasts and myofibroblasts, lymphocytes, plasma cells, and peripheral fibrosis. Some of these tumors have activating rearrangements of the *ALK* receptor tyrosine kinase gene, located on 2p23, and treatment with *ALK* inhibitors have produced sustained responses in such cases.

Tumors in the mediastinum may arise in mediastinal structures or may be metastatic from the lung or other organs. They often invade or compress the lungs. Table 15.12 lists the most common tumors in the various compartments of the mediastinum. Specific tumor types are discussed in appropriate sections of this book.

Metastatic Tumors

The lung is the most common site of metastatic neoplasms. Both carcinomas and sarcomas arising anywhere in the body may spread to the lungs via the blood or lymphatics or by direct continuity. Esophageal carcinoma and mediastinal lymphoma may also invade the lung by direct extension.

MORPHOLOGY

The pattern of metastatic spread to the lungs is quite variable. In the usual case, multiple discrete nodules (cannonball lesions) are scattered throughout all lobes, particularly in the lung periphery (Fig. 15.38). Other times spread takes the form of a solitary nodule, endobronchial or pleural involvement, pneumonic consolidation, and/or some combination thereof. Foci of lepidic growth similar to adenocarcinoma in situ are seen occasionally with metastatic carcinomas and may be associated with any of the listed patterns.

PLEURA

Most pleural disorders stem from complications of disease arising elsewhere in the body. Secondary infections and inflammations are particularly common findings at autopsy. Important primary disorders include (1) intrapleural bacterial infections, presumably the product of seeding from transient bacteremia; and (2) mesothelioma, a primary pleural neoplasm (discussed later).

Pleural Effusion

Pleural effusion is a common manifestation of both primary and secondary pleural diseases, which may be inflammatory or noninflammatory.

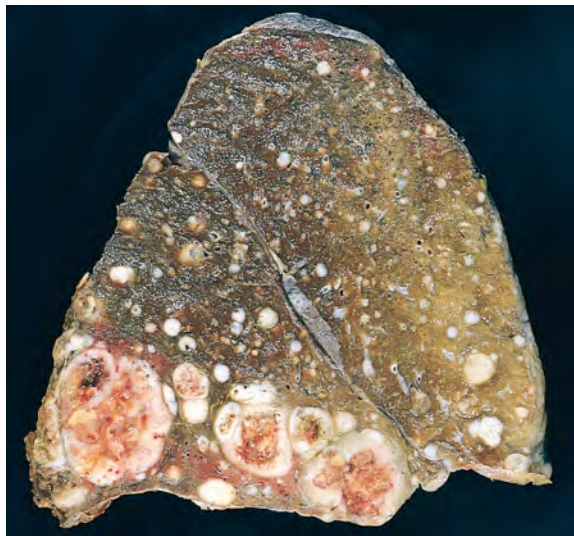


Fig. 15.38 Numerous metastases to lung from a renal cell carcinoma. (Courtesy Dr. Michelle Mantel, Brigham and Women's Hospital, Boston.)

Normally, no more than 15 mL of serous, relatively acellular, clear fluid lubricates the pleural surface. Accumulation of pleural fluid occurs in the following settings:

- *Increased hydrostatic pressure*, as in congestive heart failure
- *Increased vascular permeability*, as in pneumonia
- *Decreased osmotic pressure*, as in nephrotic syndrome
- *Increased intrapleural negative pressure*, as in atelectasis
- *Decreased lymphatic drainage*, as in mediastinal carcinomatosis

Inflammatory Pleural Effusions

Serous, serofibrinous, and fibrinous pleuritis all have an inflammatory basis, differing only in the intensity and duration of the process. The most common causes of pleuritis are disorders associated with inflammation of the underlying lung, such as tuberculosis, pneumonia, lung infarction, lung abscess, and bronchiectasis. Rheumatoid arthritis, systemic lupus erythematosus, uremia, diffuse systemic infections, and metastatic involvement of the pleura can also cause serous or serofibrinous pleuritis. Radiotherapy for tumors in the lung or mediastinum often causes a serofibrinous pleuritis. In most of these disorders, the pleural reaction is minimal, and the fluid exudate is resorbed with either resolution or organization of the fibrinous component. However, large amounts of fluid sometimes accumulate and compress the lung, causing respiratory distress.

A purulent pleural exudate (*empyema*) usually results from bacterial or fungal seeding of the pleural space. Most commonly, this seeding occurs by contiguous spread of organisms from intrapulmonary infection, but occasionally it occurs through lymphatic or hematogenous dissemination from a more distant site. Rarely, infections below the diaphragm, such as the subdiaphragmatic or liver abscess, may extend through the diaphragm into the pleural spaces, more often on the right side.

Empyema is characterized by loculated, yellow-green, creamy pus composed of masses of neutrophils admixed with other leukocytes. Although empyema may accumulate to large volumes (up to 500 to 1000 mL), usually the volume is small, and the pus becomes walled off by fibrosis. Empyema may resolve, but more often the exudate organizes into dense, tough fibrous adhesions that frequently obliterate the pleural space or envelop the lungs; either can seriously restrict pulmonary expansion.

Hemorrhagic pleuritis manifested by sanguineous inflammatory exudates is infrequent and is most often associated with hemorrhagic diatheses, rickettsial infections, and neoplastic involvement of the pleural cavity. The sanguineous exudate must be differentiated from hemothorax (discussed later). When hemorrhagic pleuritis is encountered, a careful search should be made for the presence of tumor cells.

Noninflammatory Pleural Effusions

Noninflammatory collections of serous fluid within the pleural cavities are called *hydrothorax*. The fluid is clear and straw colored. Hydrothorax may be unilateral or bilateral, depending on the underlying cause. The most common cause of hydrothorax is heart failure, and for this reason it is usually accompanied by pulmonary congestion and edema. Hydrothorax may also be seen in any other systemic disease associated with generalized edema, such as patients with renal failure or cirrhosis of the liver.

The escape of blood into the pleural cavity is known as *hemothorax*. It is most commonly a complication of trauma or less commonly surgery, but can also accompany rupture of an aortic aneurysm, a setting in which it is almost invariably fatal.

Chylothorax is an accumulation of milky fluid, usually of lymphatic origin, in the pleural cavity. Chyle is milky white because it contains finely emulsified fats. Chylothorax is most often caused by thoracic duct trauma or by obstruction of a major lymphatic duct, usually by a malignancy. Such cancers most commonly arise within the thoracic cavity and invade the lymphatics locally, but occasionally more distant cancers metastasize via the lymphatics and grow within the right lymphatic or thoracic duct, producing obstruction.

Pneumothorax

Pneumothorax refers to air or gas in the pleural cavities and is most commonly associated with emphysema, asthma, and tuberculosis. It may be spontaneous, traumatic, or therapeutic. Spontaneous pneumothorax may complicate any form of pulmonary disease that causes emphysematous changes. An abscess cavity that communicates either directly with the pleural space or with the lung interstitial tissue may also lead to the escape of air. In the latter circumstance the air may dissect through the lung substance or back through the mediastinum (interstitial emphysema), eventually entering the pleural cavity. Traumatic pneumothorax is usually caused by some perforating injury to the chest wall, but sometimes the trauma pierces the lung and thus provides two avenues for the accumulation of air within the pleural spaces. Resorption of the air in the pleural space occurs in spontaneous and traumatic pneumothorax, provided that the original communication seals itself.

Of the various forms of pneumothorax, the one that attracts the most clinical attention is *spontaneous idiopathic pneumothorax*. This entity is encountered in relatively young people; seems to be due to rupture of small, peripheral, usually apical subpleural blebs; and usually subsides spontaneously as the air is resorbed. Recurrent attacks are common and can be quite disabling.

Pneumothorax may cause marked respiratory distress due to collapse and atelectasis of the lung. In some instances, the pleural defect acts as a flap valve and permits the entrance of air during inspiration but fails to permit its escape during expiration. The result is called a *tension pneumothorax*, in which progressively increasing intrapleural pressure may compress vital mediastinal structures and the contralateral lung (Fig. 15.39).

Pleural Tumors

The pleura may be involved by primary or secondary tumors. Secondary metastatic involvement is far more common than are primary tumors. The most frequent metastatic malignancies arise from primary



Fig. 15.39 Tension pneumothorax (right-sided) with shifting of the mediastinum to the contralateral thorax.

neoplasms of the lung and breast. In addition to these cancers, malignancy from any organ of the body may spread to the pleural spaces. Ovarian carcinomas, for example, tend to cause widespread implants in both the abdominal and thoracic cavities. Most metastatic implants produce a serous or serosanguineous effusion that often contains neoplastic cells. For this reason, careful cytologic examination of the sediment is of considerable diagnostic value.

Solitary Fibrous Tumor

Solitary fibrous tumor is a soft tissue tumor with a propensity to occur in the pleura and, less commonly, in the lung, as well as other sites. It is highly associated with a cryptic inversion of chromosome 12 involving the genes *NAB2* and *STAT6*. This rearrangement creates a *NAB2::STAT6* fusion gene that appears to be virtually unique to solitary fibrous tumor. It encodes a chimeric transcription factor that is hypothesized to be a key driver of tumor development.

MORPHOLOGY

Grossly, solitary fibrous tumor consists of dense fibrous tissue with occasional cysts filled with viscid fluid. The tumor is often attached to the pleural surface by a pedicle. It may be small (1 to 2 cm in diameter) or may reach an enormous size, but it tends to remain confined to the surface of the lung (Fig. 15.40). Microscopically, the tumor shows whorls of reticulin and collagen fibers among which are interspersed spindle cells resembling fibroblasts. Rarely, this



Fig. 15.40 Solitary fibrous tumor. Cut surface is solid with a whorled appearance. (Courtesy Dr. Justine A. Barletta, Department of Pathology, Brigham and Women's Hospital, Boston.)

tumor may be malignant, marked by pleomorphism, mitotic activity, necrosis, and large size (>10 cm). The tumor cells are positive for CD34 and STAT6 and negative for keratin by immunostaining, features that help to distinguish this lesion from malignant mesothelioma (which has the opposite phenotype). The solitary fibrous tumor has no relationship to asbestos exposure.

Malignant Mesothelioma

Malignant mesothelioma, although rare, has assumed great importance in the past few decades because of its increased incidence among people with heavy exposure to asbestos (see pneumoconioses). Thoracic mesothelioma arises from either the visceral or the parietal pleura. In coastal areas with shipping industries in the United States and Great Britain, as well as in Canadian, Australian, and South African mining areas, as many as 90% of mesotheliomas are asbestos-related. The lifetime risk of developing mesothelioma in heavily exposed individuals is as high as 7% to 10%. There is a long latent period of 25 to 45 years for the development of asbestos-related mesothelioma, and there seems to be no increased risk of mesothelioma in asbestos workers who smoke. This is in contrast to the risk of asbestos-related lung carcinoma, which is markedly magnified by smoking. Thus, asbestos workers (particularly those who smoke) are at much higher risk of dying of lung carcinoma than mesothelioma.

Asbestos bodies (see Fig. 15.18A) are found in increased numbers in the lungs of patients with mesothelioma. Another marker of asbestos exposure, the *asbestos plaque*, has been previously discussed (see Fig. 15.18B).

Although several cytogenetic abnormalities have been detected, the most common is homozygous deletion of chromosome 9p leading to loss of the tumor suppressor gene *CDKN2A*, which occurs in about 80% of mesotheliomas. Sequencing of mesothelioma genomes has shown that driver mutations are also common in the *NF2* (neurofibromatosis-2) gene, which encodes a cell signaling regulator; and *BAP1*, which encodes a protein that interacts with the *BRCA1* tumor suppressor and appears to function as a chromatin regulator. Of note, individuals with germline mutations in *BAP1* have a markedly elevated risk of mesothelioma, further implicating this tumor suppressor gene in the pathogenesis of the disease.

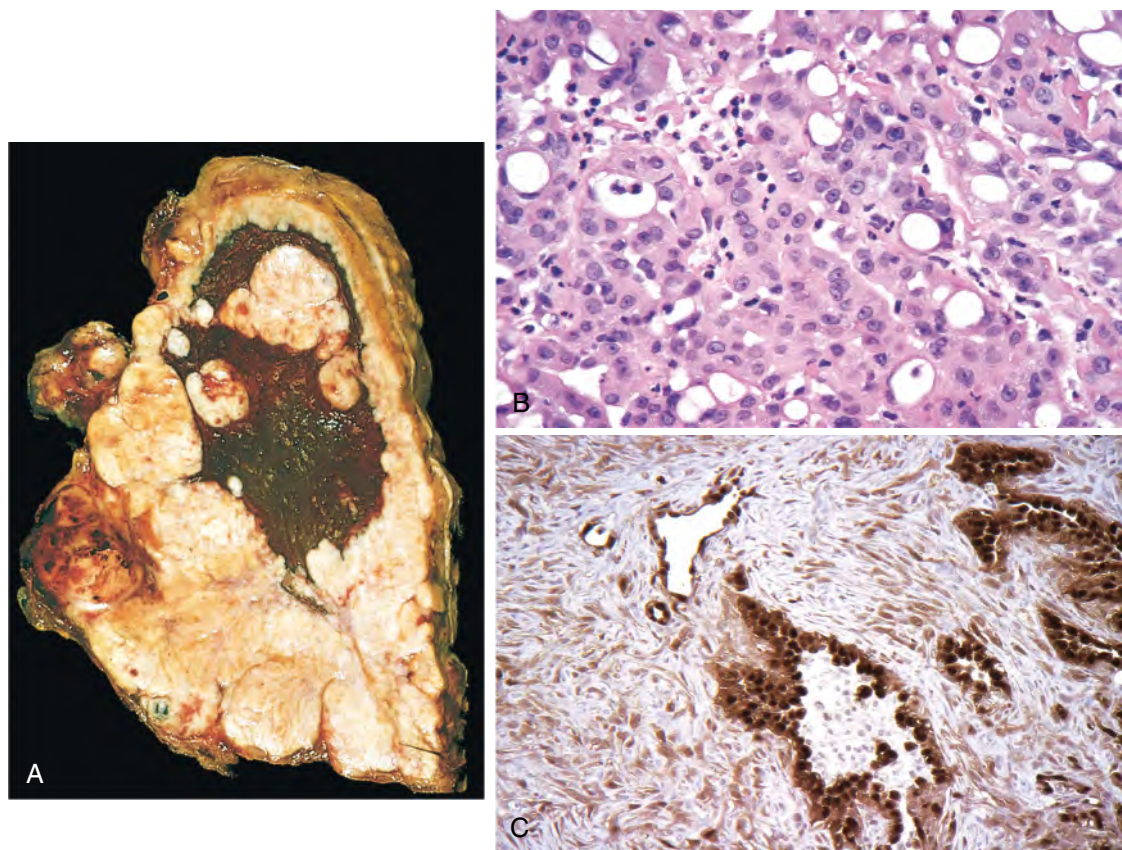


Fig. 15.41 Malignant mesothelioma. Note the thick, firm, white pleural tumor tissue that ensheathes the lung (A). Histologic variants of malignant mesothelioma. (B) Epithelioid type. (C) Biphasic type, stained for calretinin (immunoperoxidase method). The epithelial component is strongly positive (dark brown), while the sarcomatoid component is less so. (Courtesy Dr. Thomas Krausz, Department of Pathology, The University of Chicago, Pritzker School of Medicine.)

MORPHOLOGY

Malignant mesothelioma is a diffuse lesion arising from either the visceral or the parietal pleura that spreads widely in the pleural space and is usually associated with extensive pleural effusion and direct invasion of thoracic structures. The affected lung becomes ensheathed by a thick layer of soft, gelatinous, grayish pink tumor tissue (Fig. 15.41A).

Microscopically, malignant mesothelioma may be epithelioid (60% to 80%), sarcomatoid (10% to 12%), or biphasic (10% to 15%). This is in keeping with the fact that mesothelial cells have the potential to develop as epithelium-like cells or mesenchymal stromal cells.

The **epithelioid type** of mesothelioma consists of cuboidal, columnar, or flattened cells forming tubular or papillary structures resembling adenocarcinoma (Fig. 15.41B). Immunohistochemical stains are very helpful in differentiating it from pulmonary adenocarcinoma. Most mesotheliomas show strong positivity for keratin proteins, calretinin (Fig. 15.41C), Wilms tumor 1 (WT-1), cytokeratin 5/6, and podoplanin, and unlike adenocarcinomas are negative for markers of epithelial differentiation like Claudin-4 and BerEp4. This panel of antibodies is diagnostic in most cases when interpreted in the context of morphology and clinical presentation. The **sarcomatoid type** of mesothelioma has an appearance resembling fibrosarcoma and is usually extensively fibrotic and paucicellular. Sarcomatoid mesotheliomas tend to have lower expression of many of the markers described above and some may be positive only for keratin. The **biphasic type** of mesothelioma contains both epithelioid and sarcomatoid patterns in varying proportions and tends to behave similarly to sarcomatoid type (see Fig. 15.41C).

Clinical Features

The presenting symptoms are chest pain, dyspnea, and, as noted, recurrent pleural effusions. Concurrent pulmonary asbestosis (fibrosis) is present in only 20% of individuals with pleural mesothelioma. The lung is invaded directly, and there is often metastatic spread to the hilar lymph nodes and, eventually, to the liver and other distant organs. Fifty percent of patients die within 12 months of diagnosis, and few survive longer than 2 years. Aggressive therapy (extrapleural pneumonectomy, chemotherapy, radiation therapy) seems to improve this poor prognosis in some patients.

Mesotheliomas also arise in the peritoneum, pericardium, tunica vaginalis, and genital tract (benign adenomatoid tumor) (see Chapter 21). *Peritoneal mesotheliomas* are related to heavy asbestos exposure in 60% of male patients (the number is much lower in females). Although in about half of cases the disease remains confined to the abdominal cavity, intestinal involvement frequently leads to death from intestinal obstruction or inanition.

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